# The working (6,7): a complete named derivation

99 graphical steps | Four explicit sphere moves | Portrait US Letter

**VERY BRIEF SUMMARY.** Remove the incident half twists, use a five-point disk, turn the remaining positive factors into FOUR exterior twists by PC lanterns, roses, daisies and Hurwitz transports, then cap their fourth power.

The previously missing geometric boundary closure is now supplied. The final equality is obtained from actual named substitutions and factor transports, rather than a whole-product action recognition. All four sphere moves retain their disk discrepancies, exact sphere certificates and separately checked genus-two lift.

**What is still open.** This detailed route has 99 retained graphical steps: two fewer than the previous 101-step named proof and shorter than the earlier 177-step archive. It does not yet meet the goal of fewer than twenty meaningful operations. Reusable named patterns and fixed-factor drags are identified, with all adjacent checks retained in the certificates.

**How to read it.** Read each factorization downward. Curves are paired with the braid representing that exact support; blue marked-point dots continue at braid separators. Only the changed interval is printed at a move, with the unchanged factors counted above and below. Every actual SPHERE MOVE has a yellow banner. Individual supports are not cumulative joint cut-system certificates.

**Logical scope.** Common frames and cyclic starts are explicit conjugations. The terminal four-exterior product is central, so reversing those conjugations recovers the source equality without assuming it was central. Marked points are not framed boundaries; signed framing balances are retained for PC substitutions. A single five-point cap has lift -I; the sixth- and fourth-power caps used here have +I.

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99 graphical steps retained. Four sphere moves labeled; full discrepancies in Section IX.

# What the steps will do

**1. Reach the five-point chart.** Split the required squares, match and absorb the incident half twists, change the exterior point explicitly, and retain then separately cap the six framing corrections.

**2. Use the existing roots.** Borrow Q14 from the actual four-point daisy, transport it to match the original half twists, return the same Q14 and remove the last half pair. No auxiliary half twists are introduced.

**3. Extract two exterior twists.** Actual cabled lanterns and borrowed-petal configurations expose two visible exterior twists. Keep both while closing the residual factors.

**4. Return the borrowed root.** The actual 15, 14 and 45 curves form a transported PC rose. Return the resulting transported root across the old outer and close its daisy; an untransported root with the same enclosed points would fail.

**5. Align the four-point twists.** Two actual pair transports align the four-point curves. Combine them in the existing chart. The redundant final rounding frame is omitted; two earlier frames have already combined to a single sigma4 frame.

**6. Match two daisies and cap.** Three explicit petal swaps and two individually matched transported daisies give four identical exterior twists. Combine their fourth power, followed by a separate labeled sphere cap with +I lift.

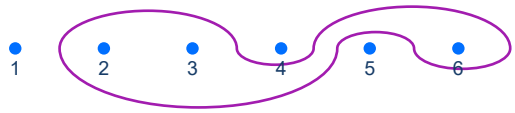
# I. The fixed-point boundary chart

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

# Complete factorization before The fixed-point boundary chart

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

**1. F1**
T3^2



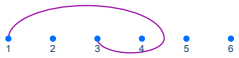

**2. F2**
H




**3. F4**
T3^2




**4. F3**
H



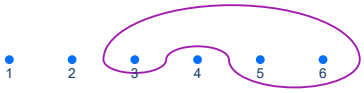

**5. F5**
T3^2




**6. F6**
H




**7. F8**
T3^2




**8. F9**
H




**9. F7**
H




**10. F10**
T3^2



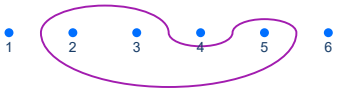

**11. F11**
H

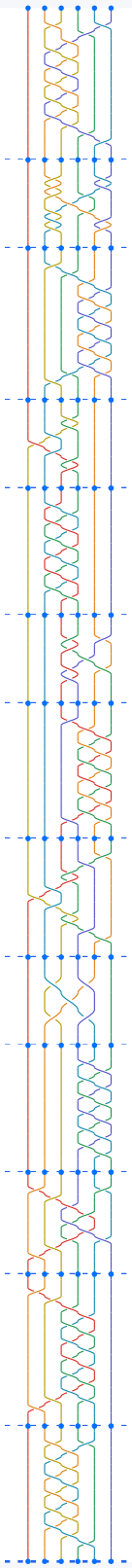



**12. F12**
T3^2




**13. F13**
T3^2



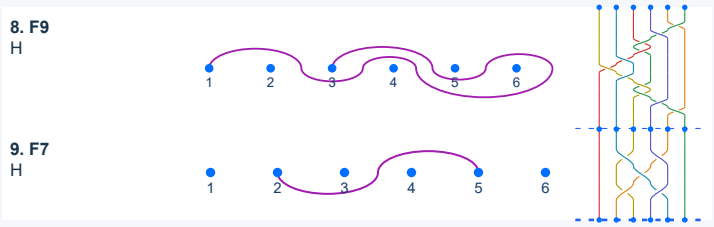
# I - step 001

Change the working ordering by an inverse Hurwitz move

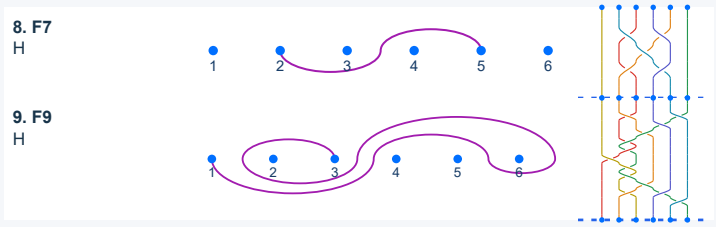
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown.

### Before




### After



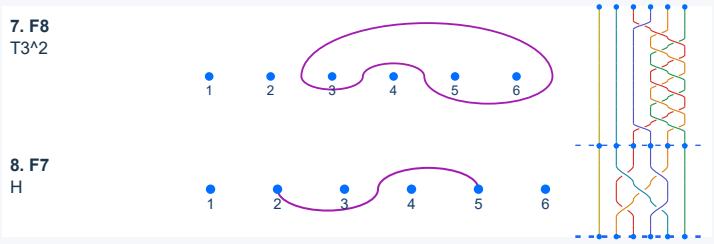

## I - step 002

Change the working ordering by an inverse Hurwitz move

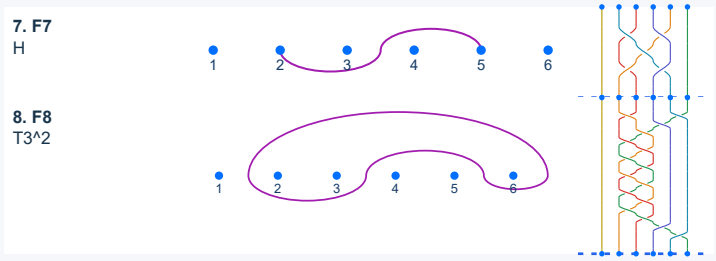
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 5 below. Only the changed interval is shown.

### Before




### After



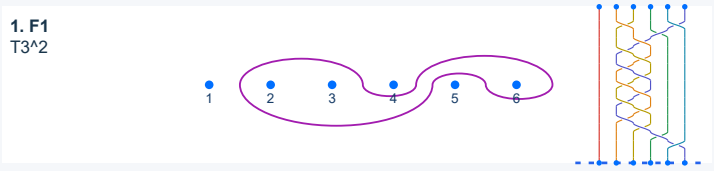

## I - step 003

Split F1 into ordinary roots

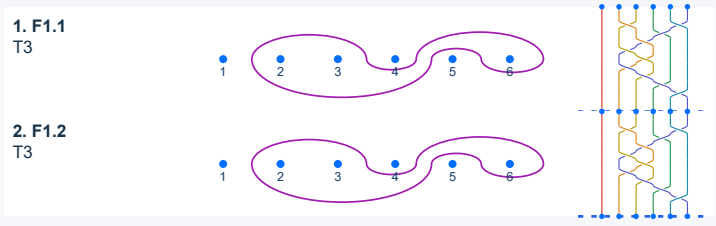
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

### Before




### After



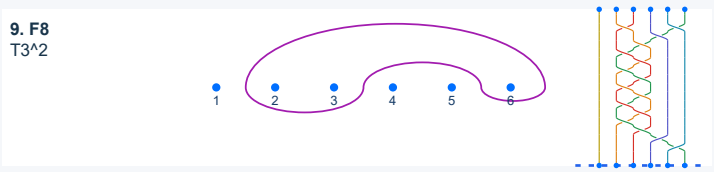
# I - step 004

Split F8 into ordinary roots

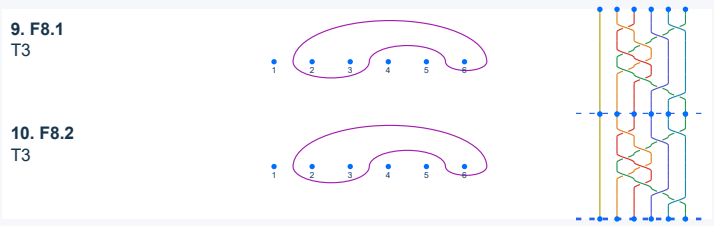
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 5 below. Only the changed interval is shown.

### Before




### After




## I - step 005

Split F10 into ordinary roots

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown.

### Before




### After



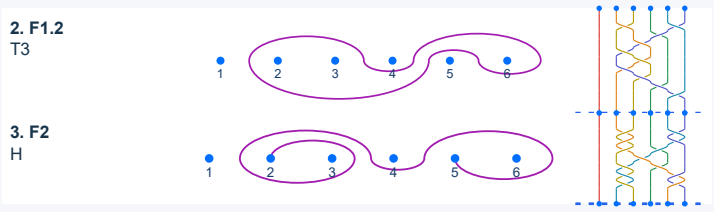

## I - step 006

F1.2 down past F2

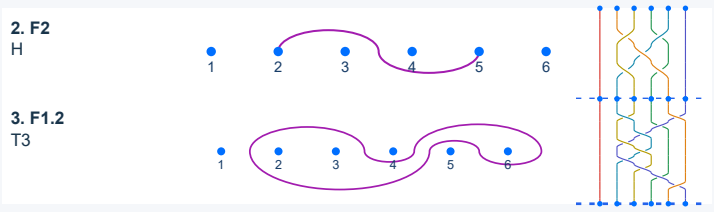
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

### Before




### After



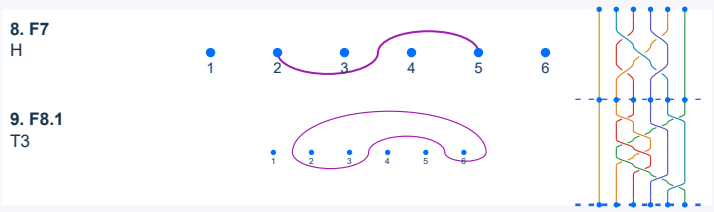
# I - step 007

F8.1 up past F7

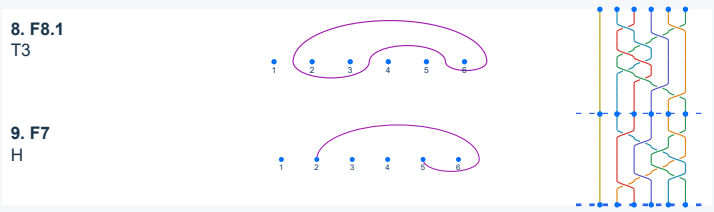
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

### Before




### After



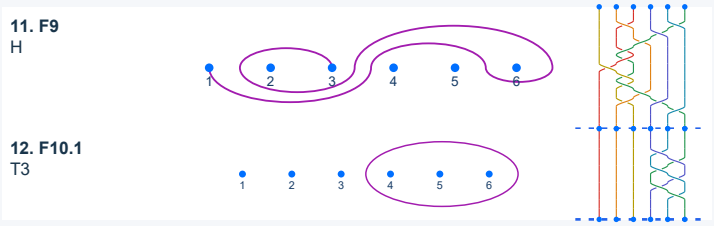

## I - step 008

F10.1 up past F9

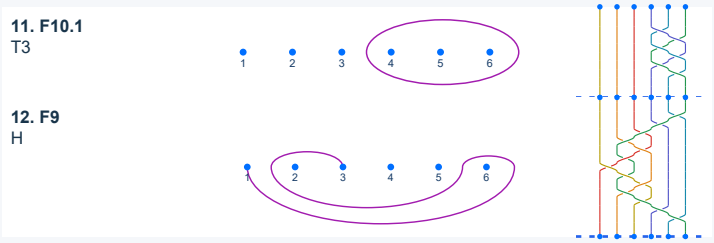
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

### Before




### After




## I - step 009

F10.1 up past F8.2

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 5 below. Only the changed interval is shown.

### Before




### After



# I - step 010

F10.1 up past F7

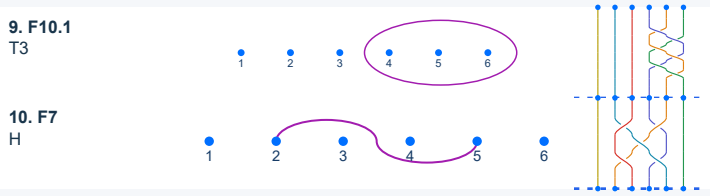
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

### Before




### After




## I - step 011

Use the exact disk equality of the matched arcs

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown.

### Before




### After



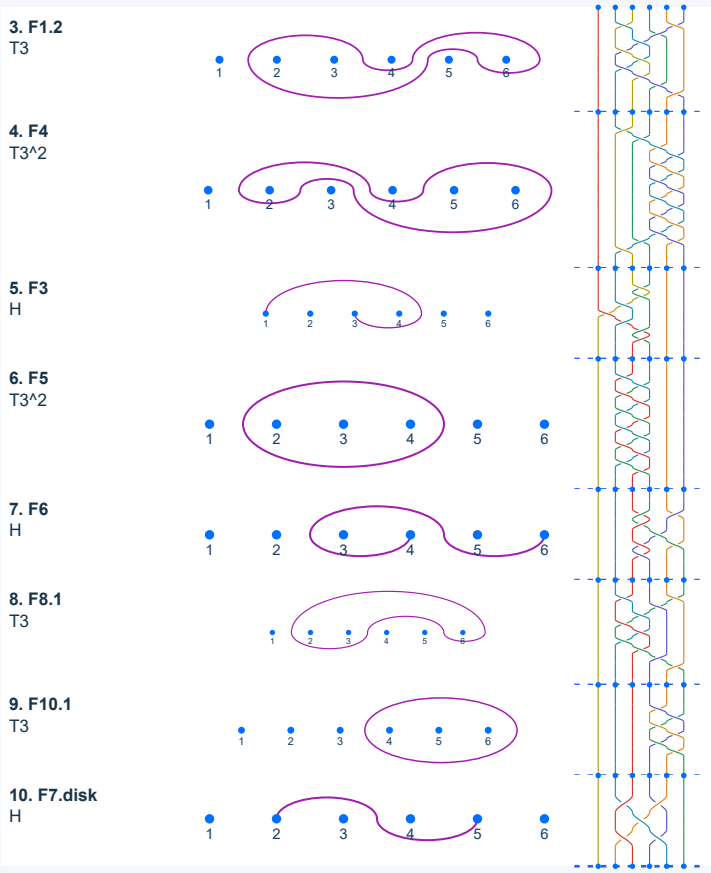
# I - step 012

Drag the matched arc up across SEVEN factors (one Hurwitz drag)

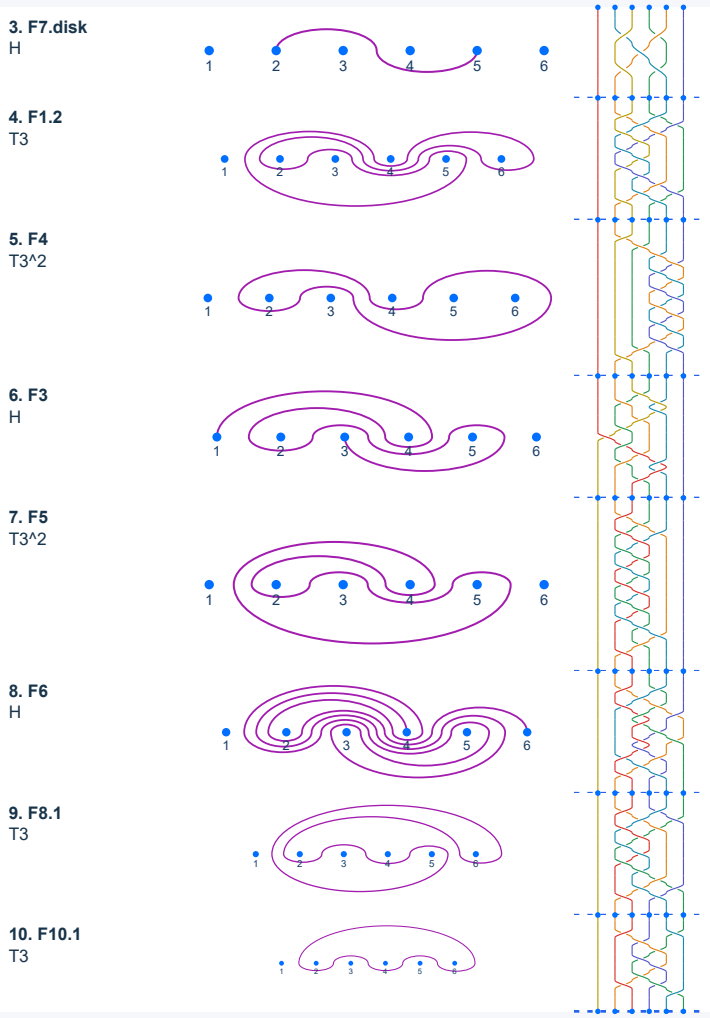
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 7.

## Before




## After



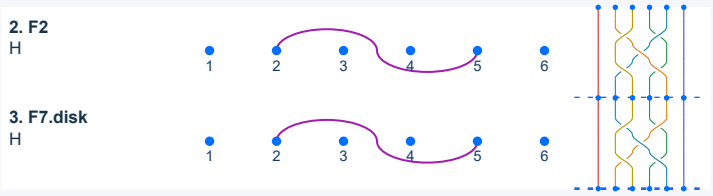
# I - step 013

Combine the two matched half twists

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown.

### Before




### After



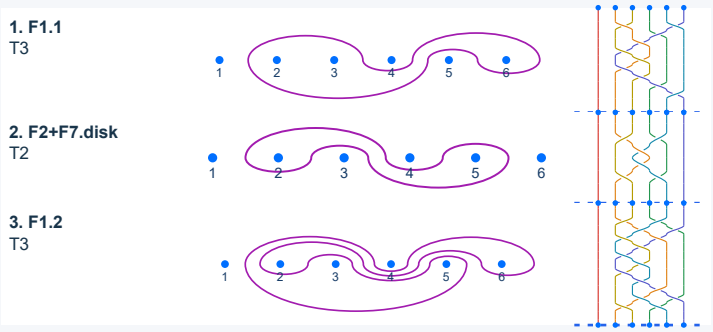

## I - step 014

SPHERE LANTERN: absorb the matched pair and the two prefix roots

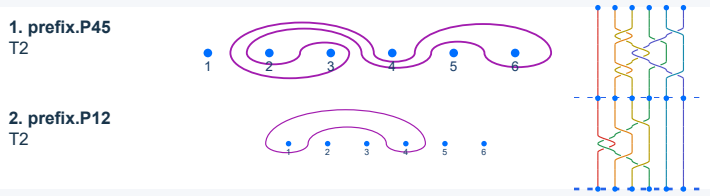
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

### Before




### After



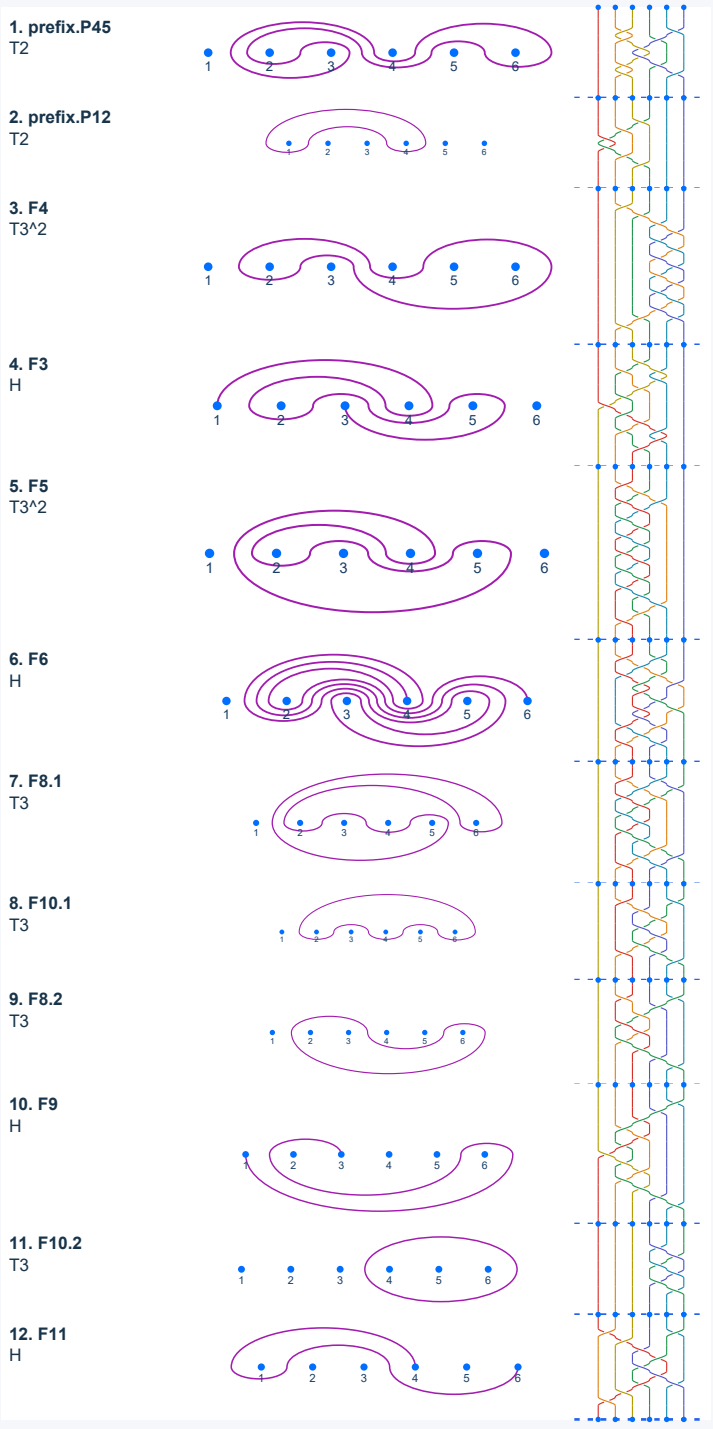
# I - step 015

Global frame: fixed working point 2 becomes displayed point 6

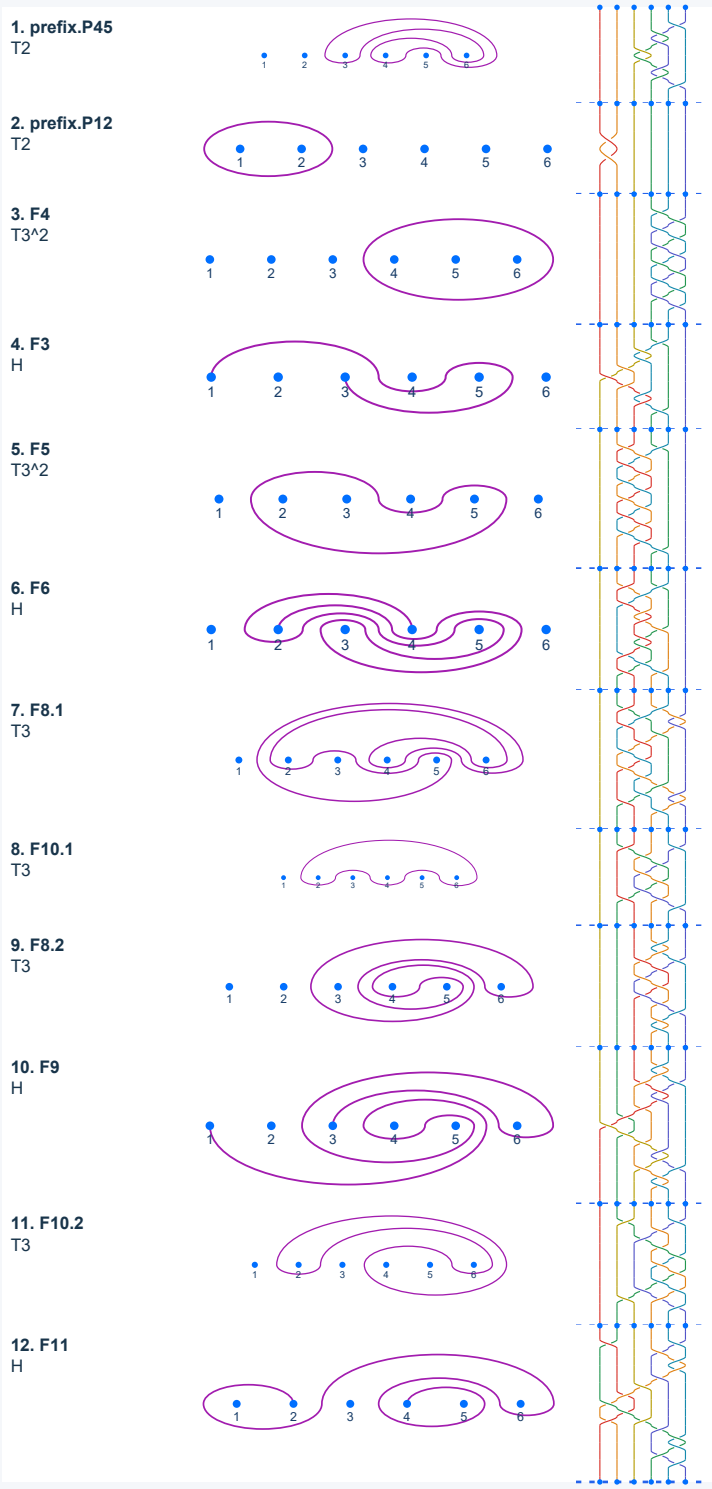
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before




## After



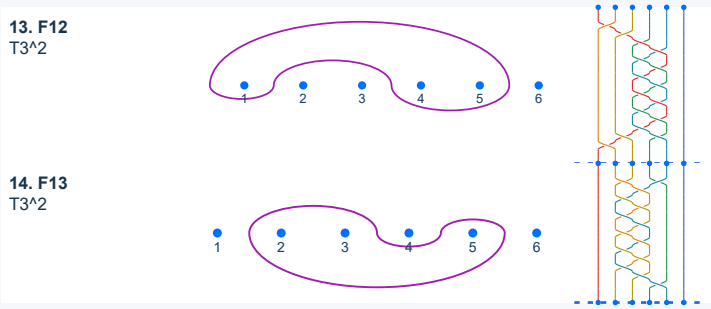
# I - step 015 - continued

Global frame: fixed working point 2 becomes displayed point 6

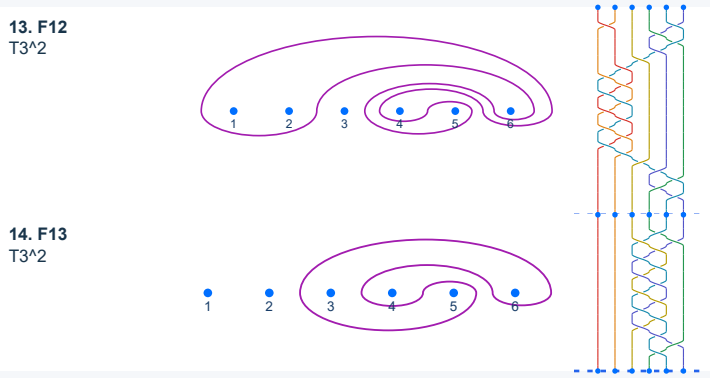
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before




## After



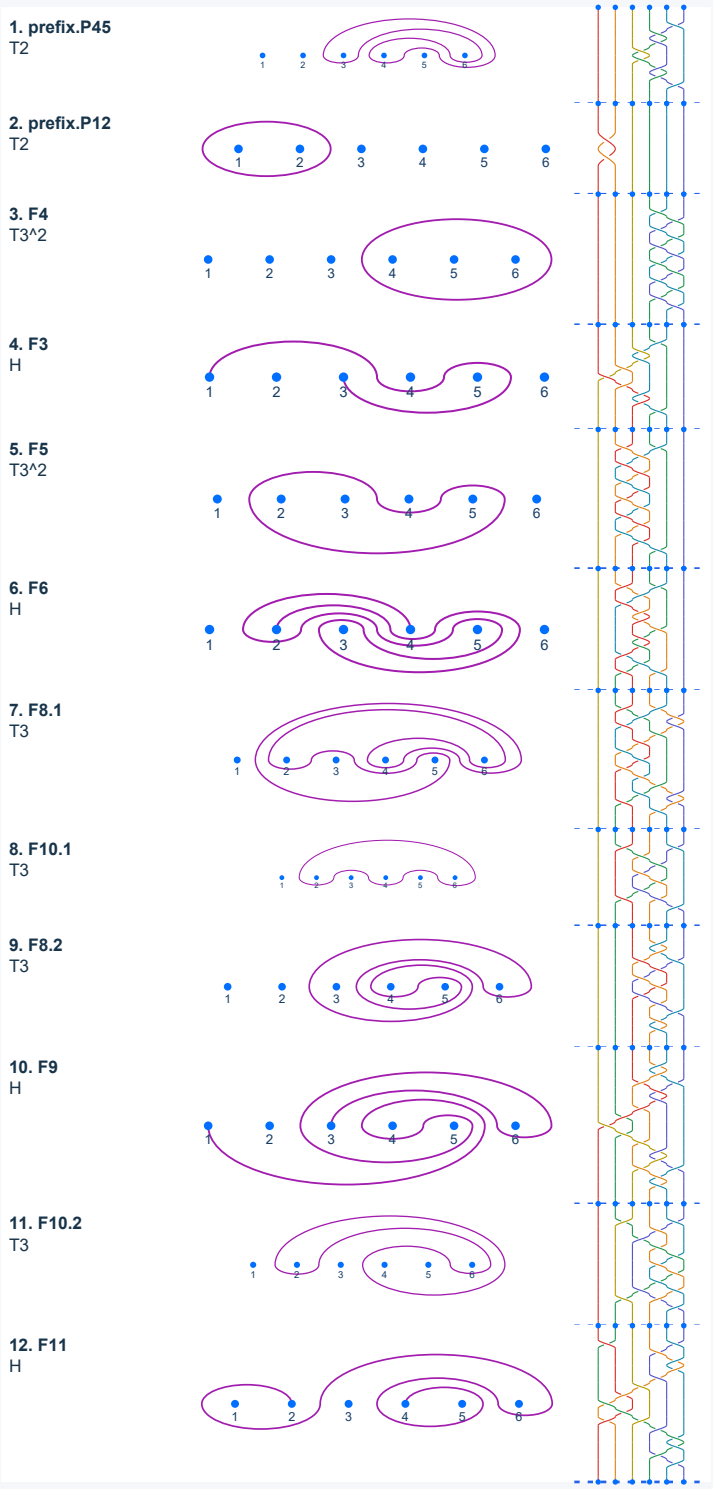
# I - step 016

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

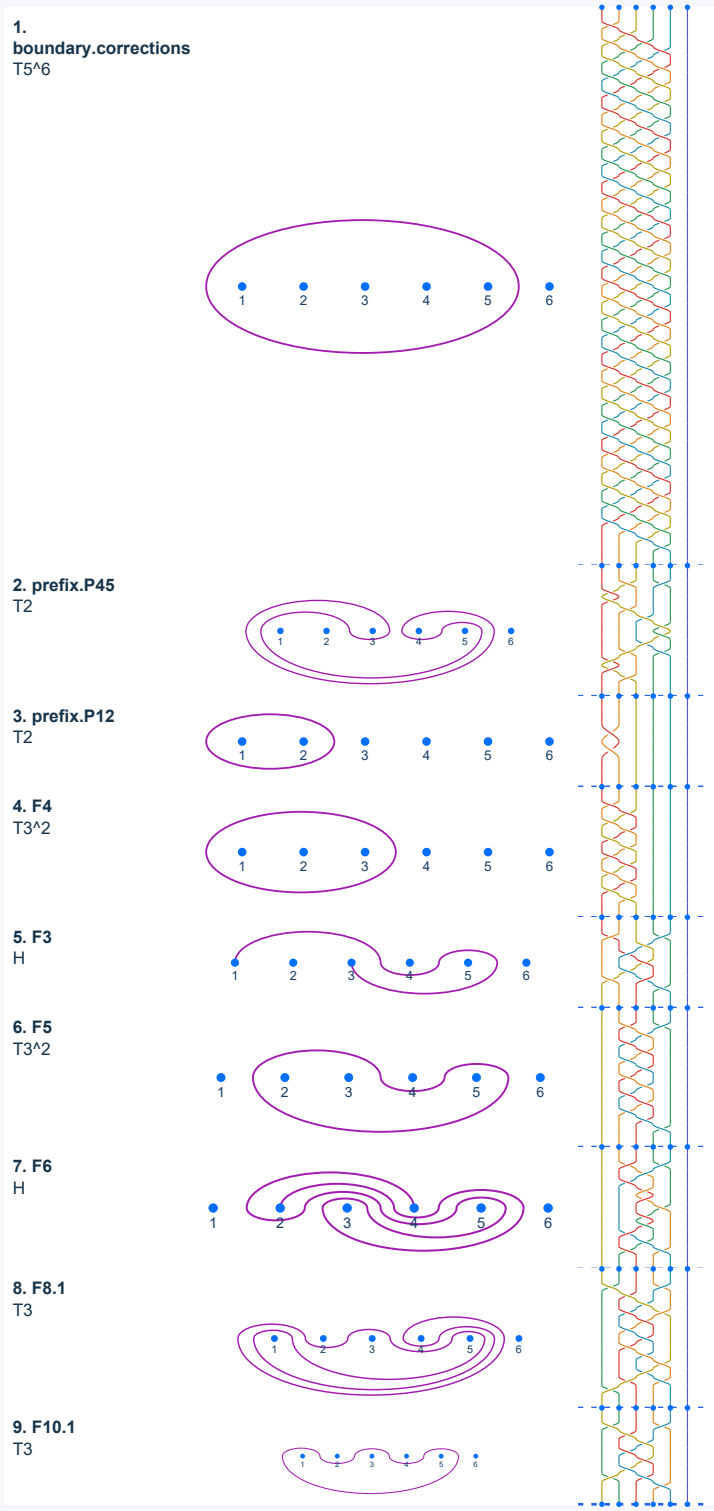
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

## Before




## After



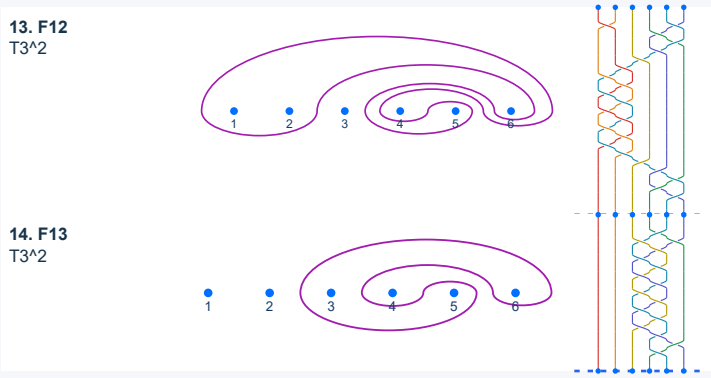
# I - step 016 - continued

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

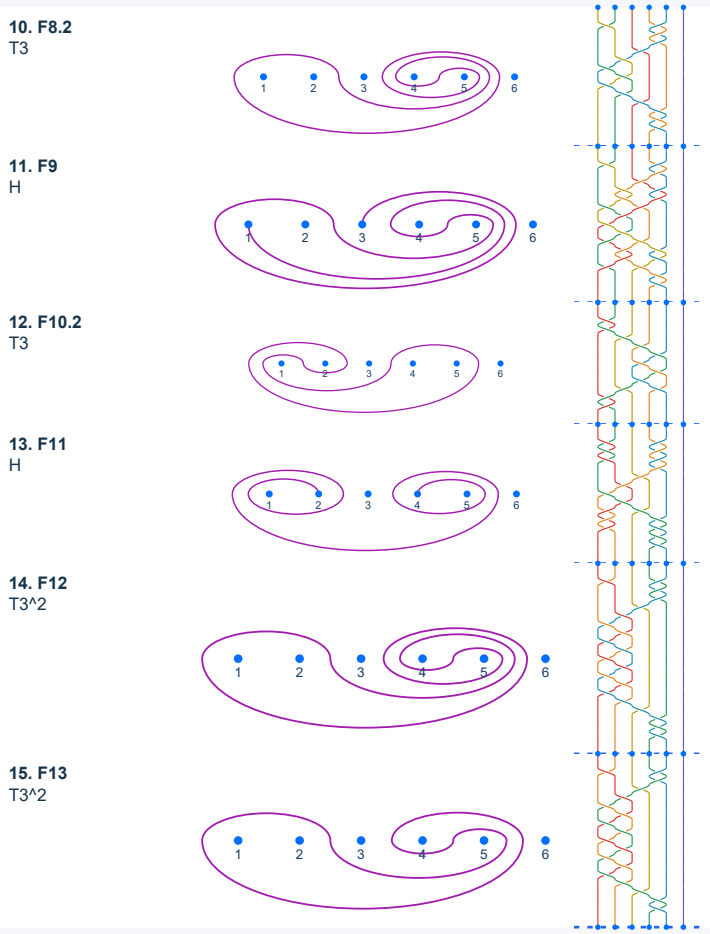
SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

### Before




### After



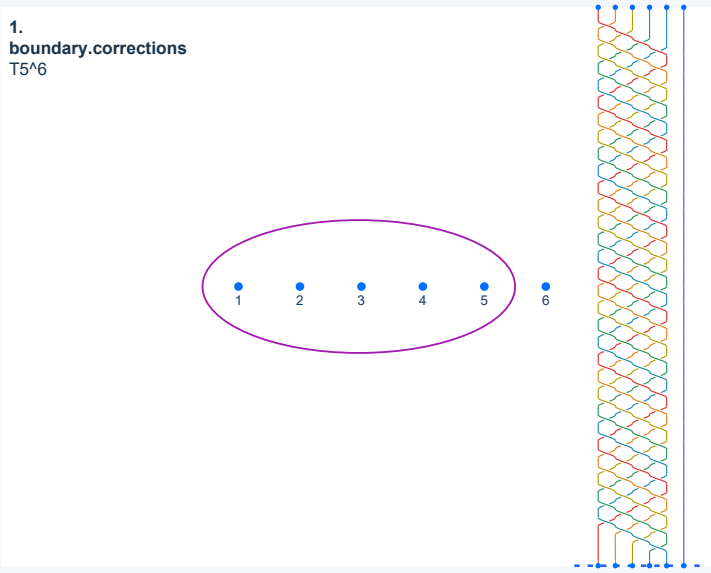
# I - step 017

SPHERE MOVE: cap the SIX framing corrections

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

## Before




## After

Identity / no factors in this interval.

# II. Existing roots remove all half twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

# II - step 018

Split F4 into its two positive units

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

### Before




### After



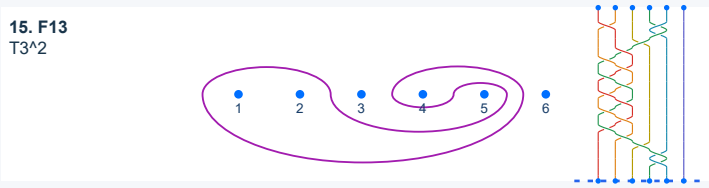

## II - step 019

Split F13 into its two positive units

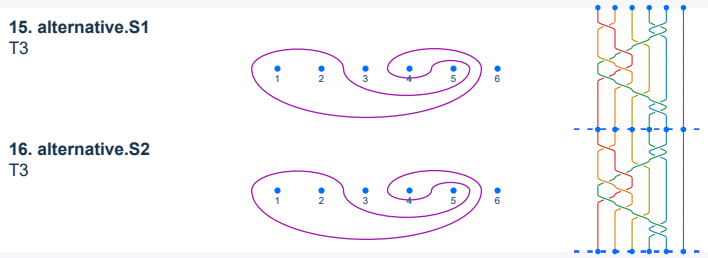
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 0 below. Only the changed interval is shown.

### Before




### After



# II - step 020

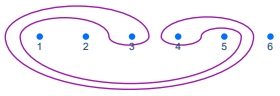
Choose the cyclic start at the existing 124 root; its whole common frame is retained

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before

**1. prefix.P45**
T2



**2. prefix.P12**
T2



**3. alternative.R1**
T3



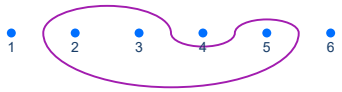
**4. alternative.R2**
T3



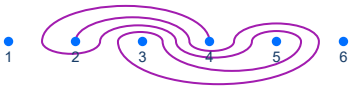
**5. F3**
H



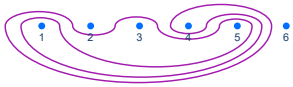
**6. F5**
T3^2



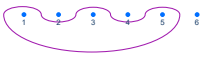
**7. F6**
H



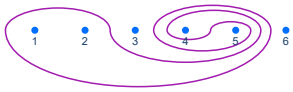
**8. F8.1**
T3



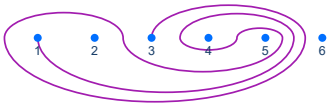
**9. F10.1**
T3



**10. F8.2**
T3

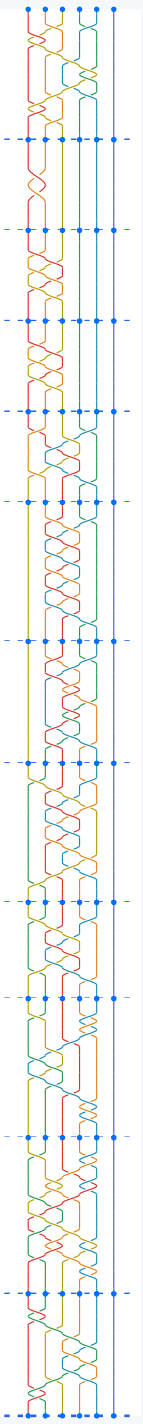


**11. F9**
H



**12. F10.2**
T3



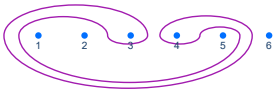


## After

**1. alternative.S2**
T3



**2. prefix.P45**
T2



**3. prefix.P12**
T2



**4. alternative.R1**
T3



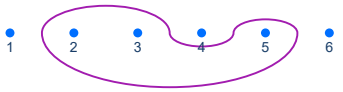
**5. alternative.R2**
T3



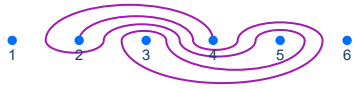
**6. F3**
H



**7. F5**
T3^2



**8. F6**
H



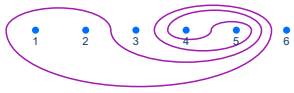
**9. F8.1**
T3



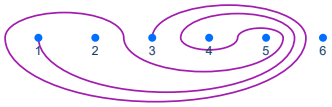
**10. F10.1**
T3

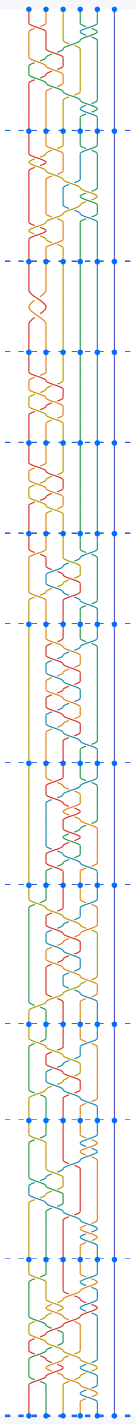


**11. F8.2**
T3



**12. F9**
H





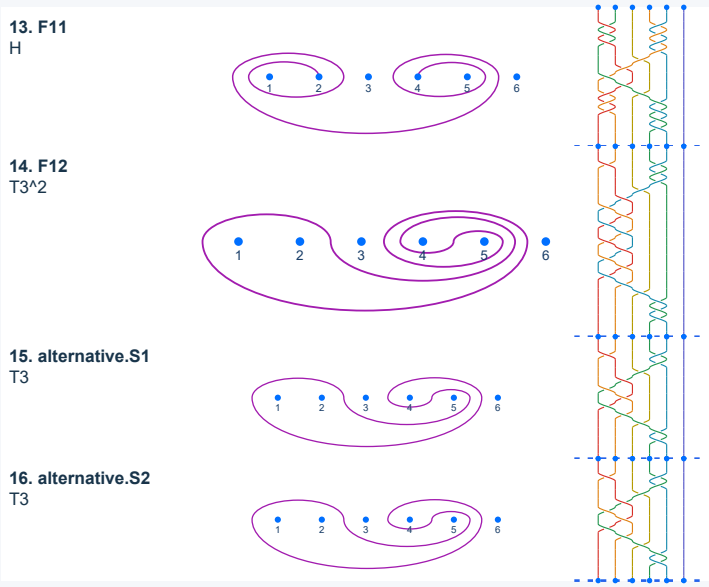
# II - step 020 - continued

Choose the cyclic start at the existing 124 root; its whole common frame is retained

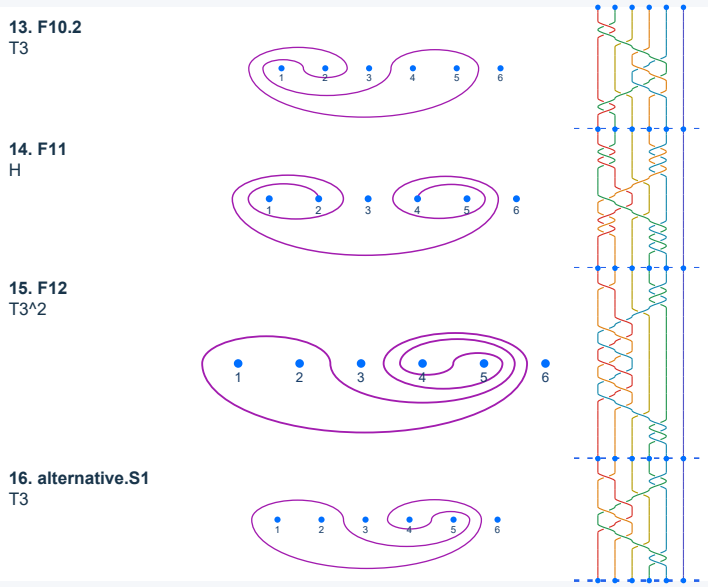
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

### Before




### After



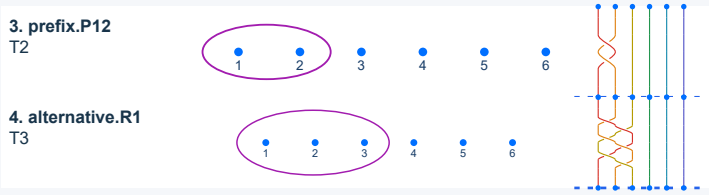

## II - step 021

Commute one round T123 root above the round T12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 12 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



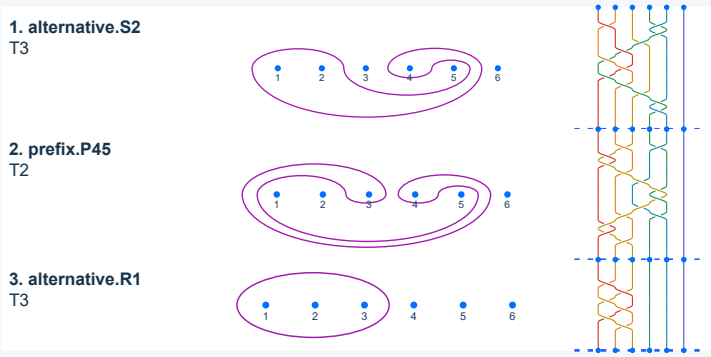
# II - step 022

PC cluster lantern on the ACTUAL 124, 34 and 123 curves

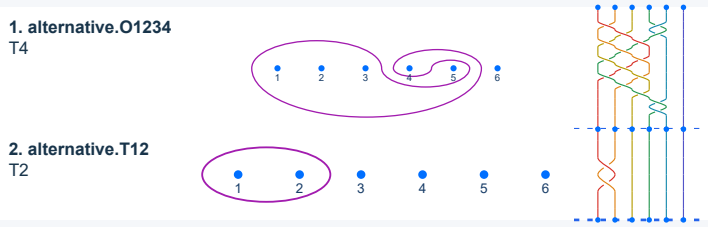
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

### Before




### After




## II - step 023

Combine the matching alternative.T12 and prefix.P12

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 12 below. Only the changed interval is shown.

### Before




### After



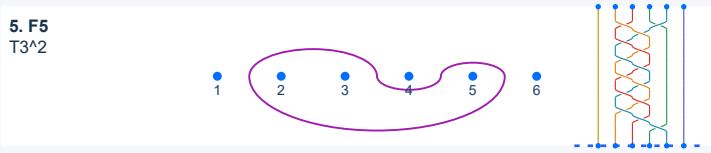

## II - step 024

Split F5 into its two positive units

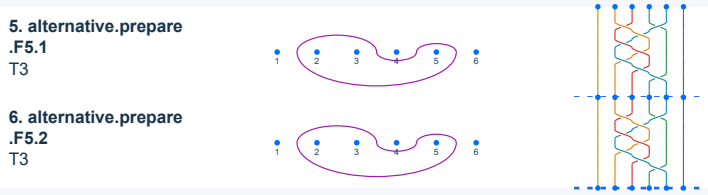
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown.

### Before




### After



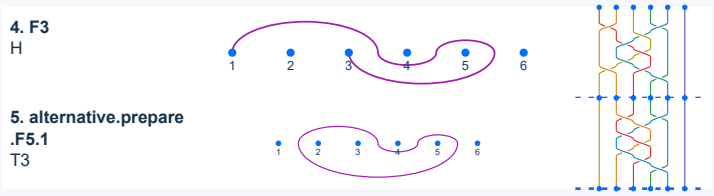
# II - step 025

One 235 root up over F3

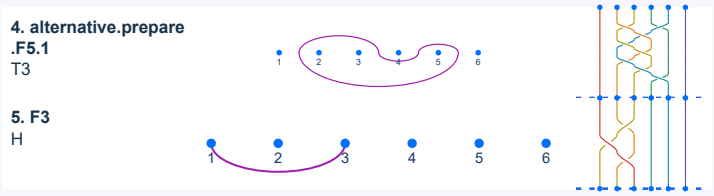
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



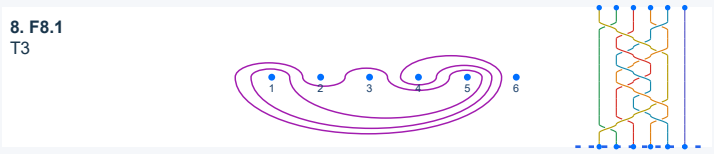

## II - step 026

Original 345 daisy supplies actual P35

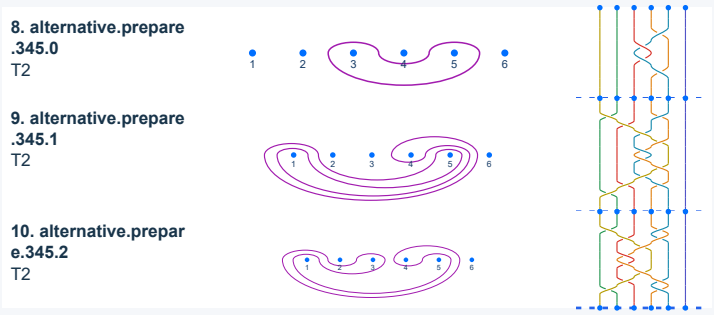
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 7 below. Only the changed interval is shown.

### Before




### After



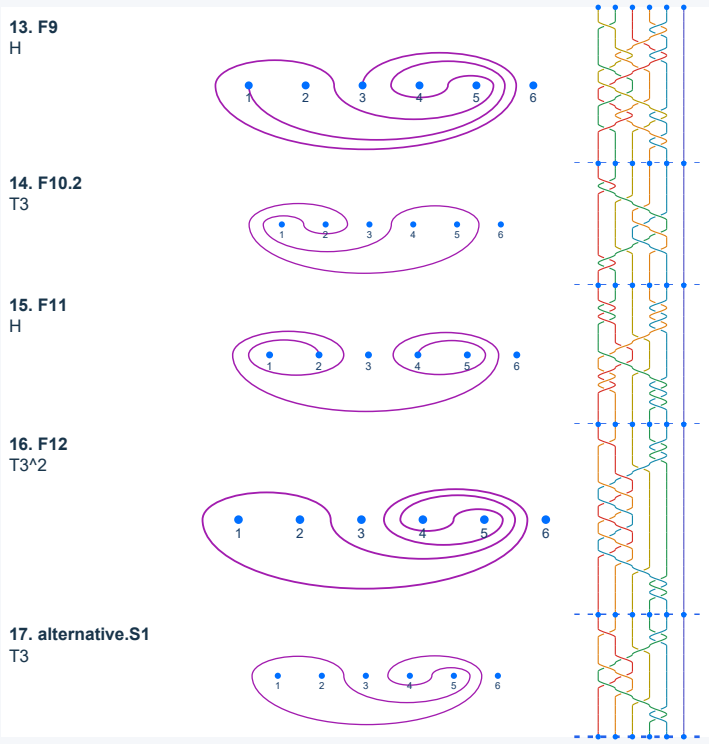
# II - step 027

Remaining 124 root up over F9

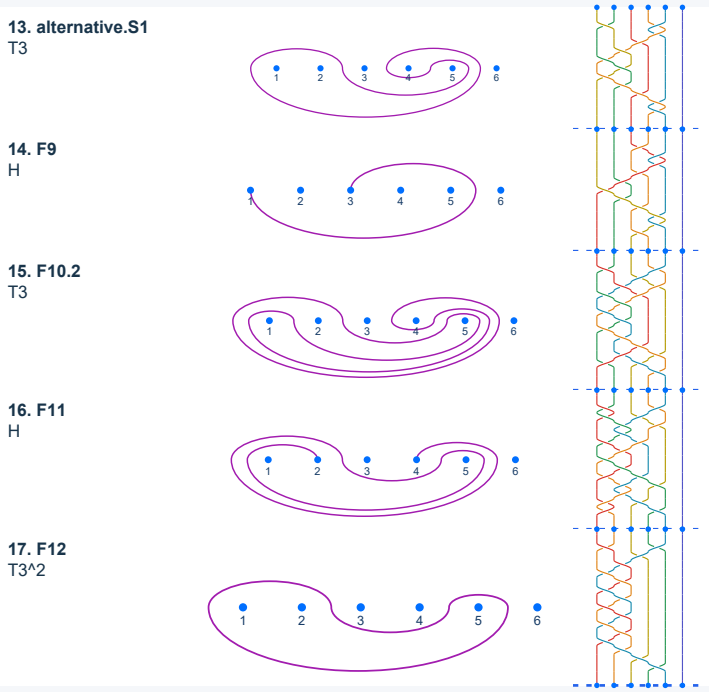
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 4.

### Before




### After



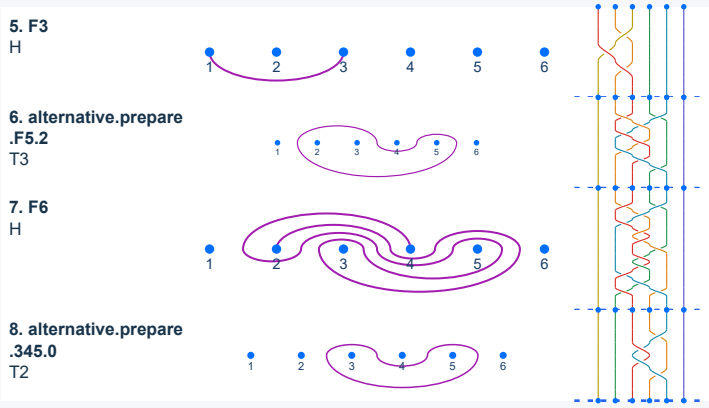

## II - step 028

Carry P35 up over F3; a Q14 transport is still needed

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 3.

### Before




### After



# II - step 029

PC daisy lends its actual Q14 petal

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 16 below. Only the changed interval is shown.

## Before

**1. alternative.O1234**
T4



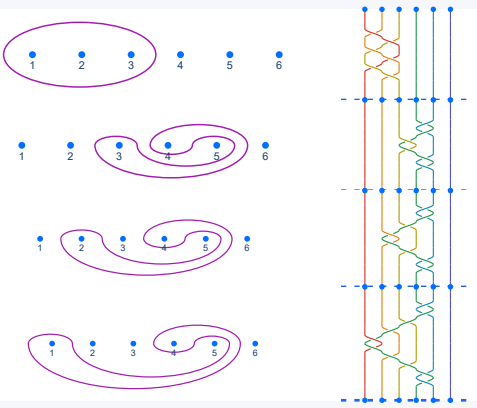

## After

**1.**
**alternative.qdaisy.0**
T3

**2.**
**alternative.qdaisy.1**
T2

**3.**
**alternative.qdaisy.2**
T2

**4.**
**alternative.qdaisy.3**
T2



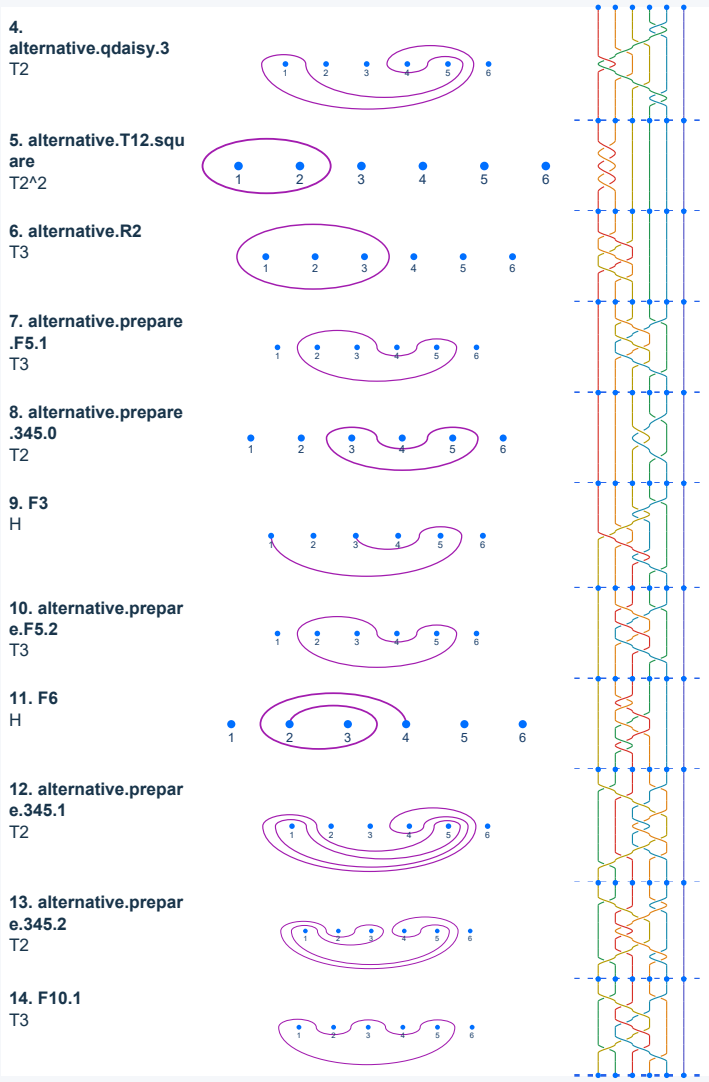
# II - step 030

Carry Q14 down across F3 and the remaining 345 petals

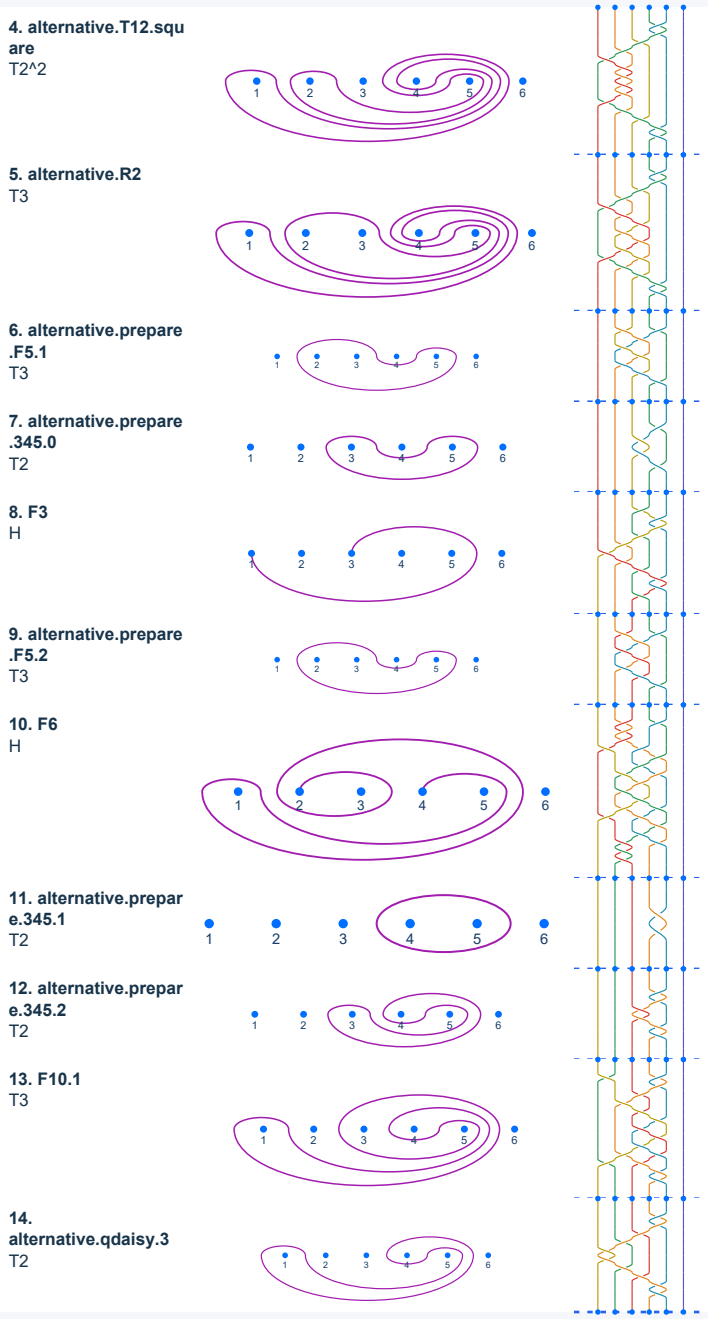
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 10.

## Before




## After



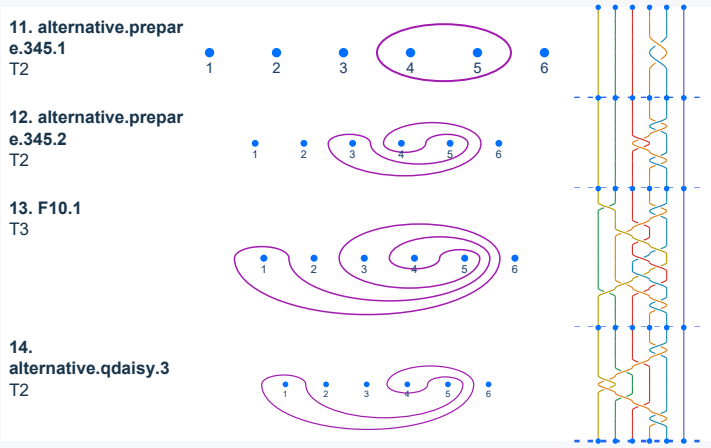
# II - step 031

PC daisy contracts the ACTUAL transported petals to O1345

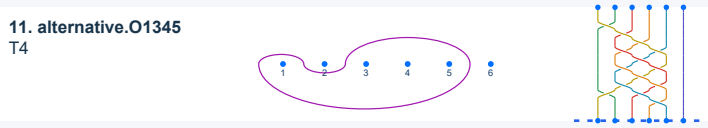
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 6 below. Only the changed interval is shown.

### Before




### After



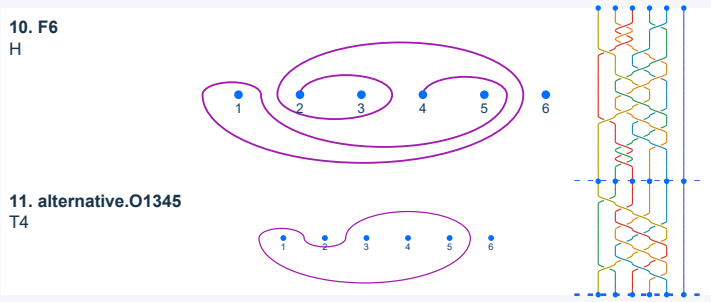

## II - step 032

Carry O1345 up across F6 before the matching-half gather

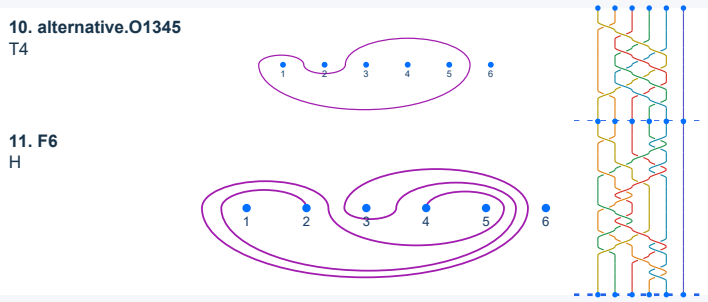
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



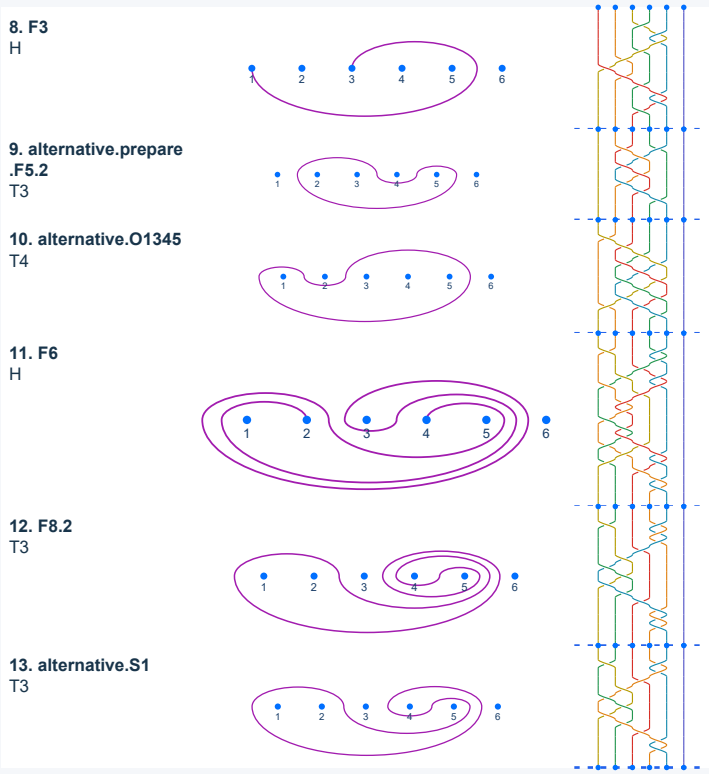
# II - step 033

Carry the matched F3 down to F9; F6 unwinds

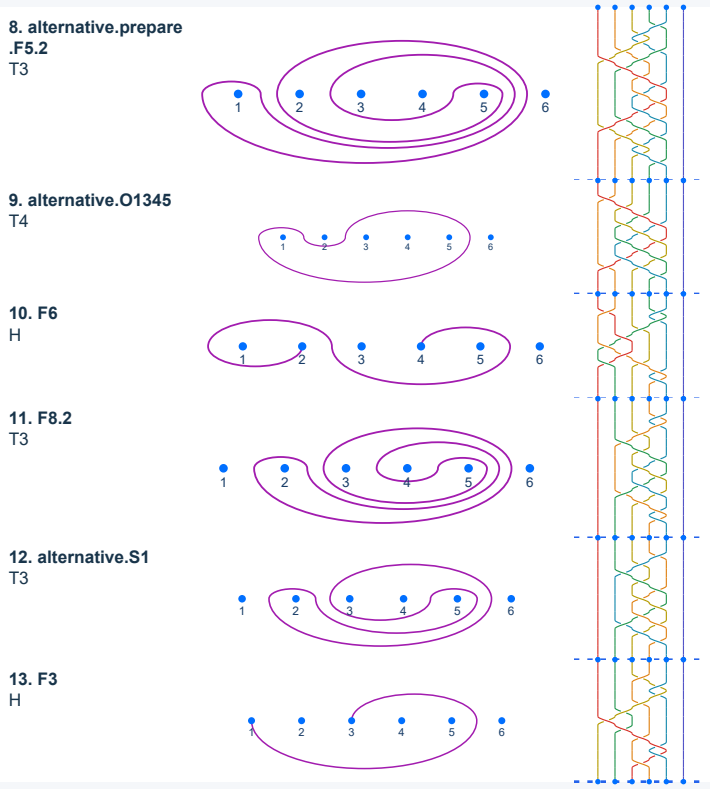
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 5.

### Before




### After



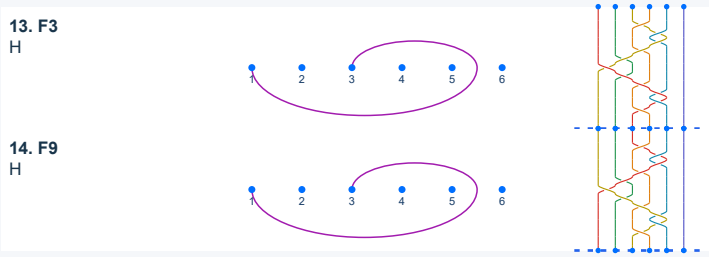

## II - step 034

Combine the matching F3 and F9

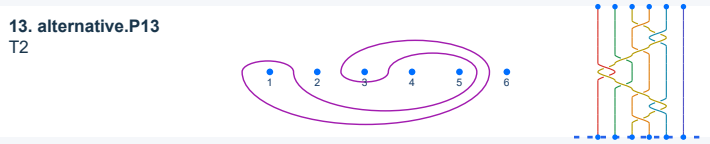
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown.

### Before




### After



# II - step 035

The O1345 daisy returns EXACTLY the borrowed Q14

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

## Before

**9. alternative.O1345**
T4



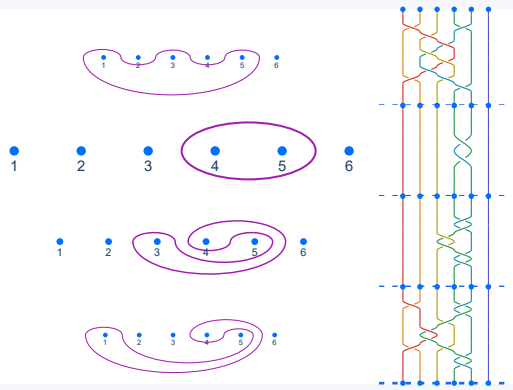

## After

**9. alternative.return.0**
T3

**10. alternative.return.1**
T2

**11. alternative.return.2**
T2

**12. alternative.return.3**
T2



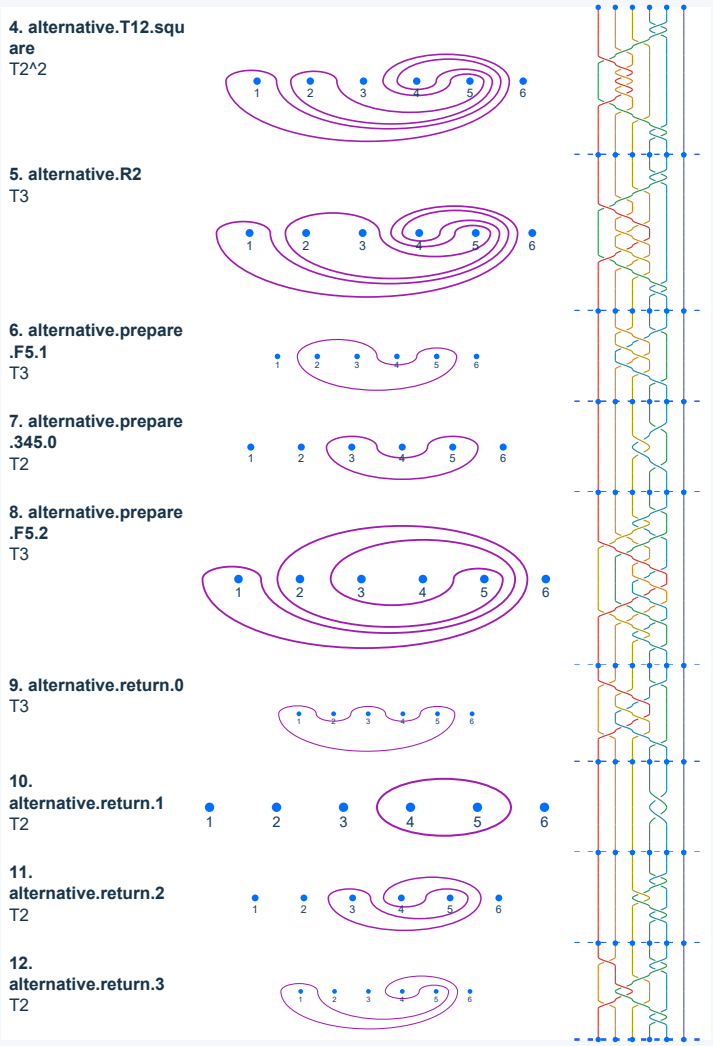
# II - step 036

Return Q14 up to restore the round T12 square and T123

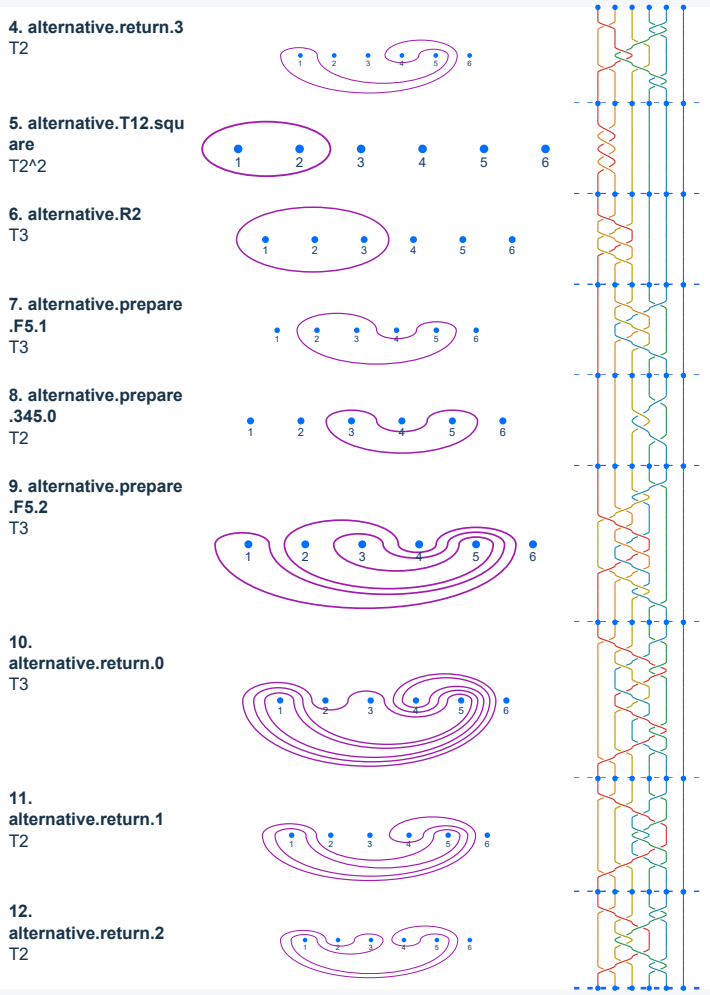
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

## Before




## After



# II - step 037

Close the original daisy after its actual Q14 returns

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown.

### Before




### After



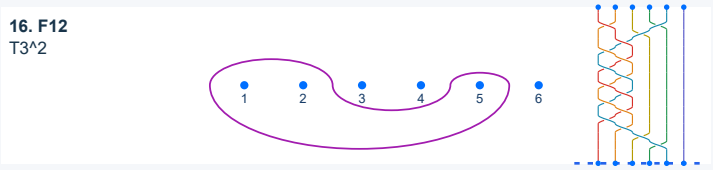

## II - step 038

Split F12 into its two positive units

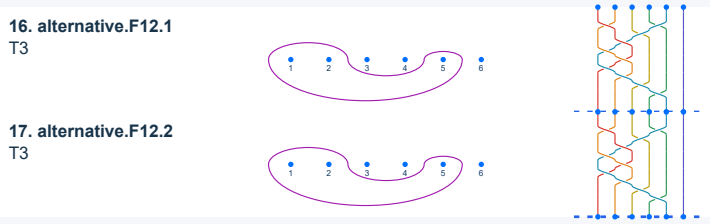
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 0 below. Only the changed interval is shown.

### Before




### After



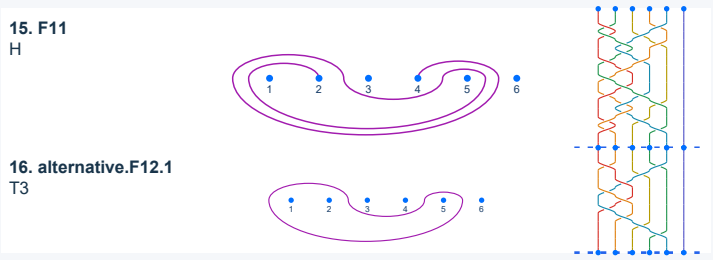

## II - step 039

One C12 root shortens F11

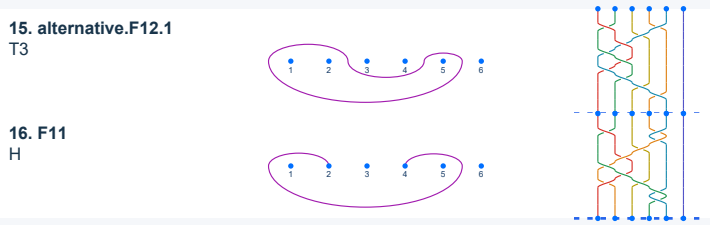
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



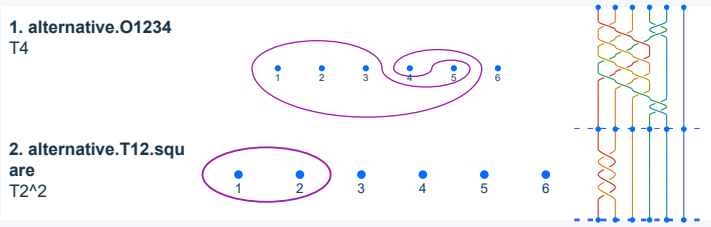
# II - step 040

Commute the recovered round square to the top

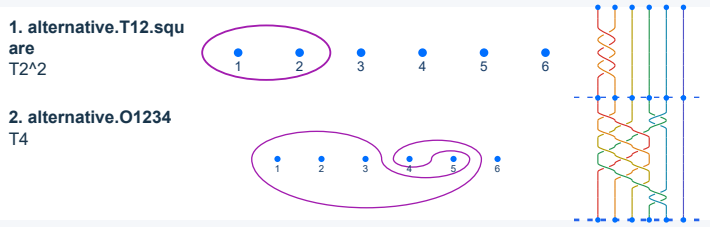
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 15 below. Only the changed interval is shown. Retained adjacent checks: 1.

## Before




## After



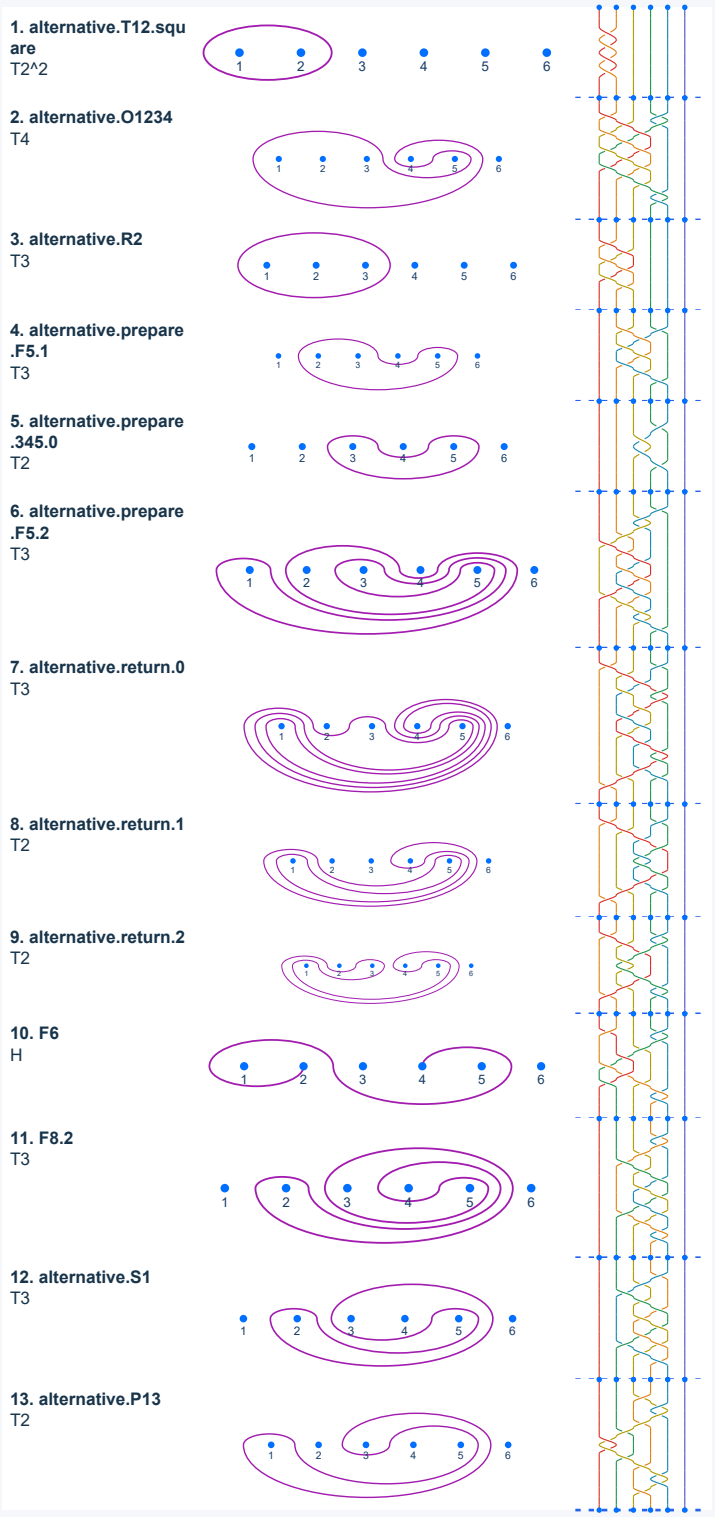
# II - step 041

Choose the cyclic start just after the round square

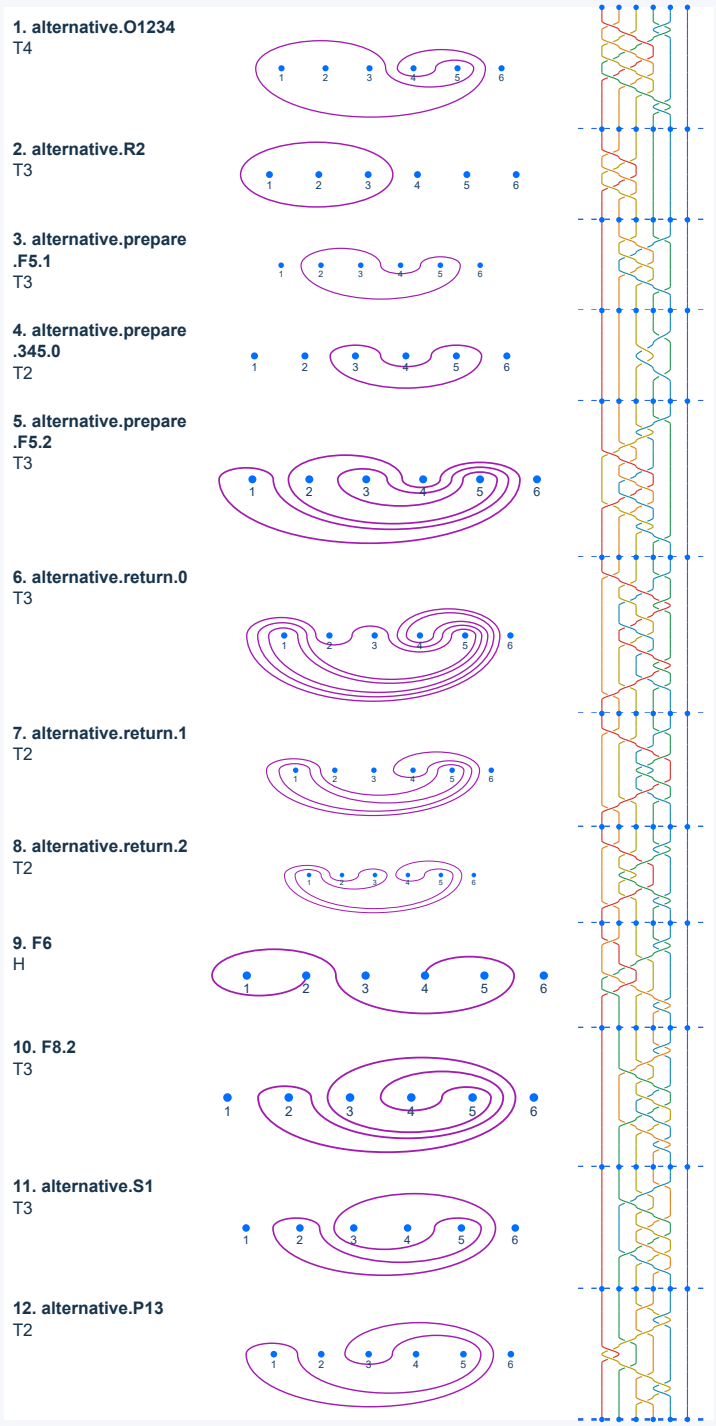
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before




## After



# II - step 041 - continued

Choose the cyclic start just after the round square

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

### Before




### After



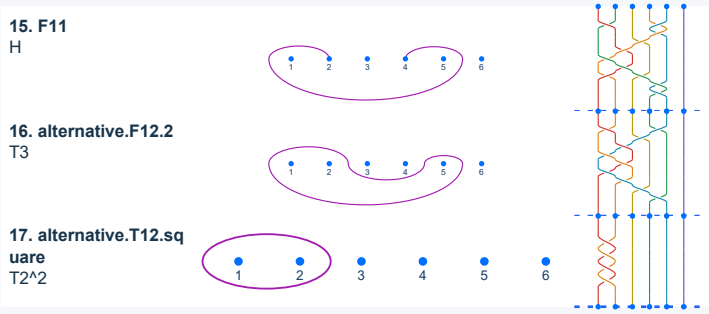

## II - step 042

Carry the square up the short side; F11 matches F6

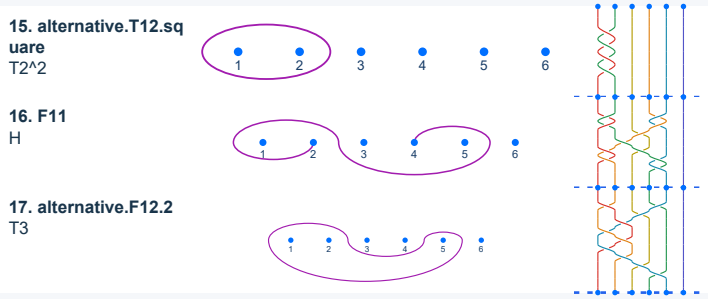
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

14 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



# II - step 043

One common sigma4 frame replaces the two old frames

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before

**1. alternative.O1234**
T4

**2. alternative.R2**
T3

**3. alternative.prepare .F5.1**
T3

**4. alternative.prepare .345.0**
T2

**5. alternative.prepare .F5.2**
T3

**6. alternative.return.0**
T3

**7. alternative.return.1**
T2

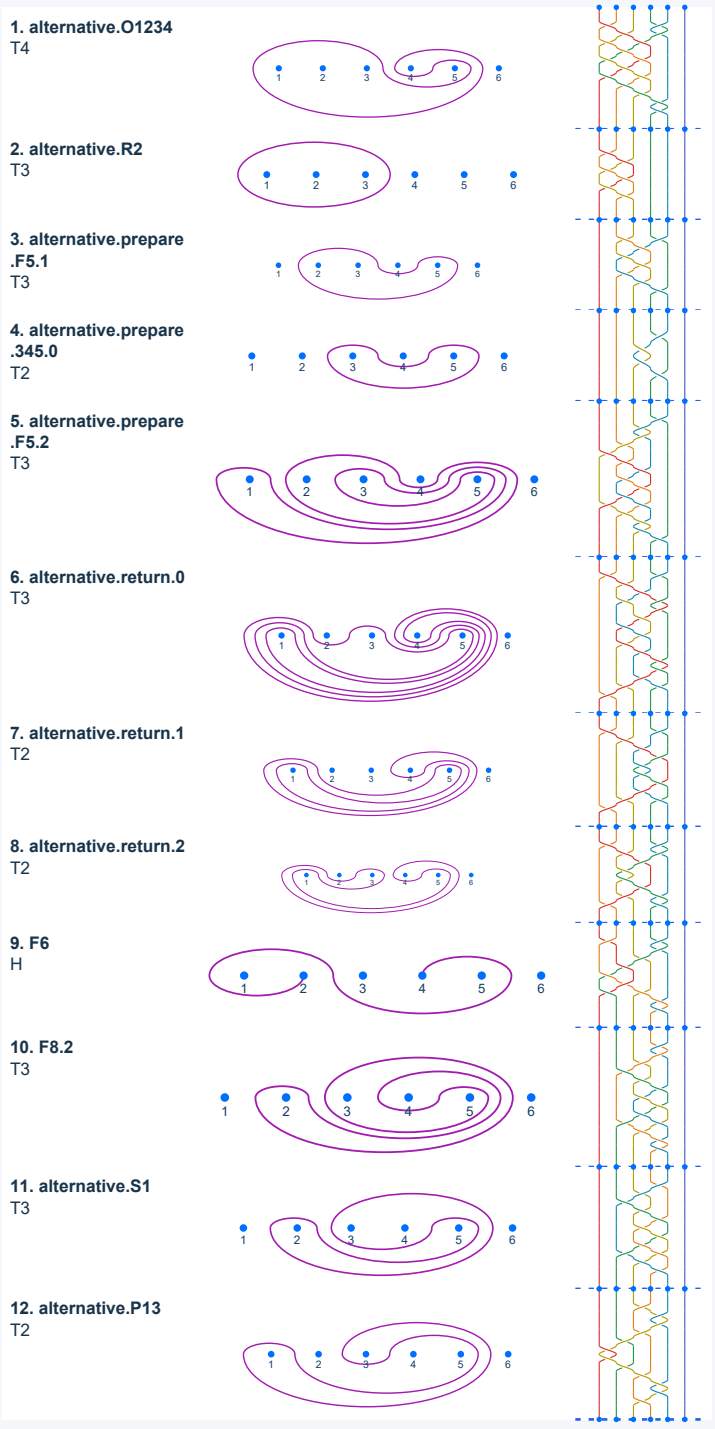
**8. alternative.return.2**
T2

**9. F6**
H

**10. F8.2**
T3

**11. alternative.S1**
T3

**12. alternative.P13**
T2



## After

**1. alternative.O1234**
T4

**2. alternative.R2**
T3

**3. alternative.prepare .F5.1**
T3

**4. alternative.prepare .345.0**
T2

**5. alternative.prepare .F5.2**
T3

**6. alternative.return.0**
T3

**7. alternative.return.1**
T2

**8. alternative.return.2**
T2

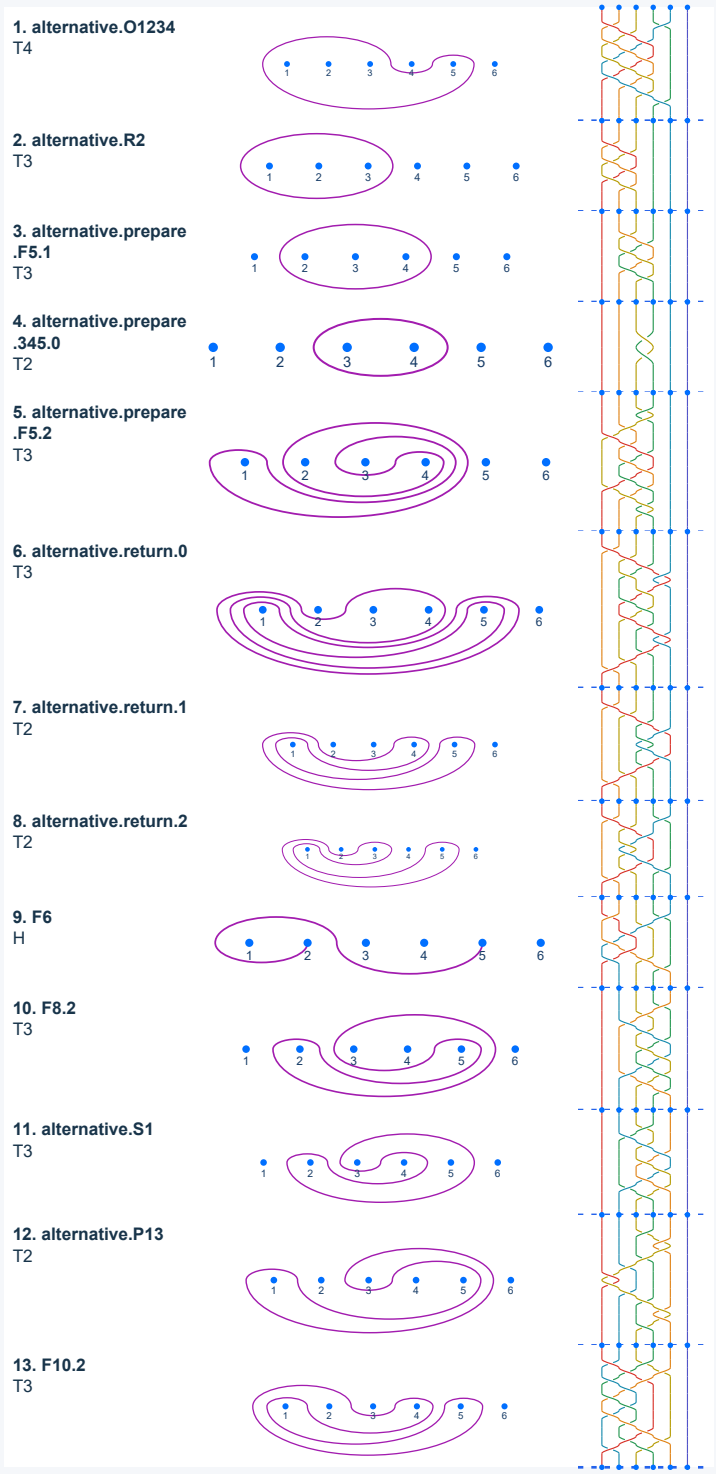
**9. F6**
H

**10. F8.2**
T3

**11. alternative.S1**
T3

**12. alternative.P13**
T2

**13. F10.2**
T3



# II - step 043 - continued

One common sigma4 frame replaces the two old frames

EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

### Before




### After



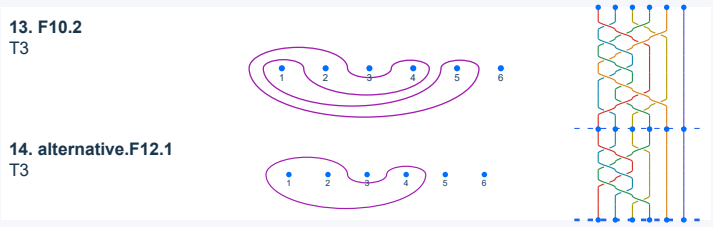

## II - step 044

Reorder C12 before the final matching-half gather

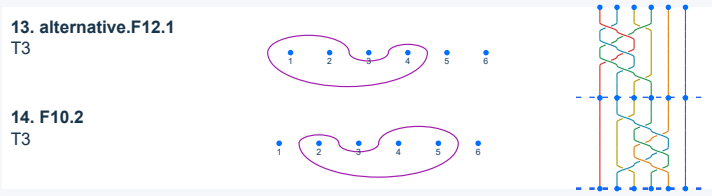
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



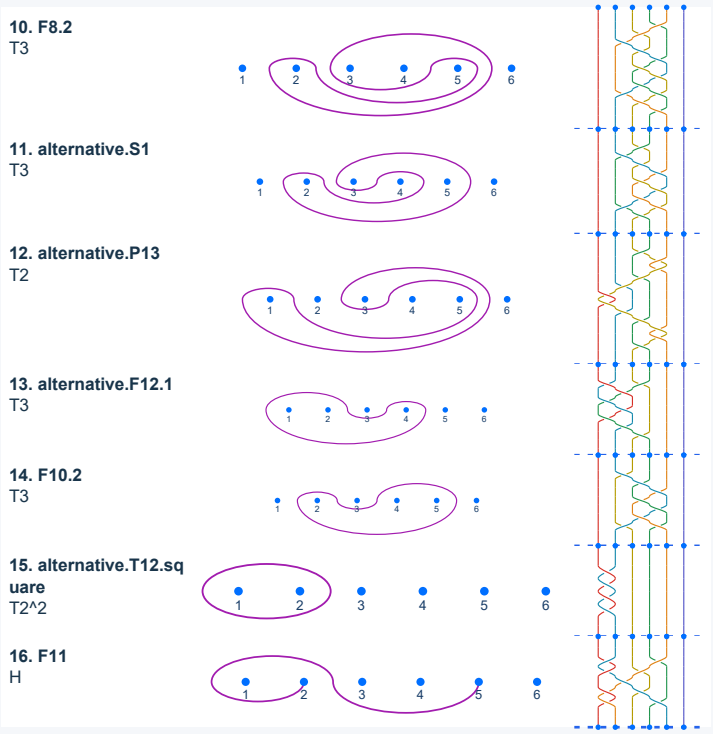
# II - step 045

Gather the final matching original halves

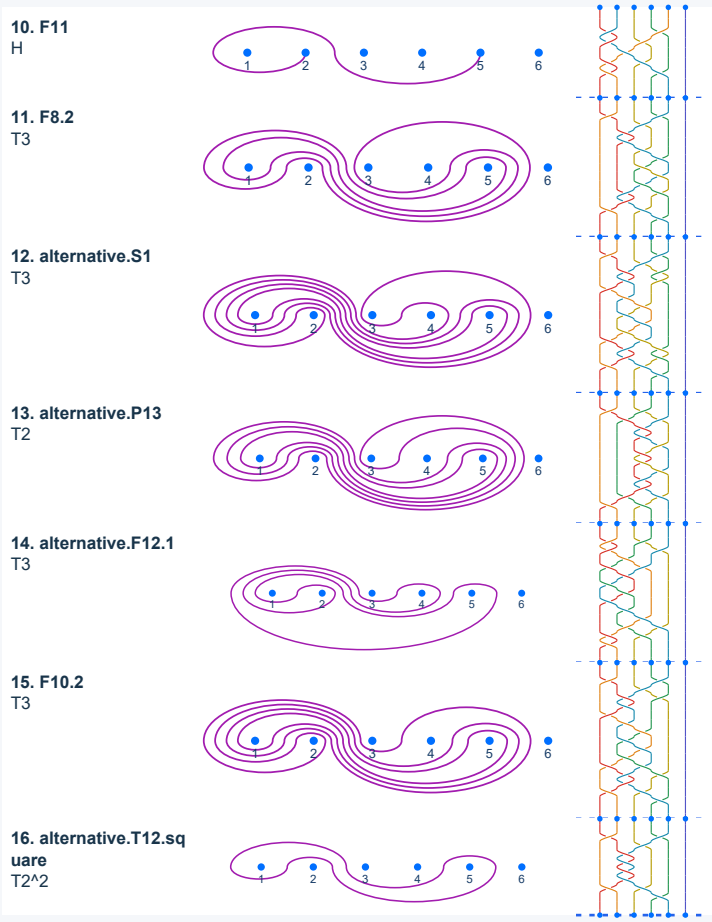
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

9 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 6.

### Before




### After



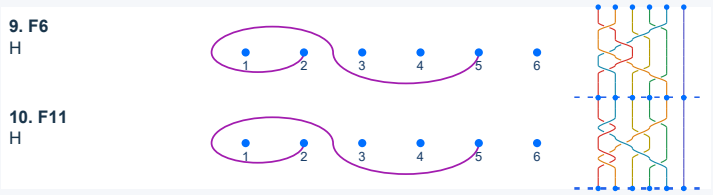

## II - step 046

Combine the matching F6 and F11

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 7 below. Only the changed interval is shown.

### Before




### After



# III. The first exterior and simpler inside daisy

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

# III - step 047

C12 daisy supplies the actual Q15

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

15 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before

**16. alternative.F12.2**
T3

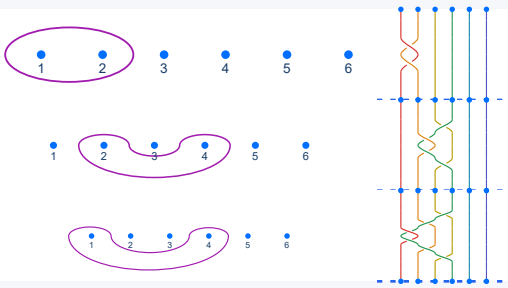



## After

**16. alternative.next.0**
T2

**17. alternative.next.1**
T2

**18. alternative.next.2**
T2



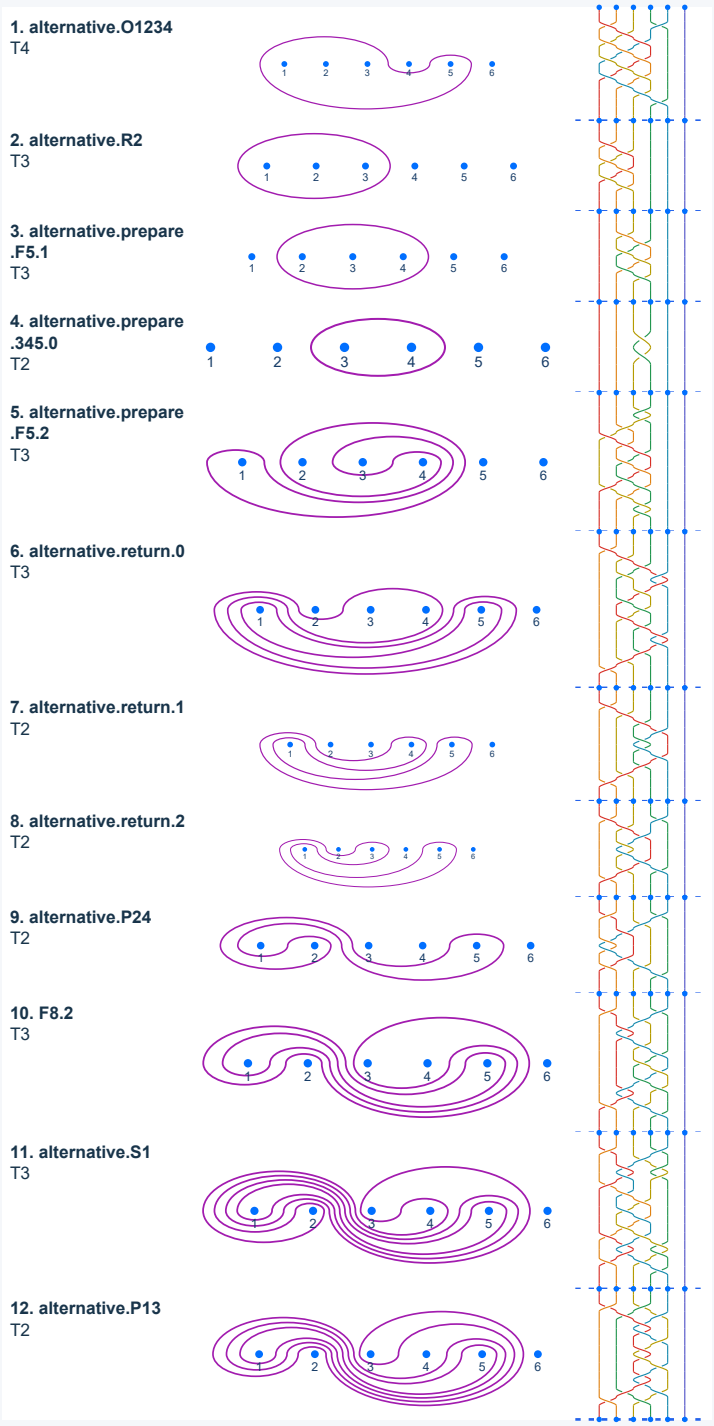
# III - step 048

Choose the cyclic start at the actual Q15

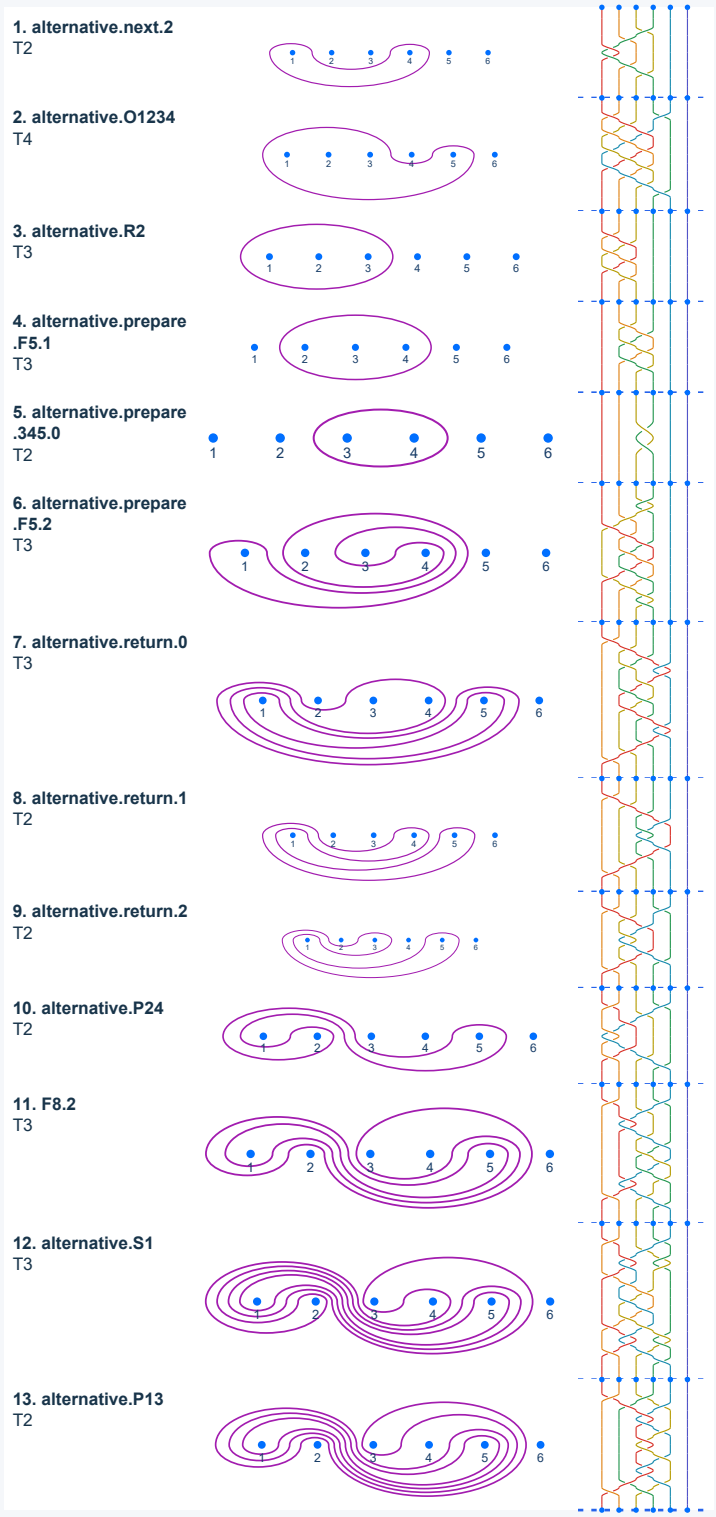
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before




## After



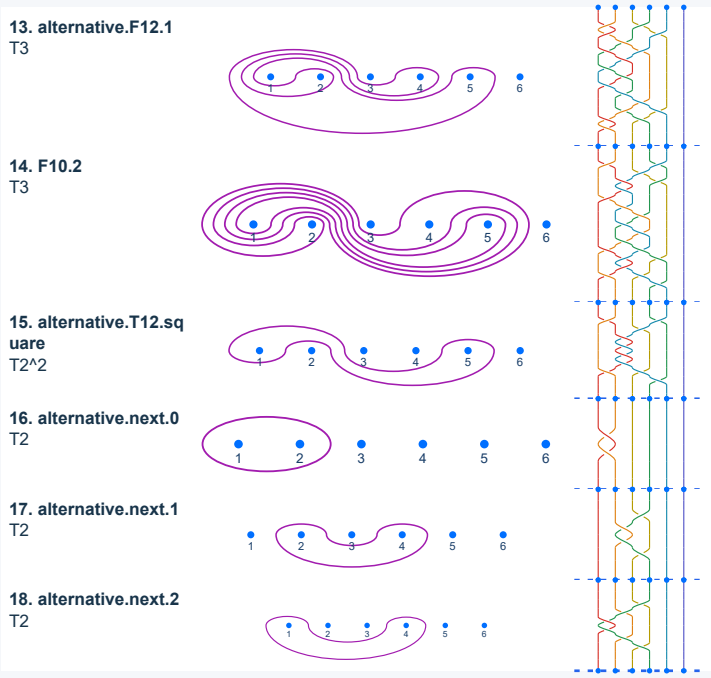
# III - step 048 - continued

Choose the cyclic start at the actual Q15

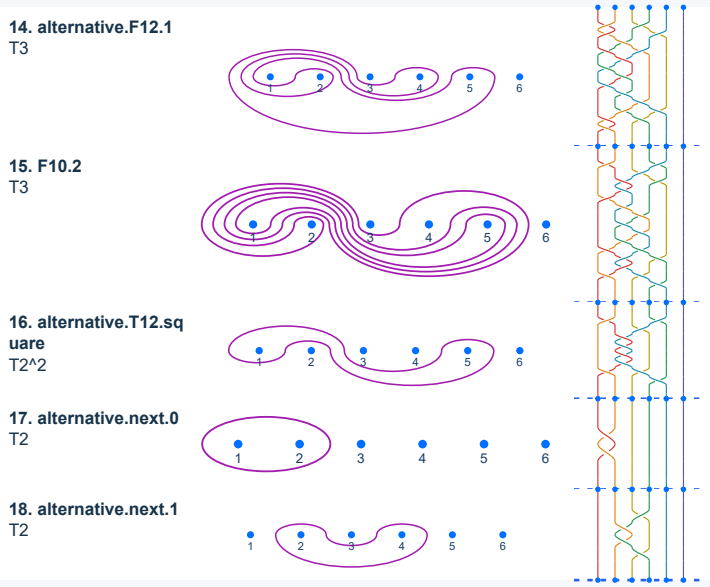
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before




## After



# III - step 049

Carry Q15 through the combined four-point prefix and the two roots

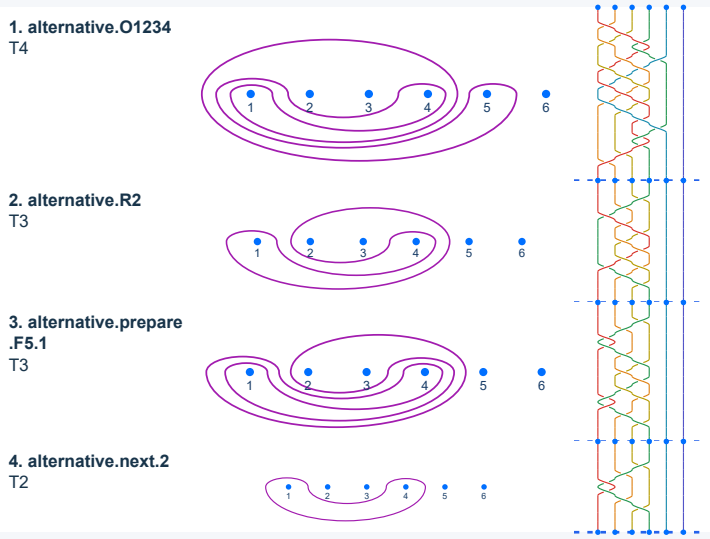
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 3.

### Before




### After



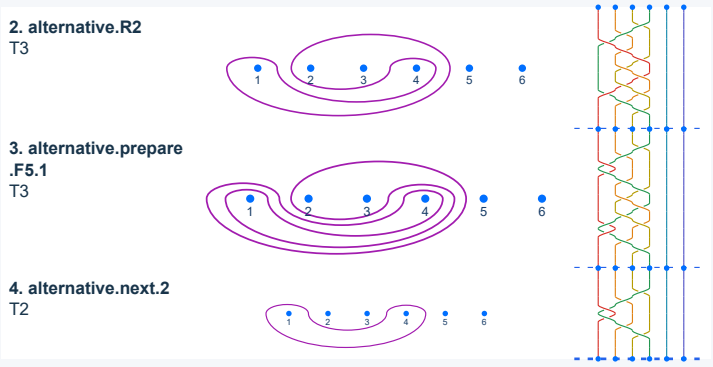

## III - step 050

PC cluster lantern gives the second four-point twist

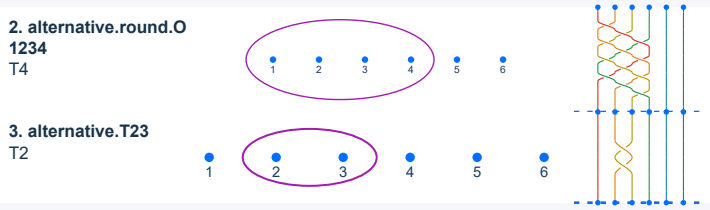
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 14 below. Only the changed interval is shown.

### Before




### After



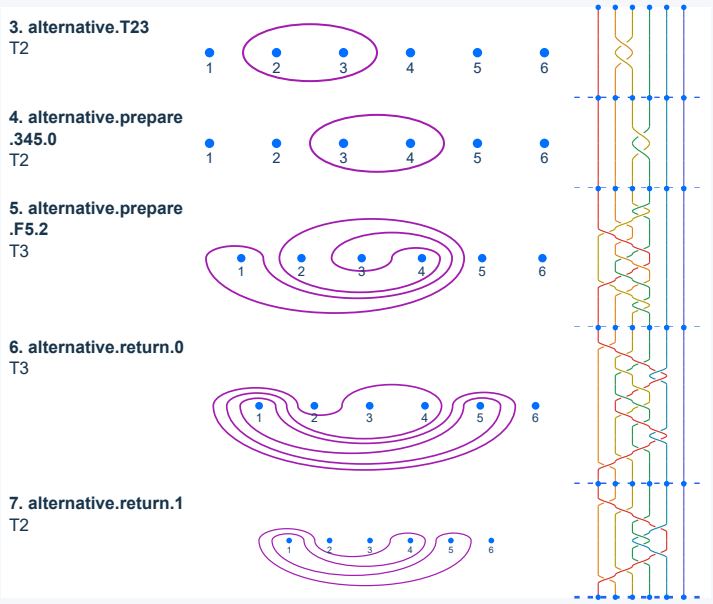
# III - step 051

Gather the ACTUAL missing 45 petal by the short side

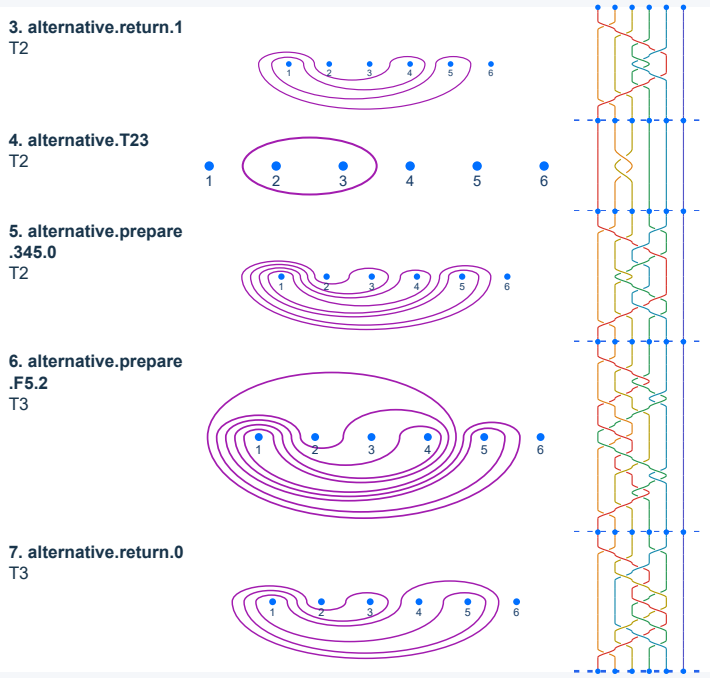
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 4.

### Before




### After




## III - step 052

Cabled PC lantern on the ACTUAL two four-point curves and supplied 45 petal

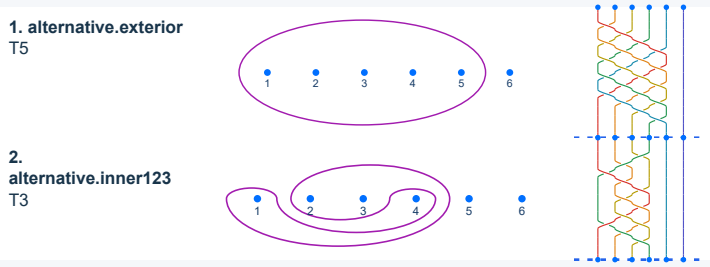
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown.

### Before




### After



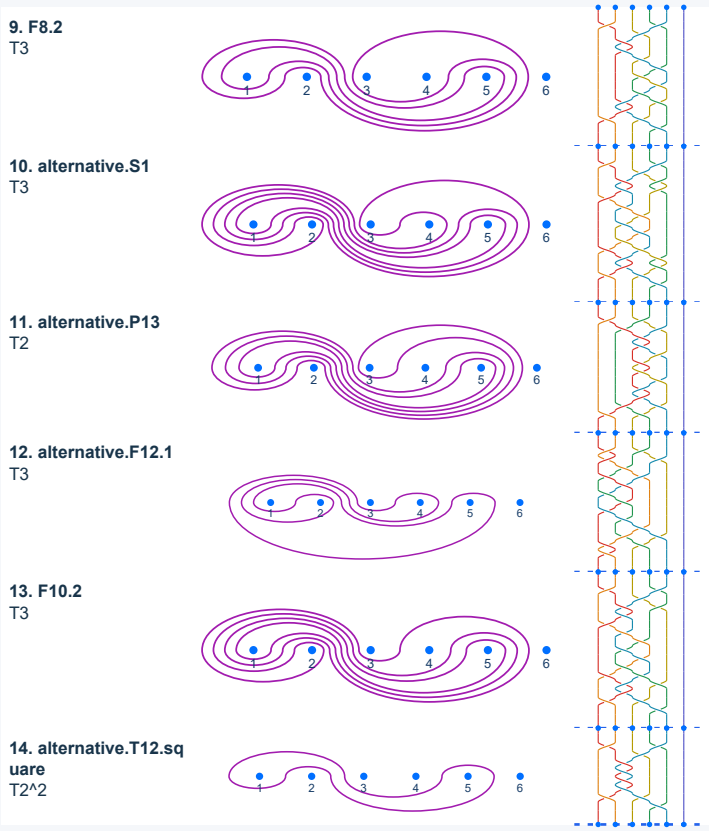
# III - step 053

Unwind five actual residual curves with the fixed positive square

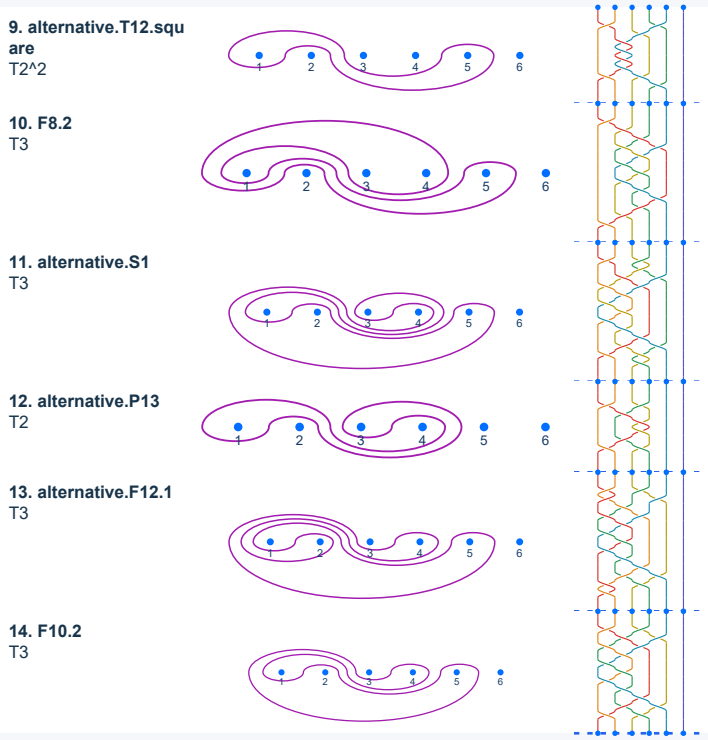
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 5.

### Before




### After



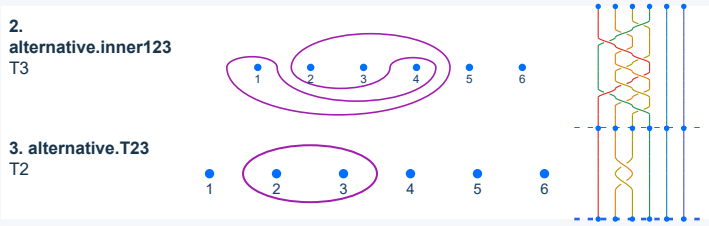

## III - step 054

Commute the 123 root past T23 to its actual 34 and 124 companions

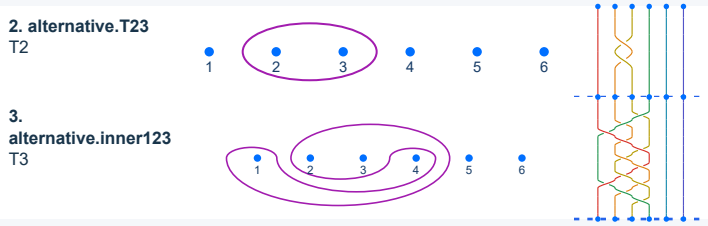
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 13 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



# III - step 055

Cabled PC lantern contracts the ACTUAL 123, 34 and 124 curves

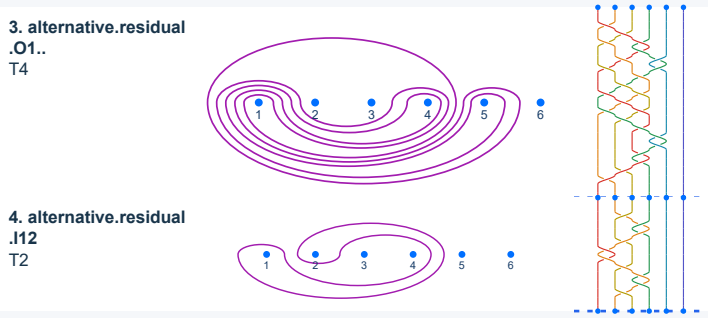
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

### Before




### After



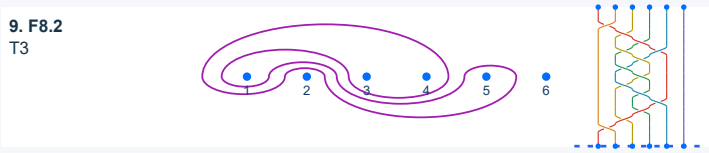

## III - step 056

The actual 345 daisy lends its round 34 petal

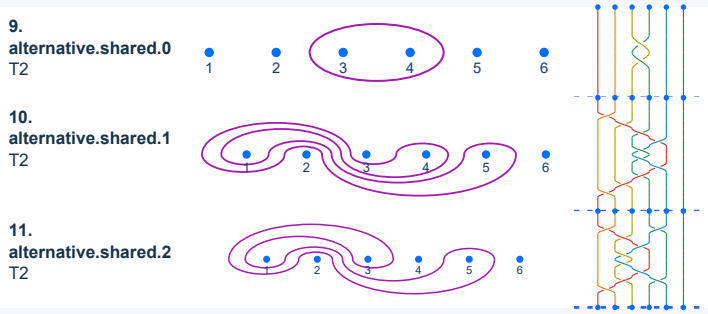
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown.

### Before




### After



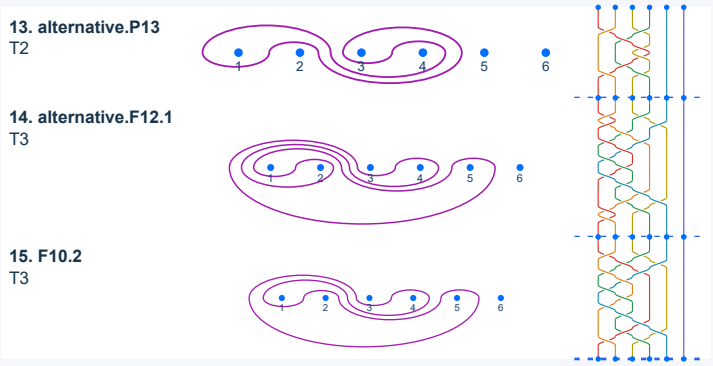
# III - step 057

Carry the 245 root up across the 145 root and the 13 pair

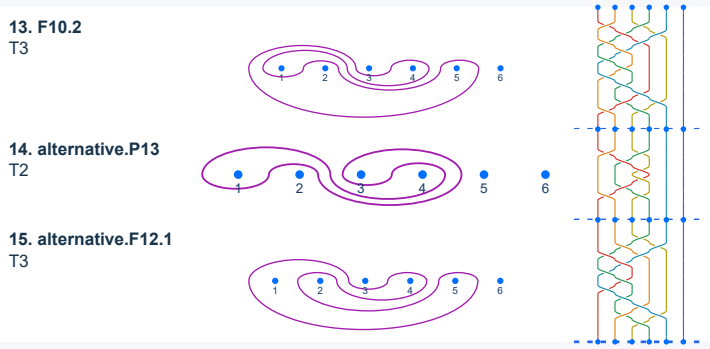
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

12 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



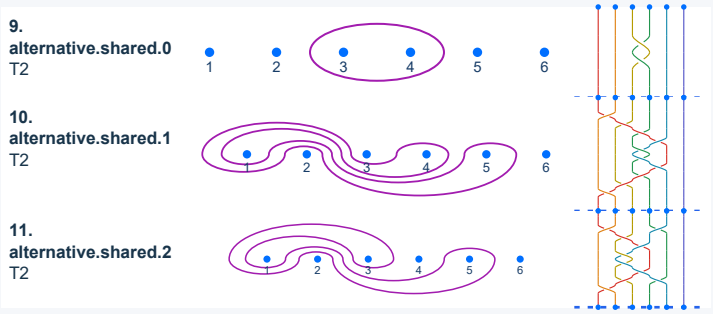

## III - step 058

Carry round 34 down through its own two remaining petals

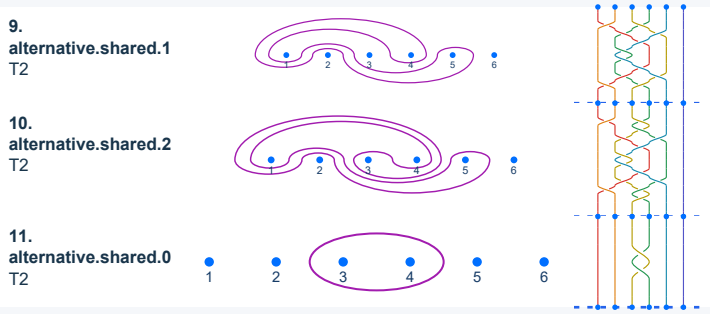
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



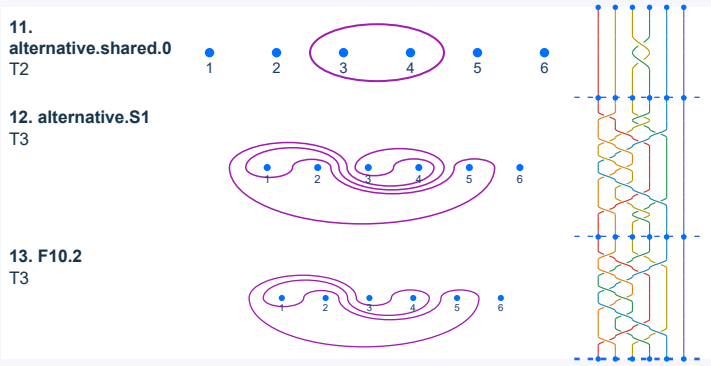
# III - step 059

Cabled PC lantern closes the actual 34, 235 and 245 cluster

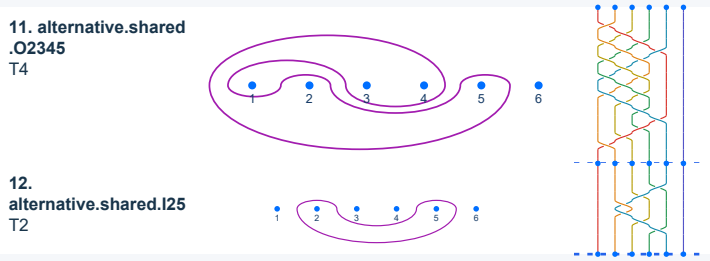
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

10 unchanged displayed factors above; 4 below. Only the changed interval is shown.

### Before




### After



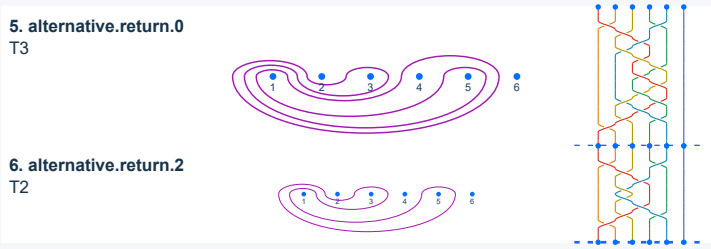

## III - step 060

The actual 35 pair unwinds the 134 root to a short chart

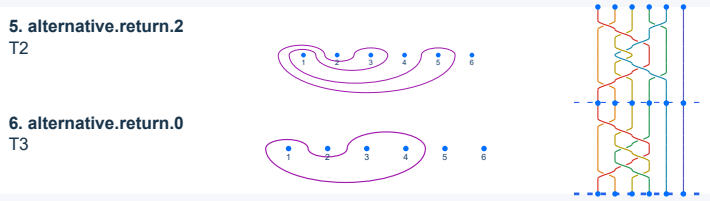
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 10 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



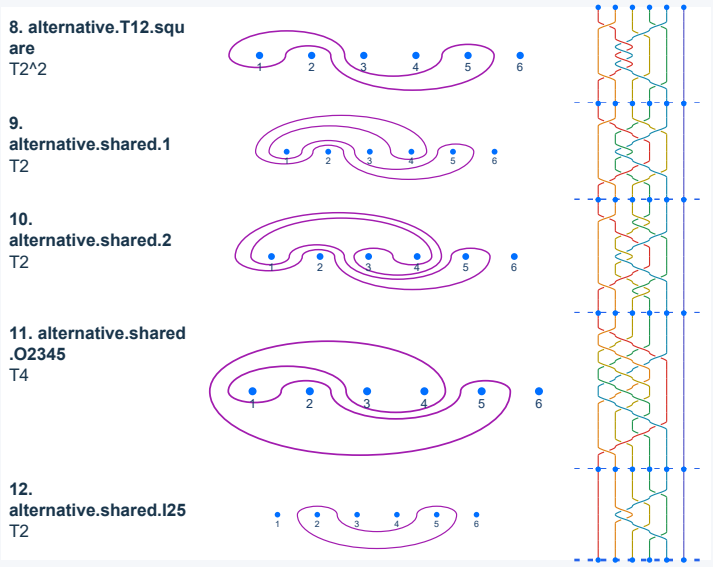
# III - step 061

Carry the shared 25 pair up; unwind the square and its two petals

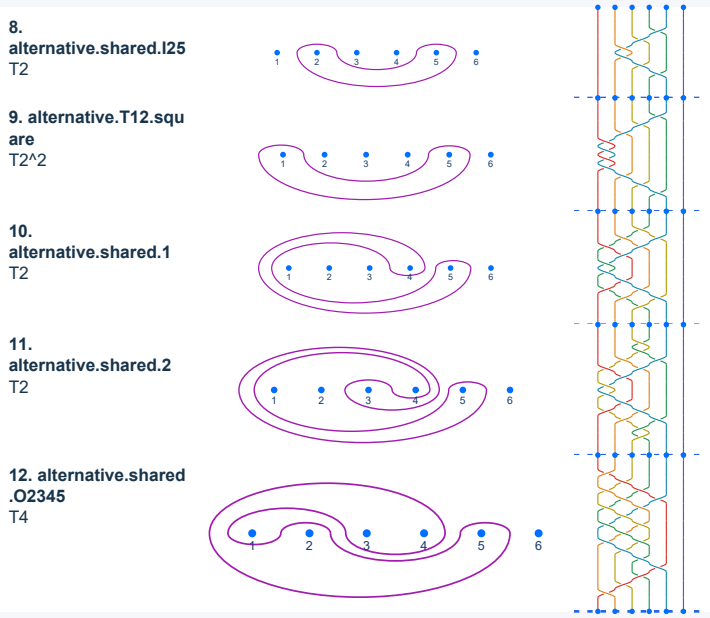
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 4.

## Before




## After



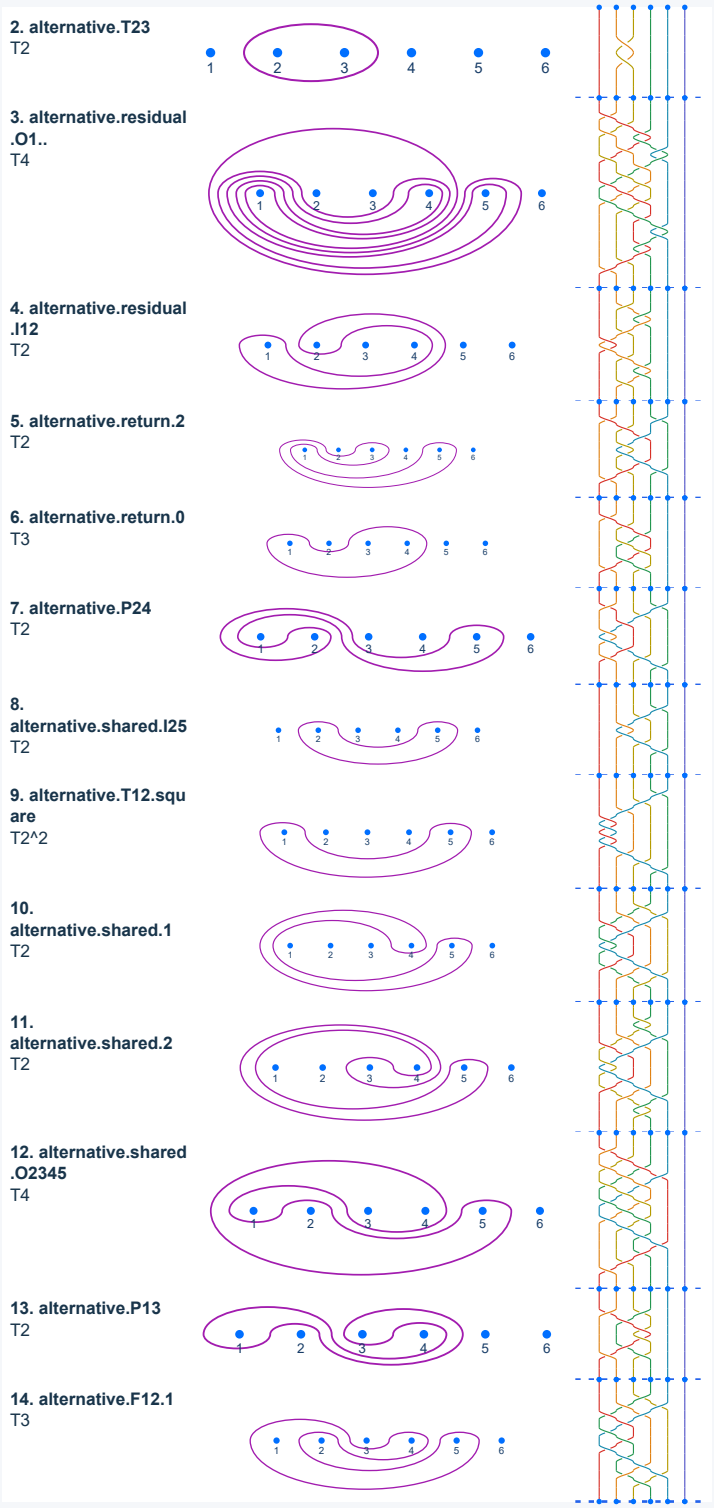
# III - step 062

Common disk frame keeps the inside-four daisy readable

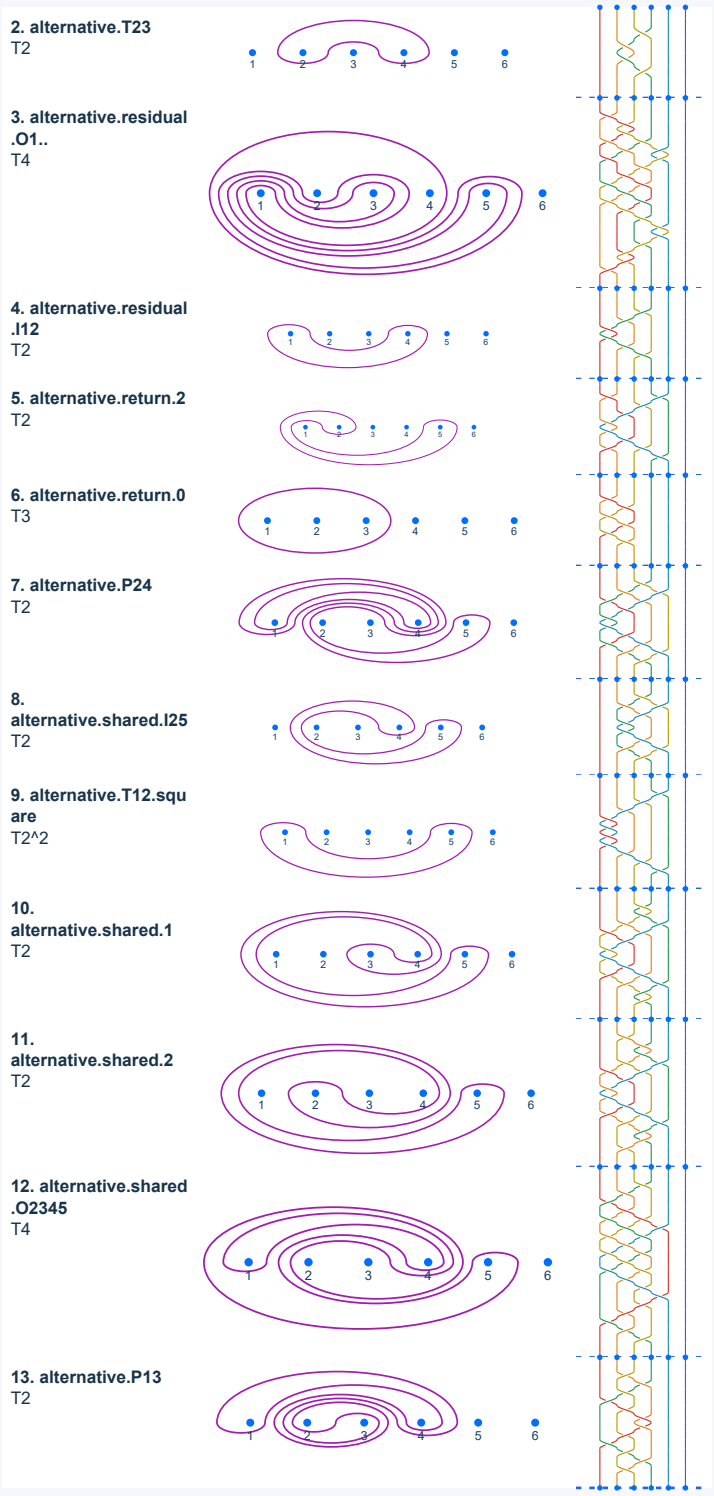
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before




## After



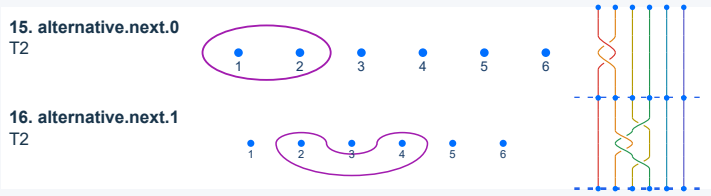
# III - step 062 - continued

Common disk frame keeps the inside-four daisy readable

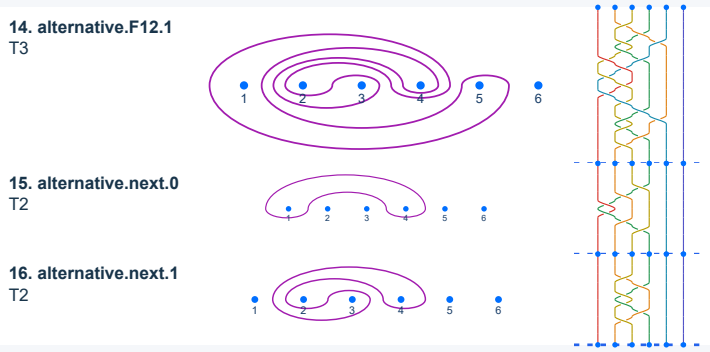
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

1 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before




## After



# III - step 063

Choose the cyclic start at the actual three-point root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before

**1. alternative.exterior**
T5

**2. alternative.T23**
T2

**3. alternative.residual .O1..**
T4

**4. alternative.residual .I12**
T2

**5. alternative.return.2**
T2

**6. alternative.return.0**
T3

**7. alternative.P24**
T2

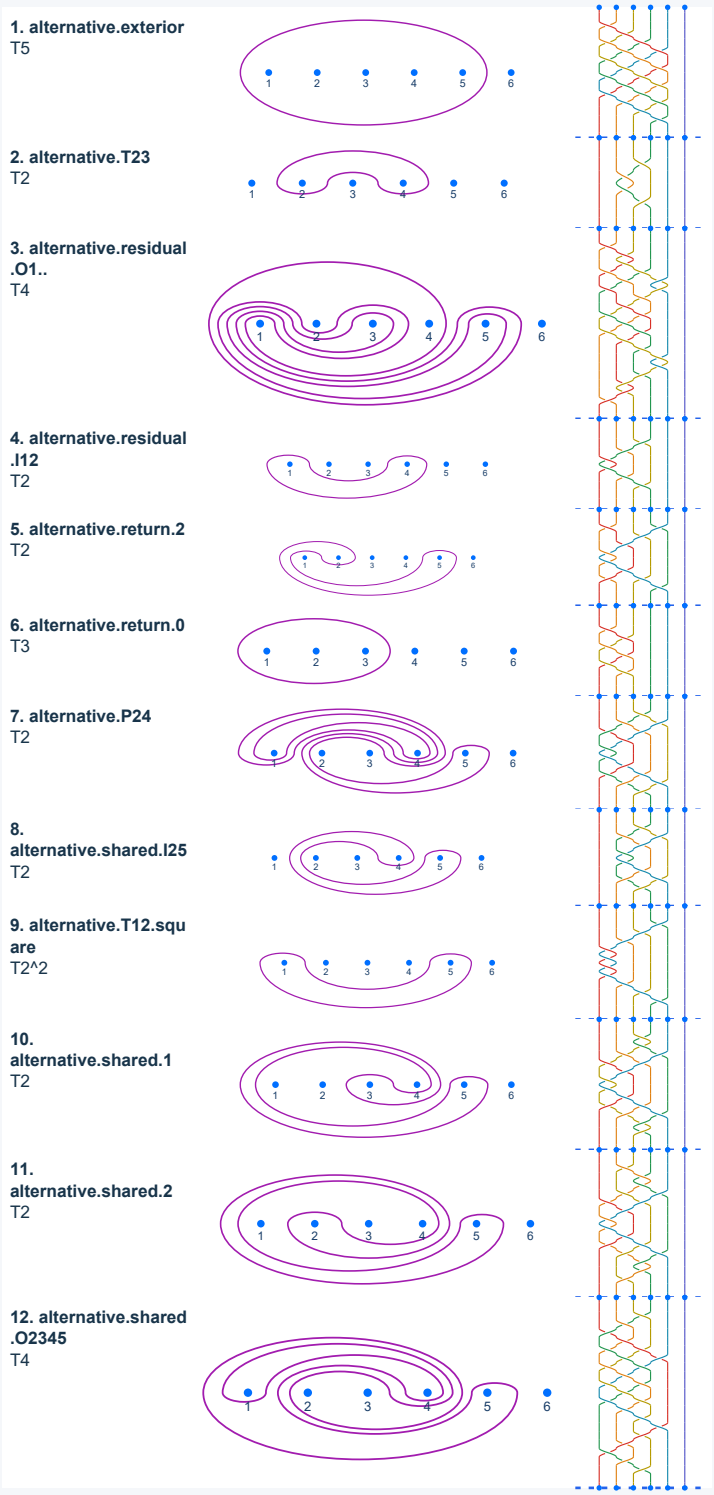
**8. alternative.shared.I25**
T2

**9. alternative.T12.square**
T2^2

**10. alternative.shared.1**
T2

**11. alternative.shared.2**
T2

**12. alternative.shared .O2345**
T4



## After

**1. alternative.return.0**
T3

**2. alternative.P24**
T2

**3. alternative.shared.I25**
T2

**4. alternative.T12.square**
T2^2

**5. alternative.shared.1**
T2

**6. alternative.shared.2**
T2

**7. alternative.shared. O2345**
T4

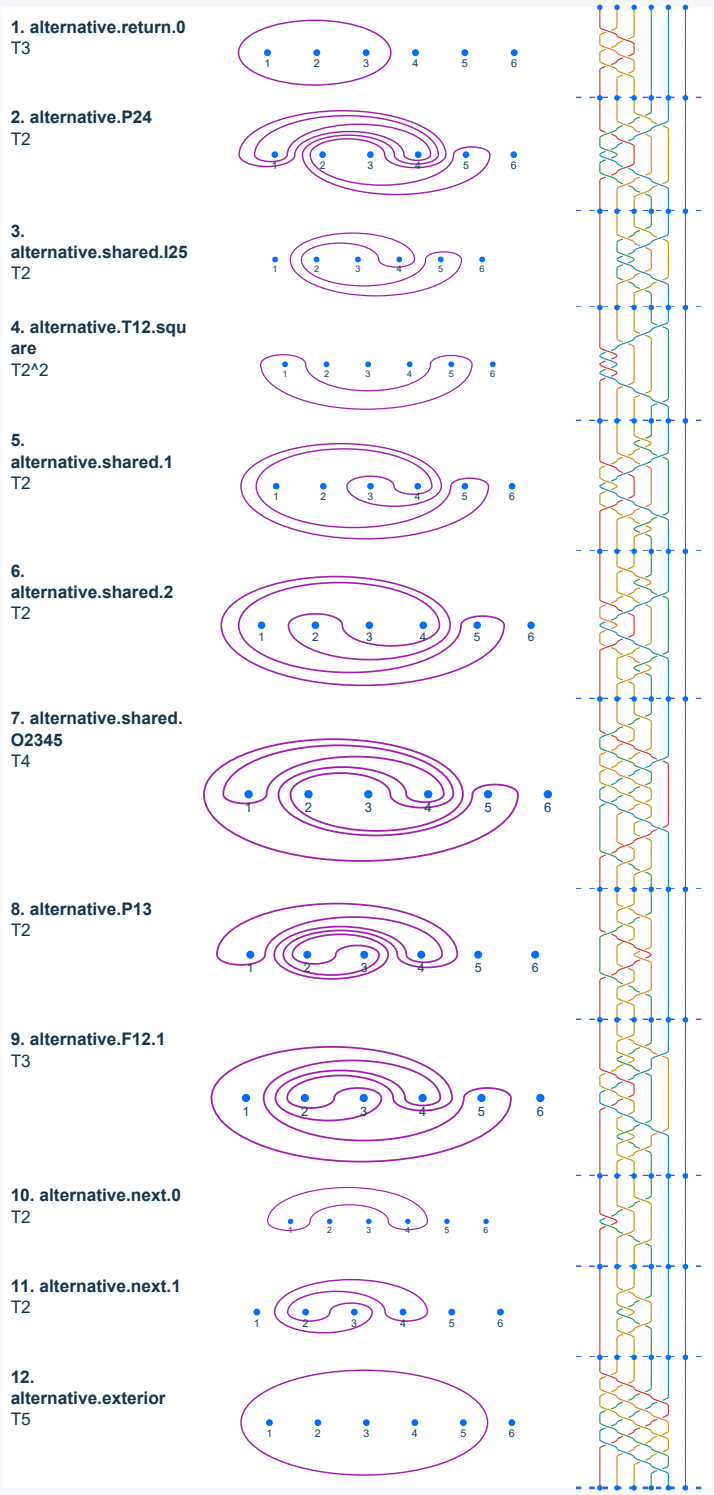
**8. alternative.P13**
T2

**9. alternative.F12.1**
T3

**10. alternative.next.0**
T2

**11. alternative.next.1**
T2

**12. alternative.exterior**
T5



# III - step 063 - continued

Choose the cyclic start at the actual three-point root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before




## After



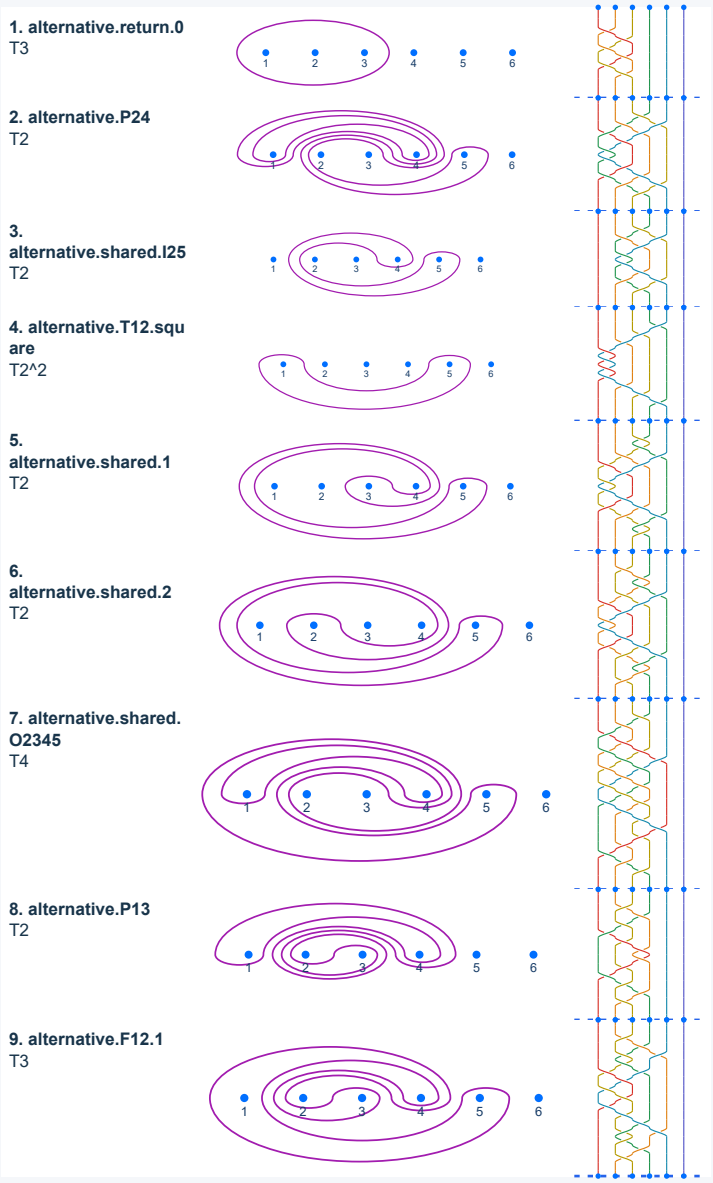
# III - step 064

Carry the three-point root down through the eight residual factors

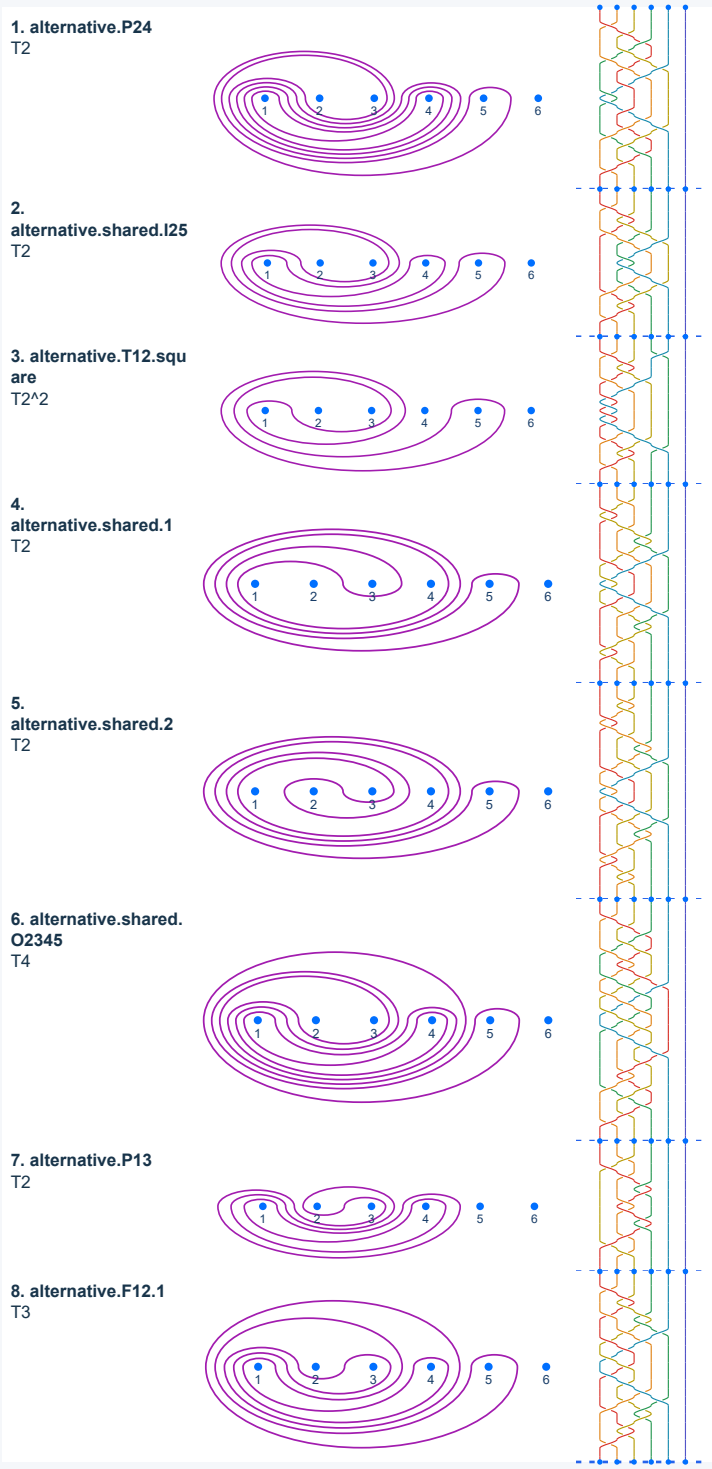
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

## Before




## After



# III - step 064 - continued

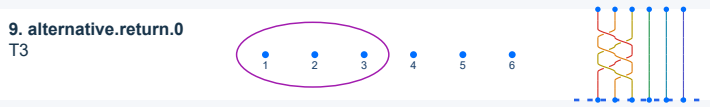
Carry the three-point root down through the eight residual factors

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 7 below. Only the changed interval is shown. Retained adjacent checks: 8.

### Before

### After



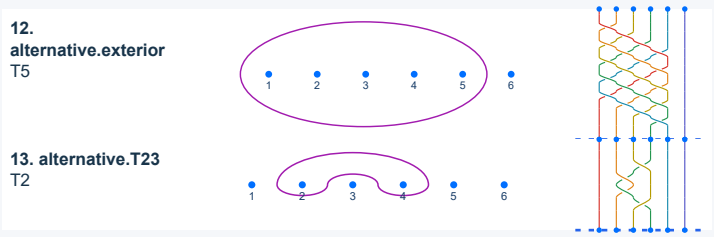

## III - step 065

Commute the final pair petal past the visible exterior

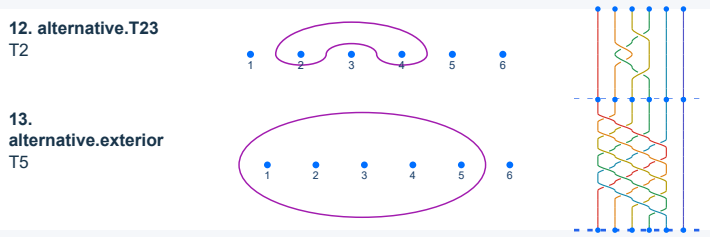
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After




## III - step 066

Transported PC daisy closes the actual root and its three pair petals

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 4 below. Only the changed interval is shown.

### Before




### After



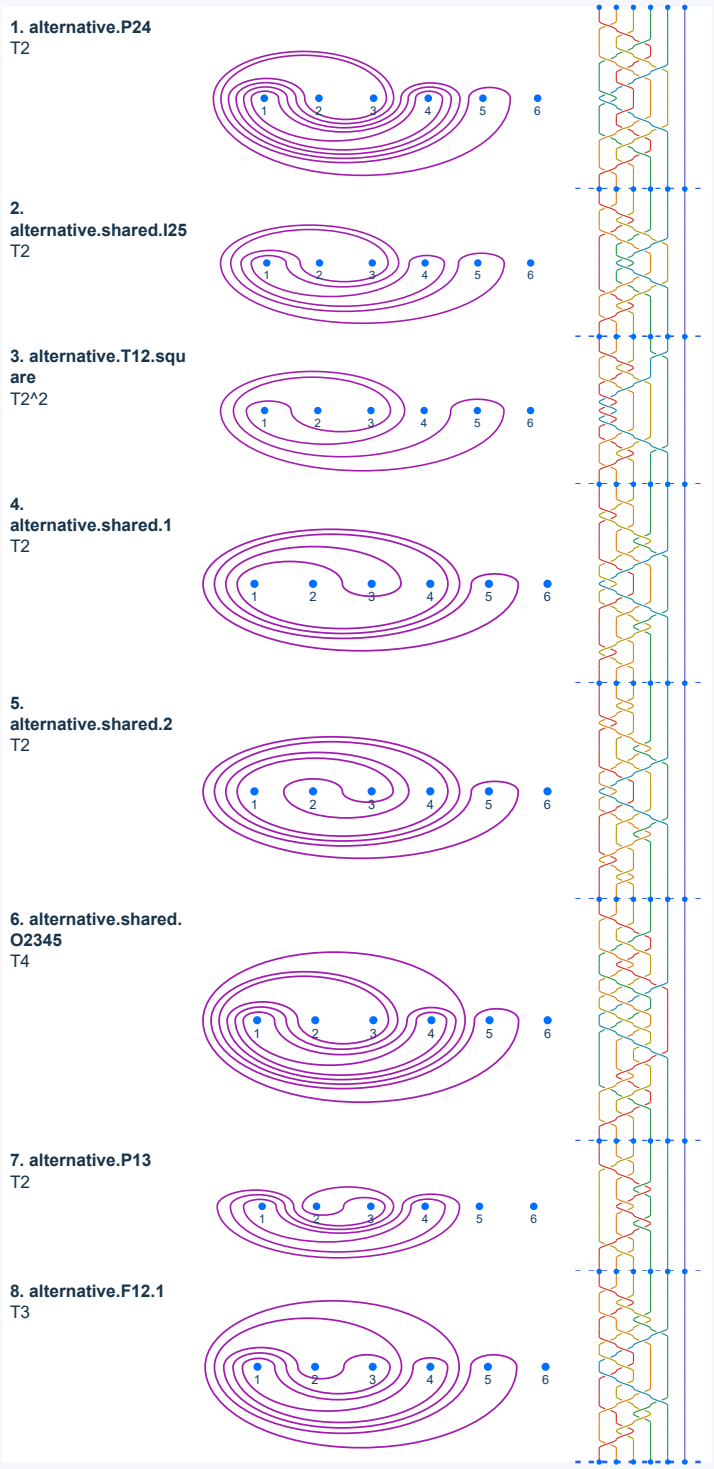
# III - step 067

Carry the new round four-point twist up to unwind the residual block

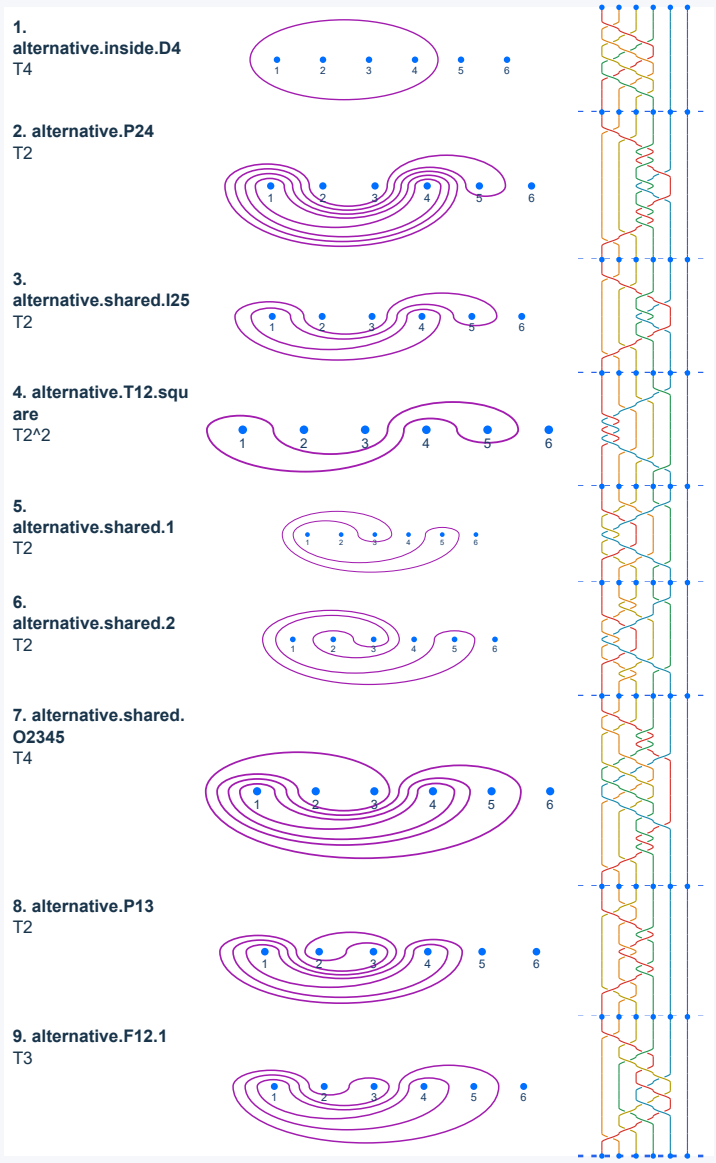
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

## Before




## After



# III - step 067 - continued

Carry the new round four-point twist up to unwind the residual block

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 8.

### Before

**9.**
**alternative.inside.D4**
T4




### After

# IV. Reach the second exterior twist

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

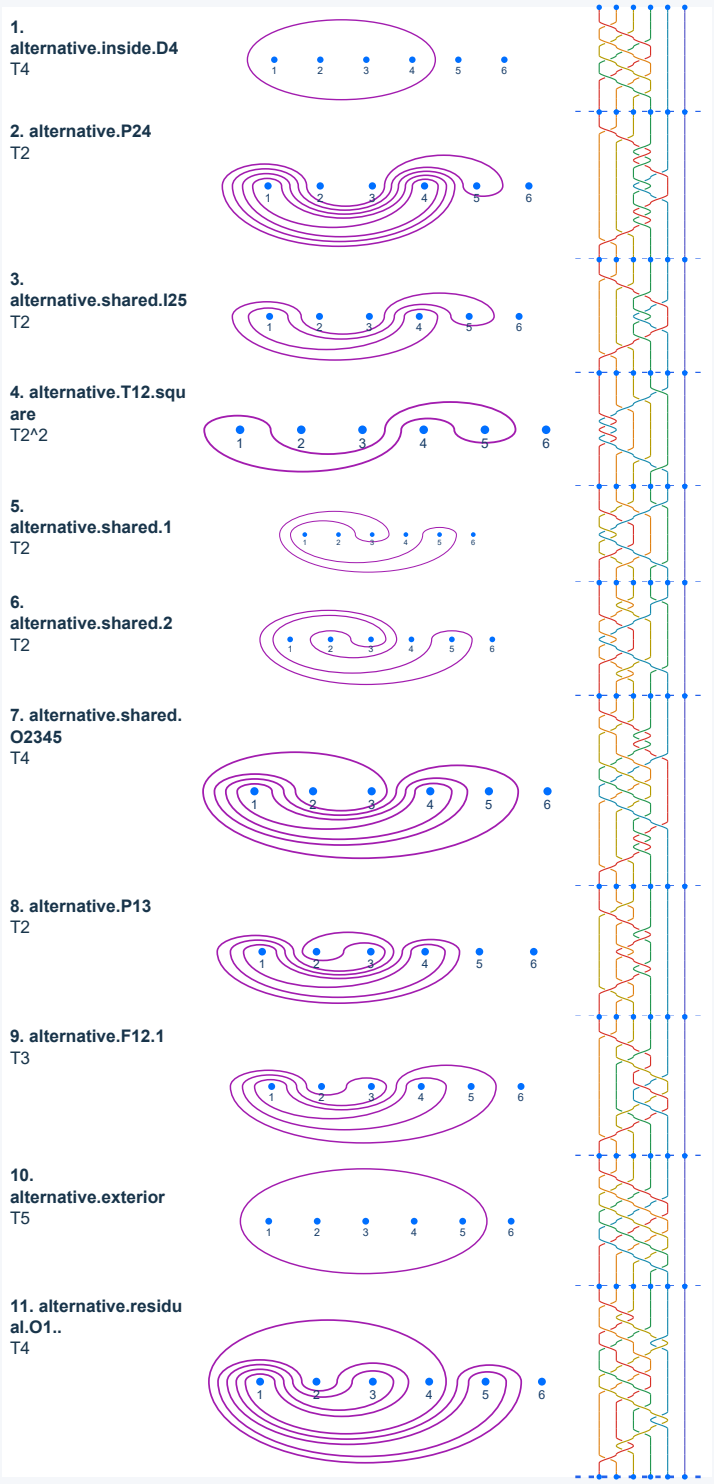
# IV - step 068

Common disk frame keeps the second exterior lantern readable

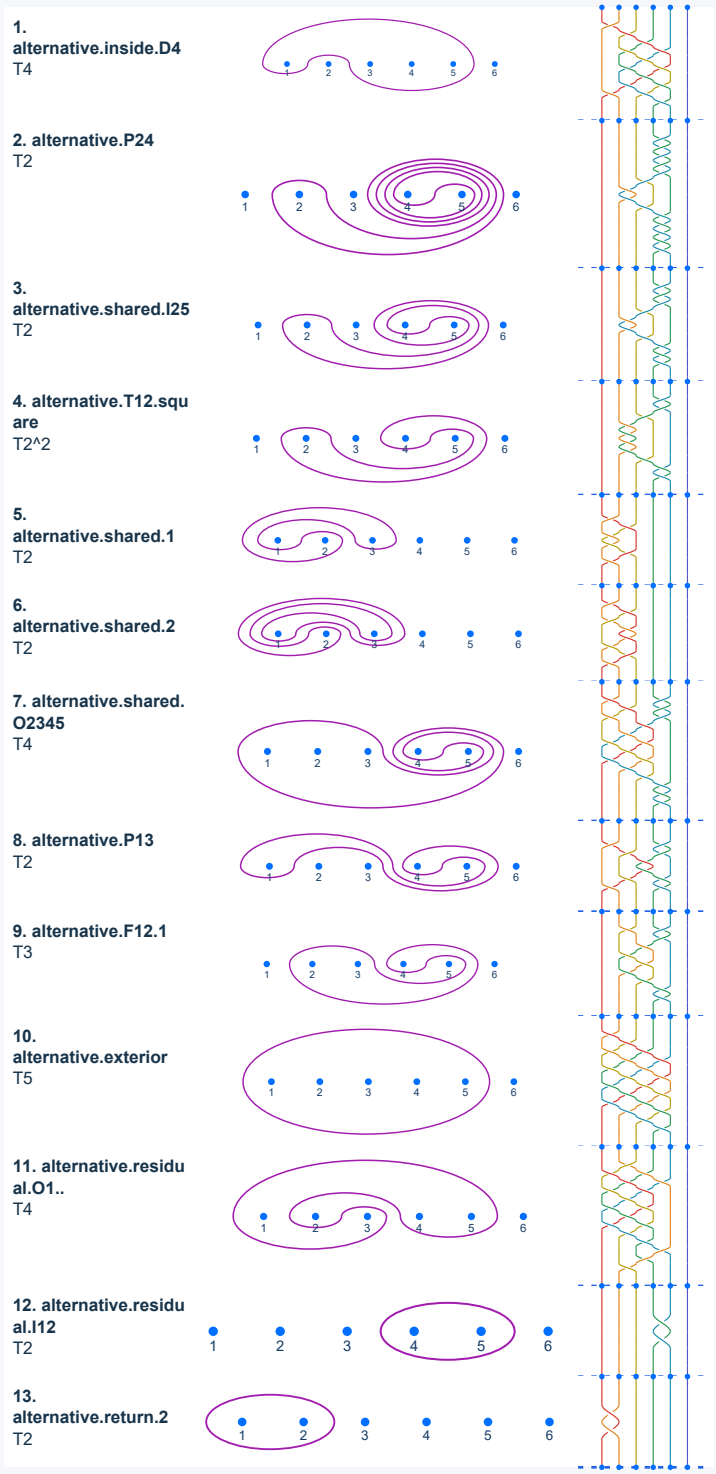
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before




## After



# IV - step 068 - continued

Common disk frame keeps the second exterior lantern readable

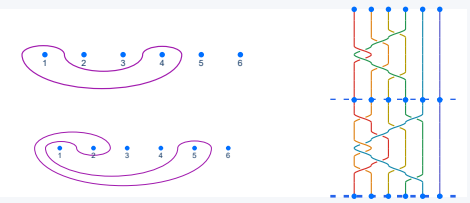
EXPLICIT GLOBAL CONJUGATION: a change of the whole product by the recorded frame.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

### Before

**12. alternative.residual.l12**
T2

**13. alternative.return.2**
T2



### After

## IV - step 069

The four-point PC daisy lends its actual three-point root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 12 below. Only the changed interval is shown.

### Before

**1. alternative.inside.D4**
T4



### After

**1. alternative.second.loan.0**
T3

**2. alternative.second.loan.1**
T2

**3. alternative.second.loan.2**
T2

**4. alternative.second.loan.3**
T2



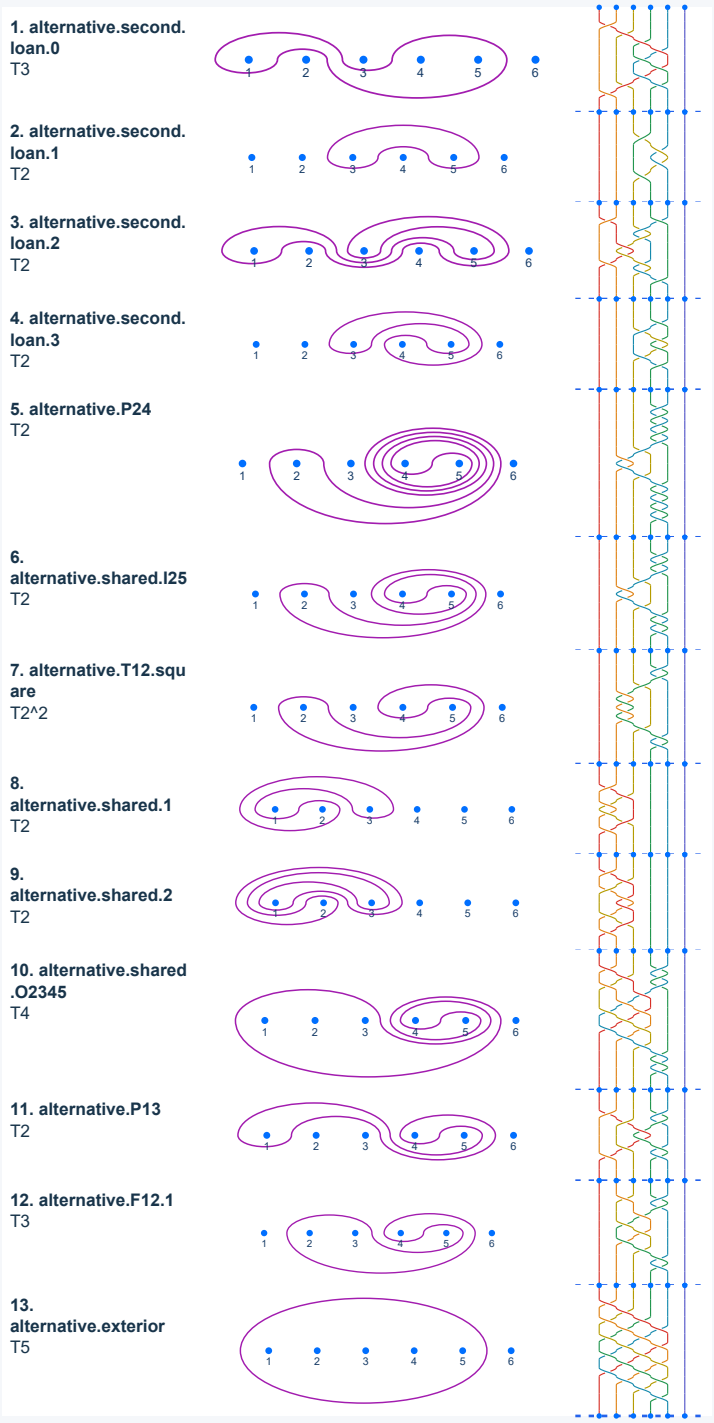
# IV - step 070

Cyclic start at the actual pair that transports the borrowed root

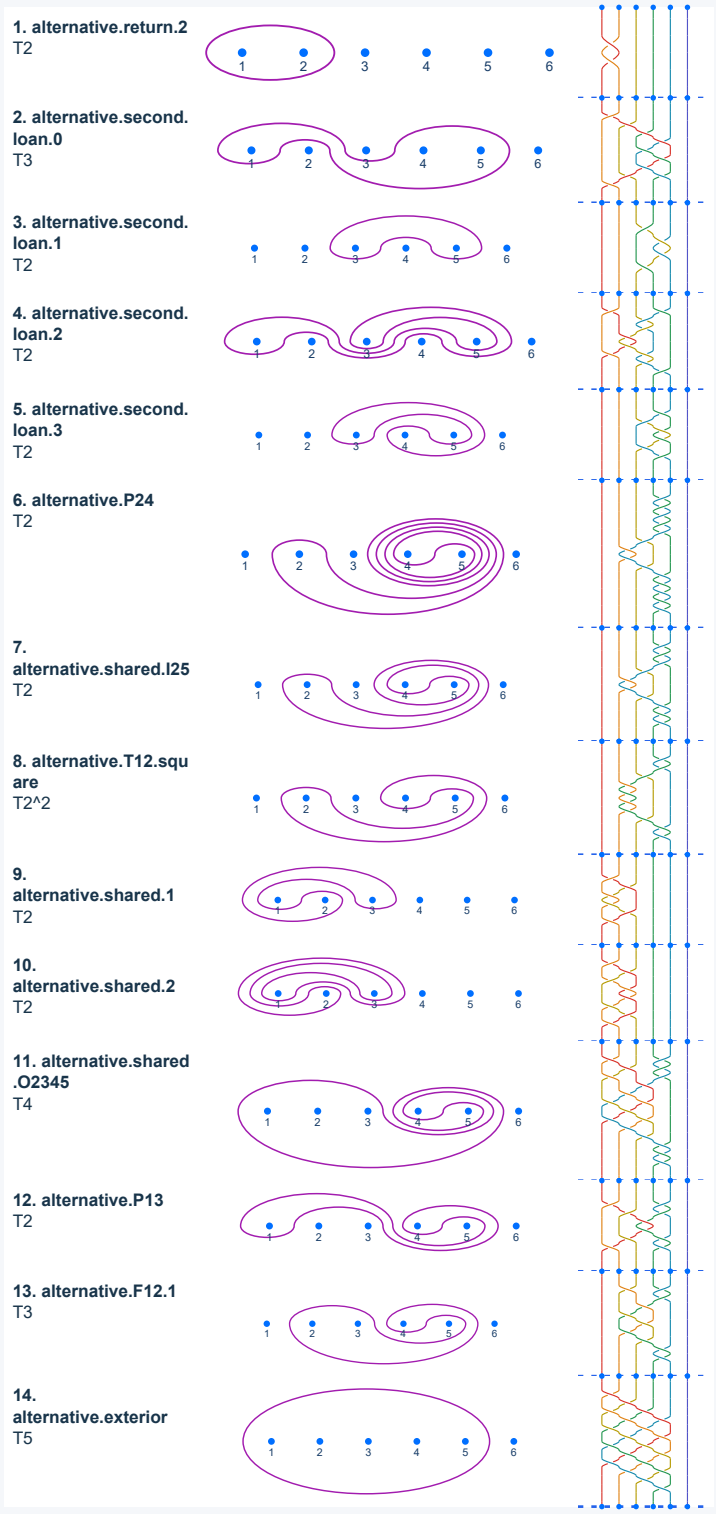
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before




## After



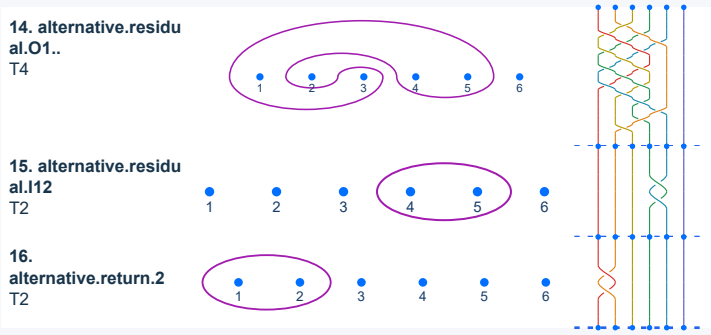
# IV - step 070 - continued

Cyclic start at the actual pair that transports the borrowed root

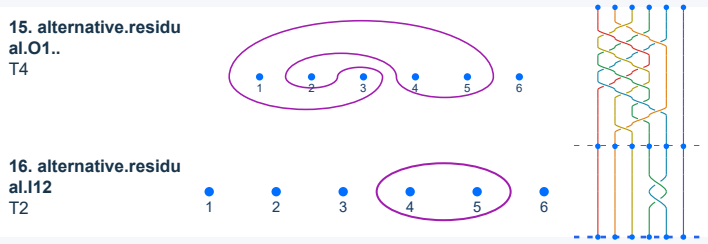
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

### Before




### After



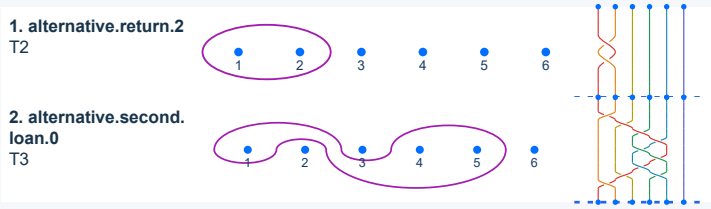

## IV - step 071

The actual pair carries the borrowed root to the exact missing class

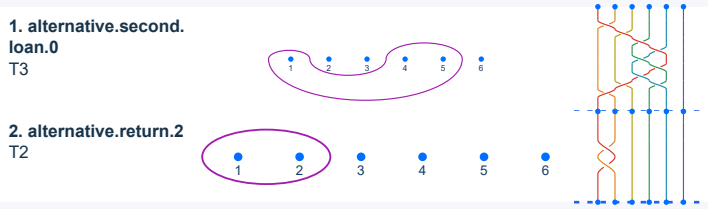
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 14 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



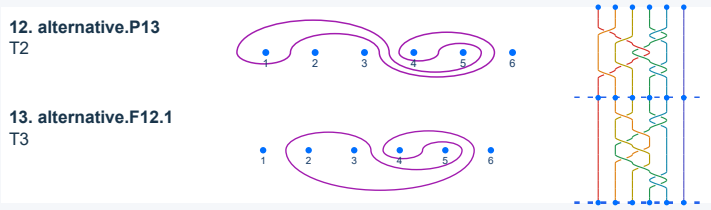

## IV - step 072

Bring the three-point curve up across one neighboring pair

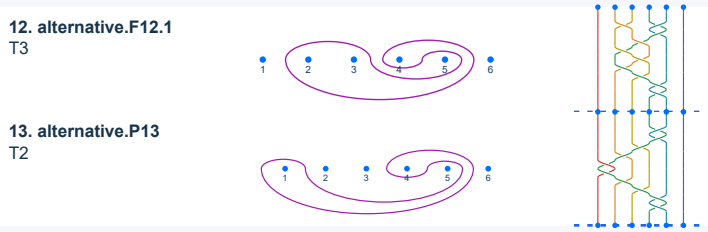
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

11 unchanged displayed factors above; 3 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



# IV - step 073

Cyclic start at the matched four-point curve

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before

**1. alternative.second.loan.0**
T3

**2. alternative.return.2**
T2



**3. alternative.second.loan.1**
T2

**4. alternative.second.loan.2**
T2

**5. alternative.second.loan.3**
T2

**6. alternative.P24**
T2

**7. alternative.shared.I25**
T2

**8. alternative.T12.square**
T2^2

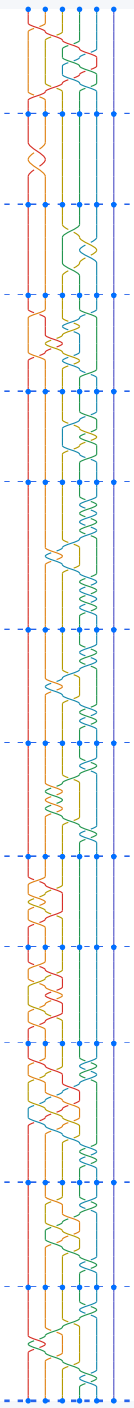
**9. alternative.shared.1**
T2

**10. alternative.shared.2**
T2

**11. alternative.shared.O2345**
T4

**12. alternative.F12.1**
T3

**13. alternative.P13**
T2



## After

**1. alternative.shared.O2345**
T4

**2. alternative.F12.1**
T3

**3. alternative.P13**
T2

**4. alternative.exterior**
T5

**5. alternative.residual.O1..**
T4

**6. alternative.residual.I12**
T2

**7. alternative.second.loan.0**
T3

**8. alternative.return.2**
T2

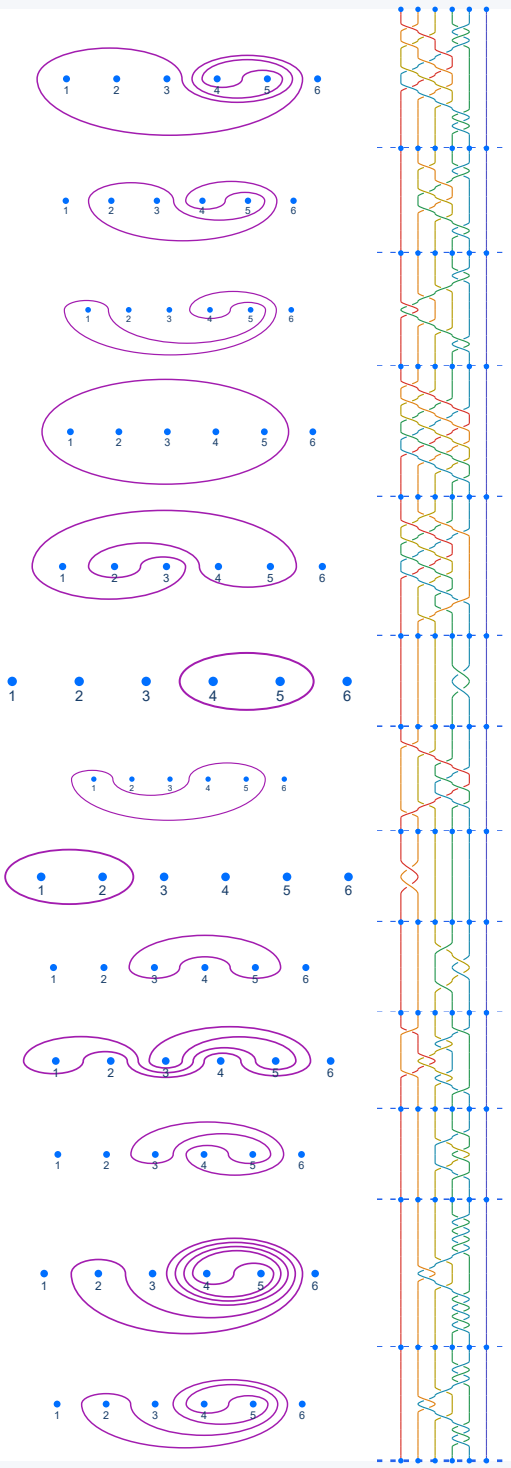
**9. alternative.second.loan.1**
T2

**10. alternative.second.loan.2**
T2

**11. alternative.second.loan.3**
T2

**12. alternative.P24**
T2

**13. alternative.shared.I25**
T2



# IV - step 073 - continued

Cyclic start at the matched four-point curve

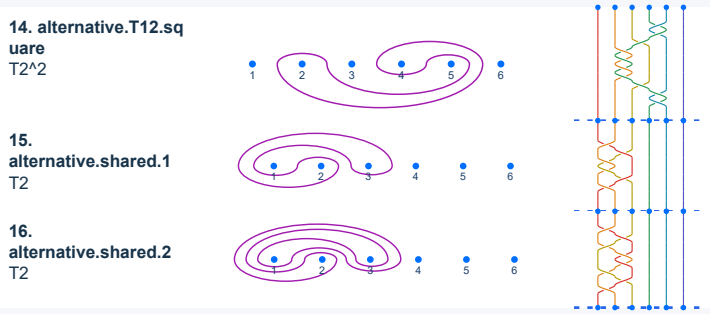
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

### Before




### After



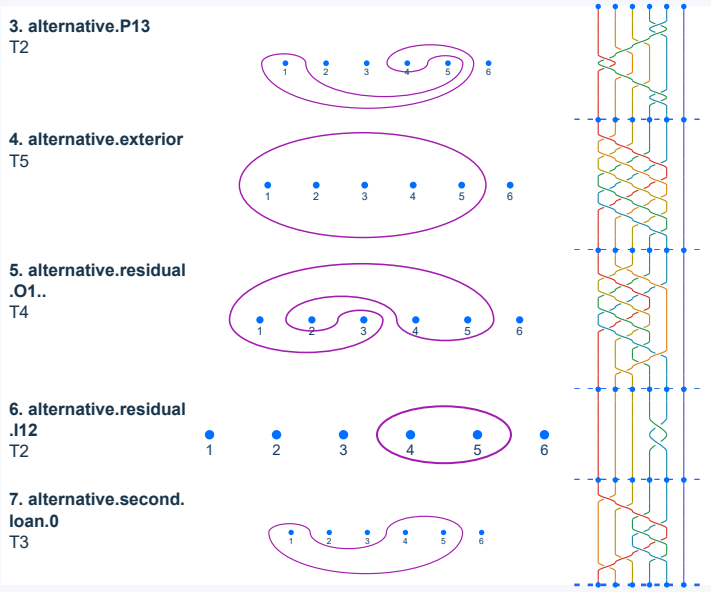

## IV - step 074

Gather the borrowed root up the four-neighbor short side

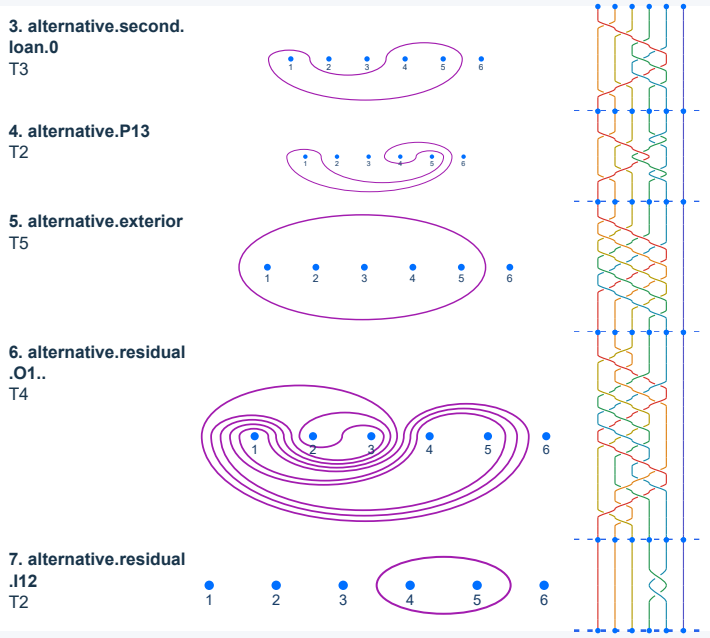
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 4.

### Before




### After



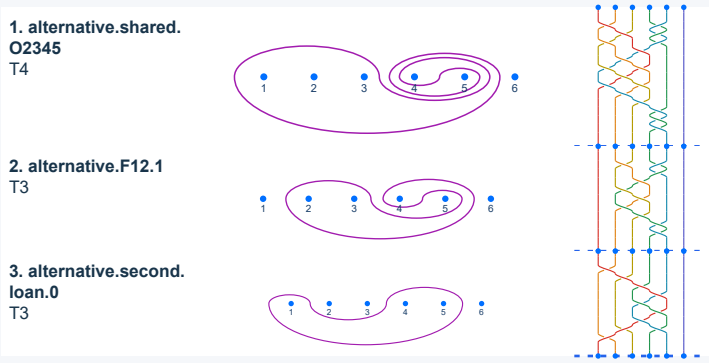
# IV - step 075

Cabled PC lantern reaches a SECOND visible exterior twist

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 13 below. Only the changed interval is shown.

## Before




## After



# V. Return the actual borrowed root

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

# Complete factorization before Return the actual borrowed root

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

**1. alternative.second.exte..**
T5



**2. alternative.second.l35**
T2



**3. alternative.second.l24**
T2



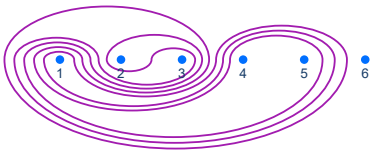
**4. alternative.P13**
T2



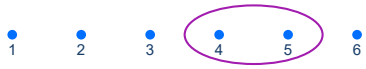
**5. alternative.exterior**
T5



**6. alternative.residual.O1..**
T4



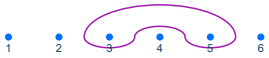
**7. alternative.residual.l12**
T2



**8. alternative.return.2**
T2



**9. alternative.second.loan.1**
T2



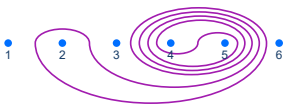
**10. alternative.second.loan.2**
T2



**11. alternative.second.loan.3**
T2



**12. alternative.P24**
T2



**13. alternative.shared.l25**
T2

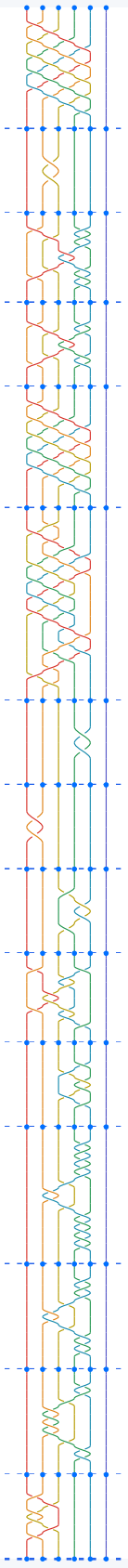


**14. alternative.T12.square**
T2^2



**15. alternative.shared.1**
T2



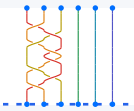


# Complete factorization before Return the actual borrowed root - continued

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

**16. alternative.shared.2**
T2



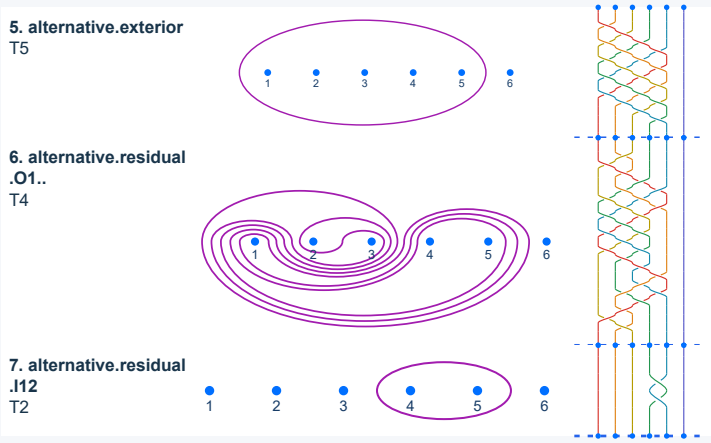
# V - step 076

Drag alternative.residual.I12 up

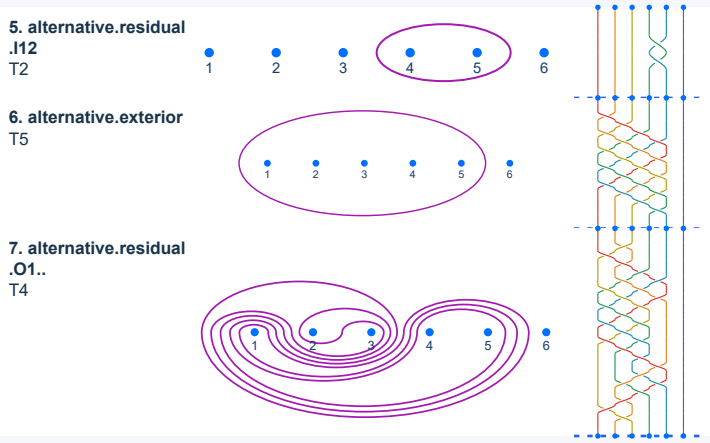
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



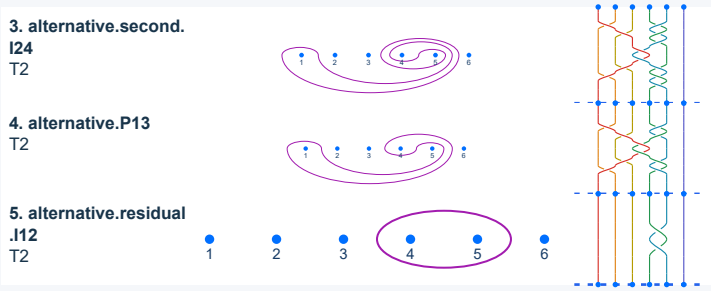

## V - step 077

A transported PC rose returns the actual borrowed root

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 11 below. Only the changed interval is shown.

### Before




### After



# V - step 078

Drag alternative.return.2 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 3.

### Before




### After



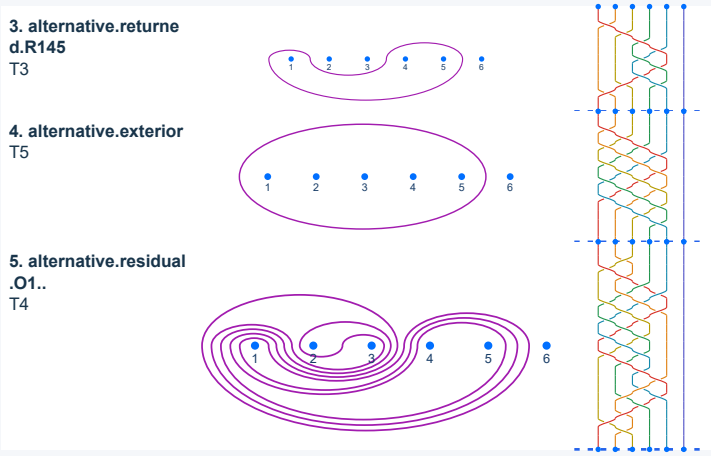

## V - step 079

Drag alternative.returned.R145 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

2 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



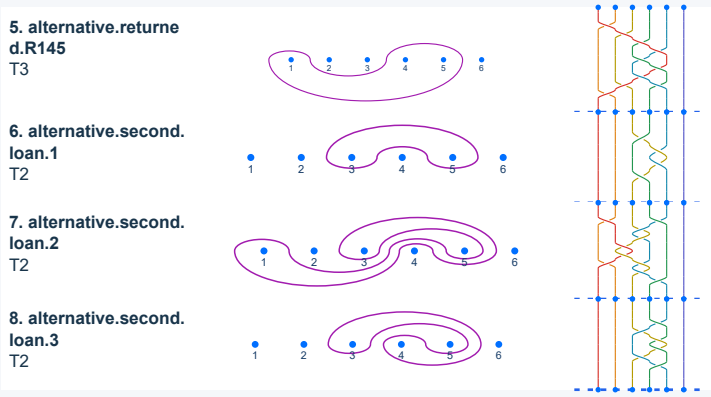
# V - step 080

Close the actual returned PC daisy

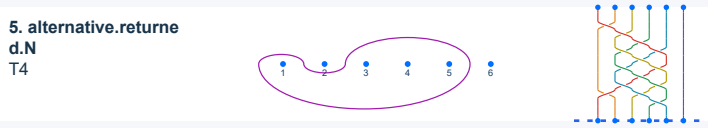
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 6 below. Only the changed interval is shown.

## Before




## After



# VI. Align the four-point twists

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

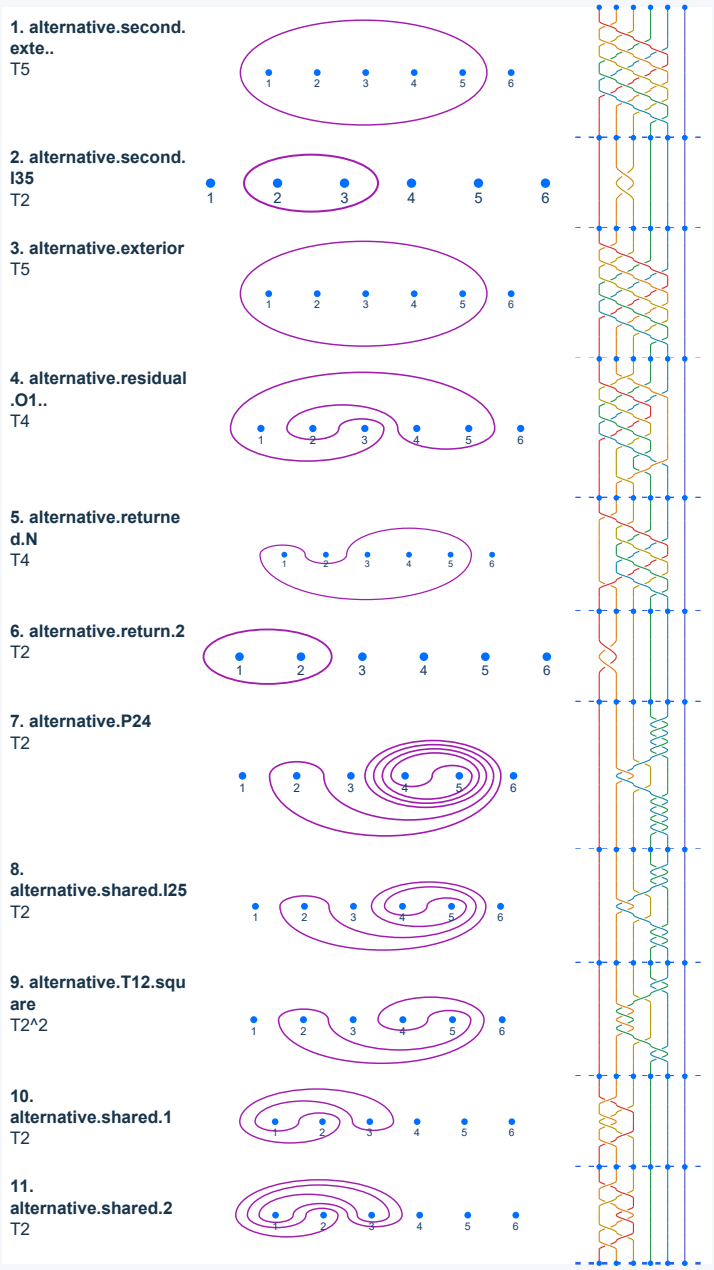
# VI - step 081

Cyclic start at the two-petal alignment supplier

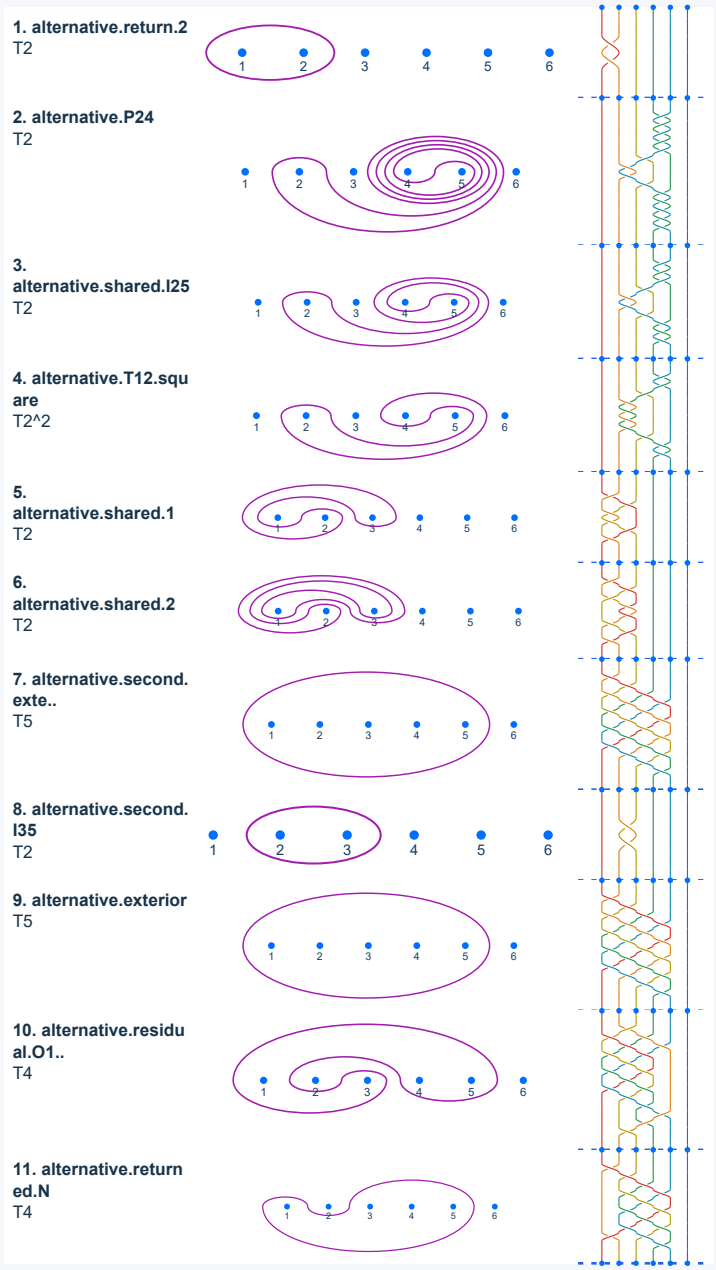
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before




## After



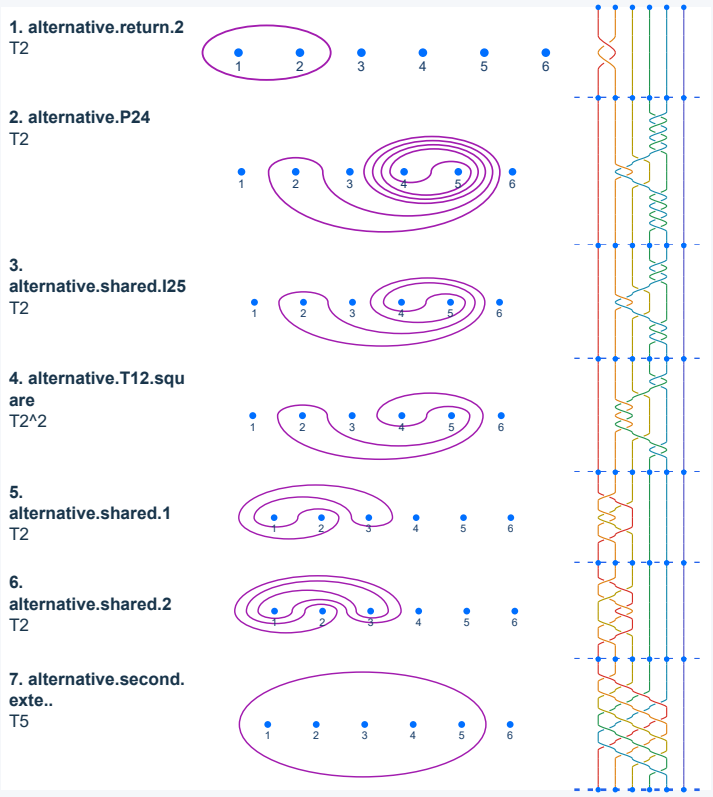
# VI - step 082

Drag alternative.return.2 down

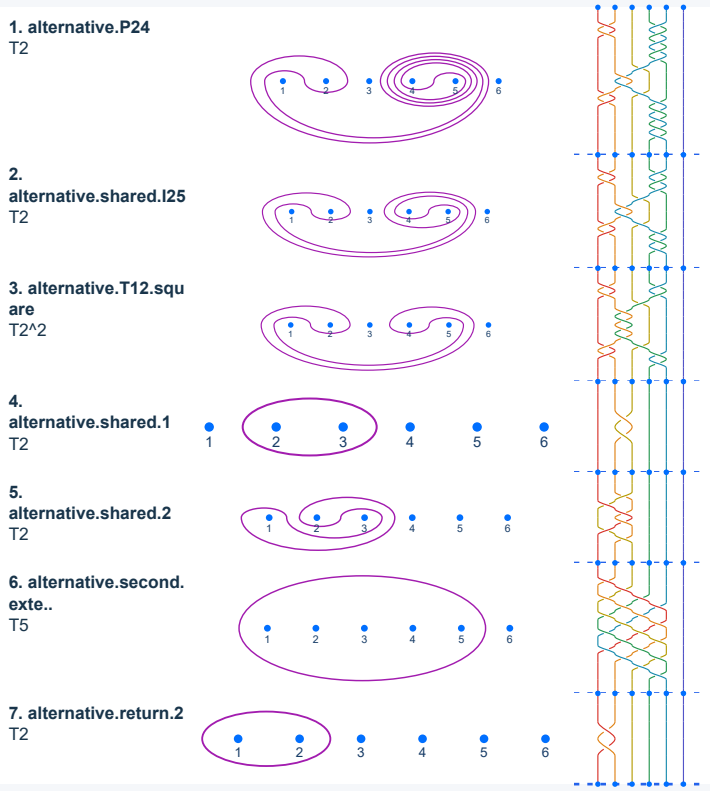
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 4 below. Only the changed interval is shown. Retained adjacent checks: 6.

### Before




### After




## VI - step 083

Drag alternative.second.I35 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 1 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



# VI - step 084

Drag alternative.return.2 down

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

6 unchanged displayed factors above; 2 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



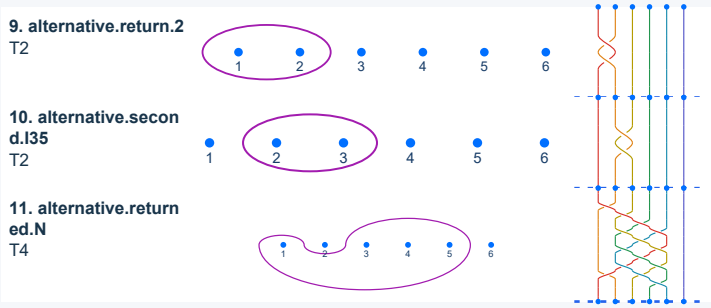

## VI - step 085

Drag alternative.returned.N up

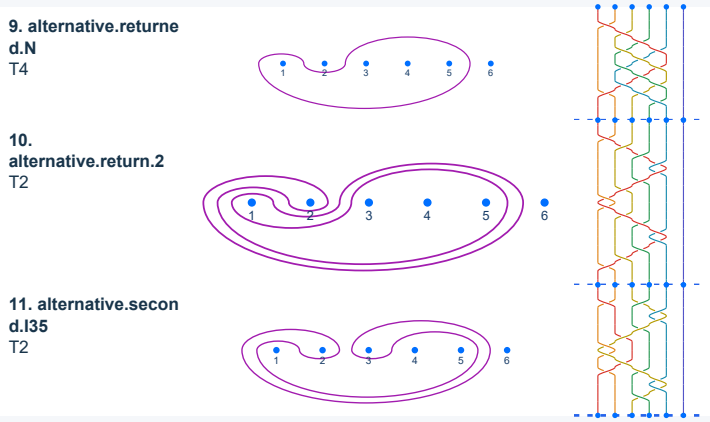
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

8 unchanged displayed factors above; 0 below. Only the changed interval is shown. Retained adjacent checks: 2.

### Before




### After



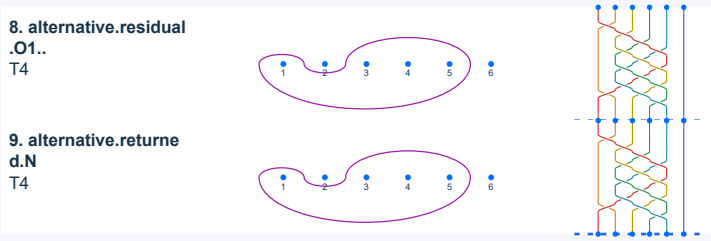
# VI - step 086

Combine the matching alternative.residual.O1234 and alternative.returned.N

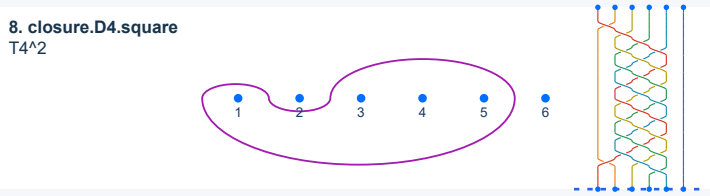
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

7 unchanged displayed factors above; 2 below. Only the changed interval is shown.

## Before




## After



# VII. Match the actual transported petals

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

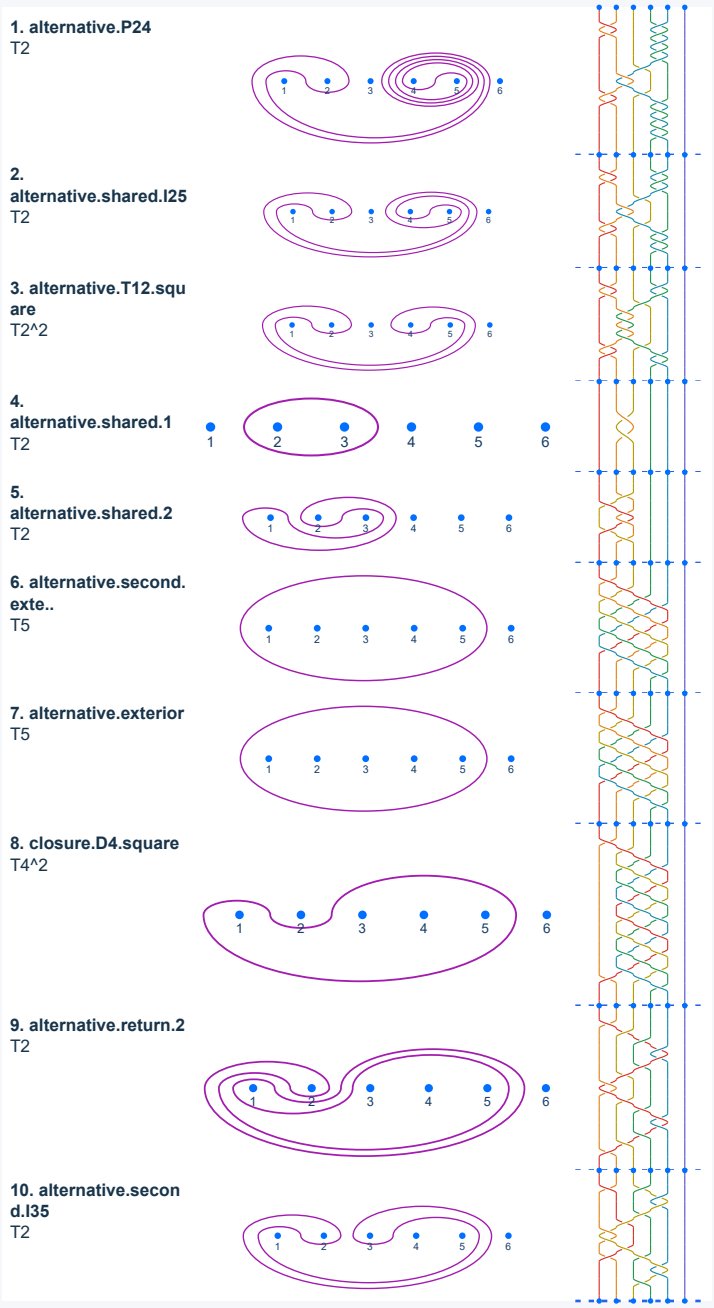
# VII - step 087

Split alternative.T12.square into its two positive units

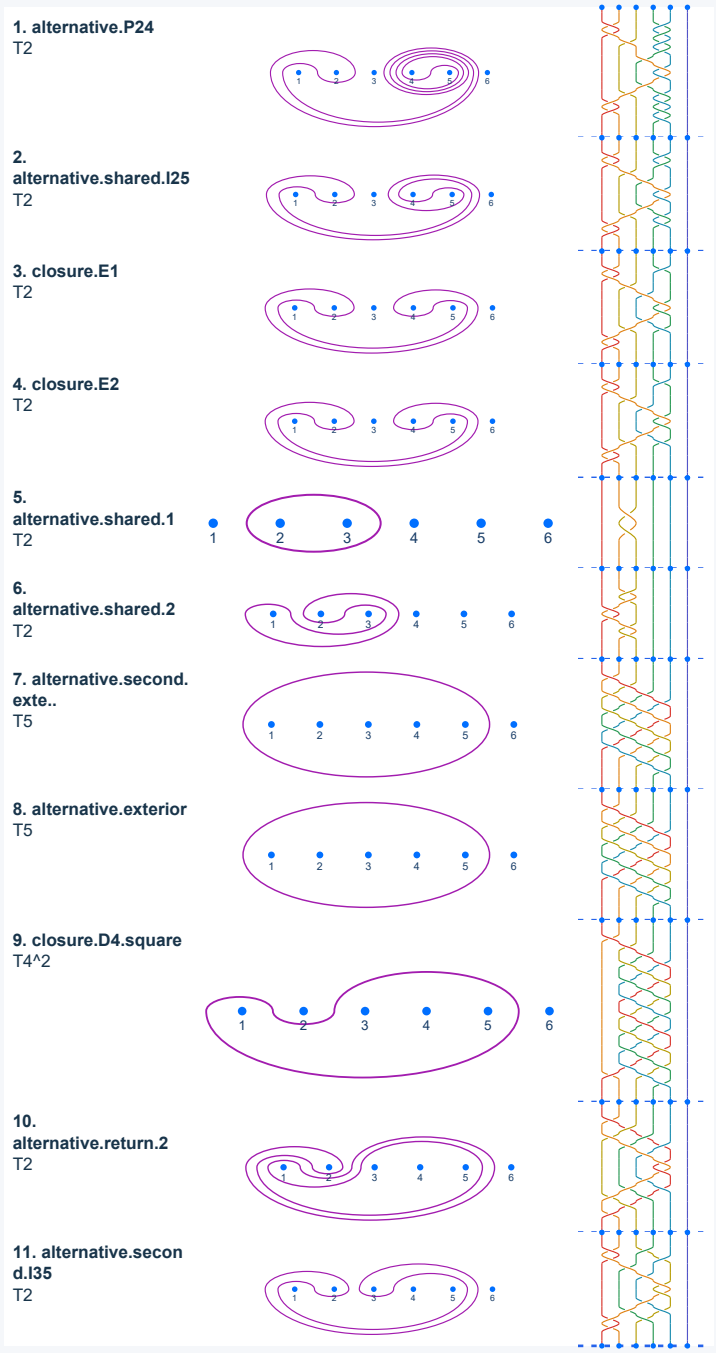
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

## Before




## After



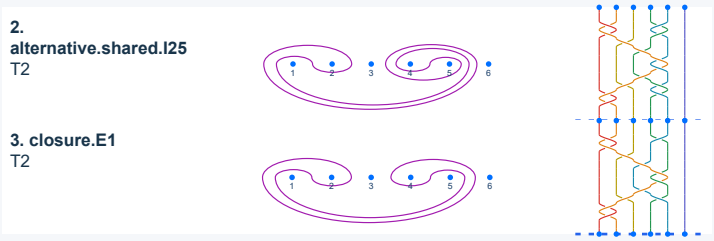
# VII - step 088

Drag alternative.shared.I25 down

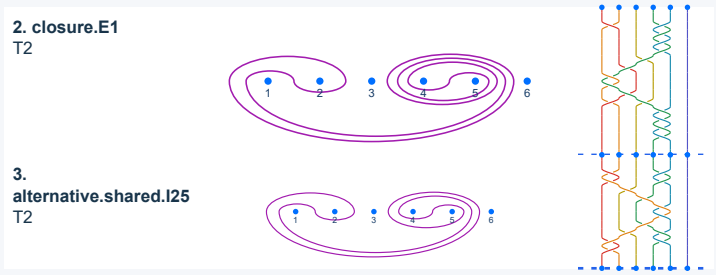
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

1 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



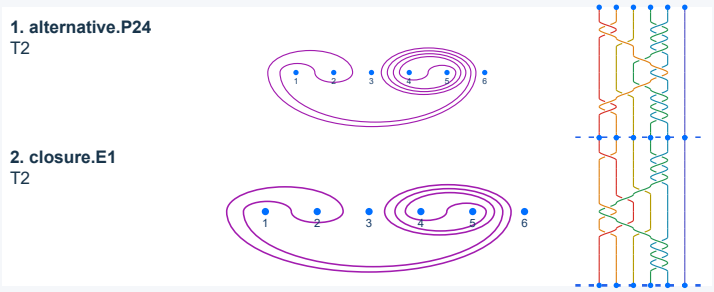

## VII - step 089

Drag closure.E1 up

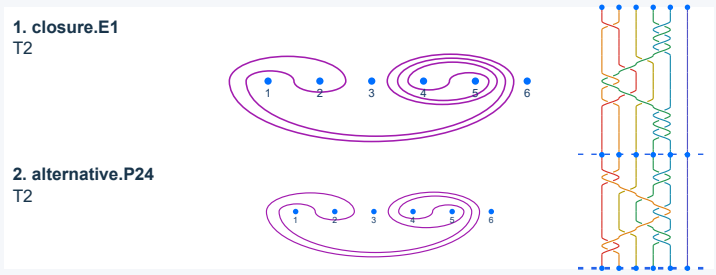
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



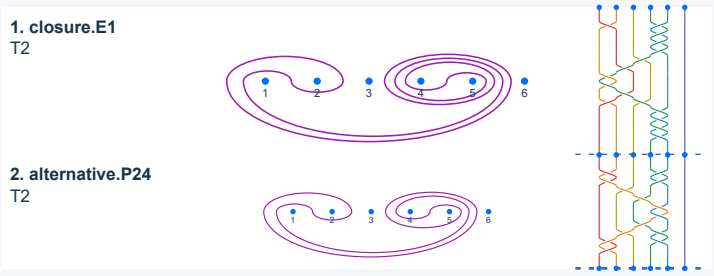

## VII - step 090

Drag alternative.P24 up

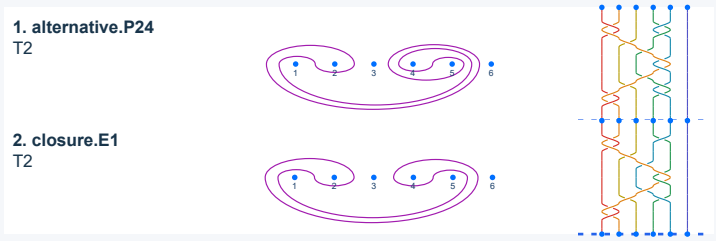
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 9 below. Only the changed interval is shown. Retained adjacent checks: 1.

### Before




### After



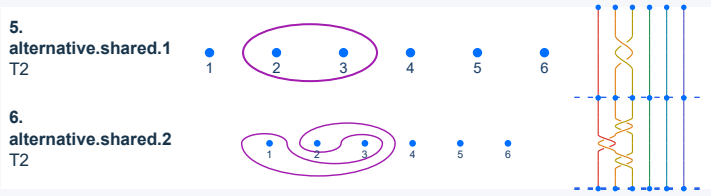
# VII - step 091

Drag alternative.shared.1 down

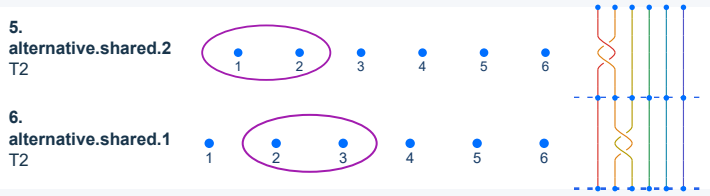
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

4 unchanged displayed factors above; 5 below. Only the changed interval is shown. Retained adjacent checks: 1.

## Before




## After



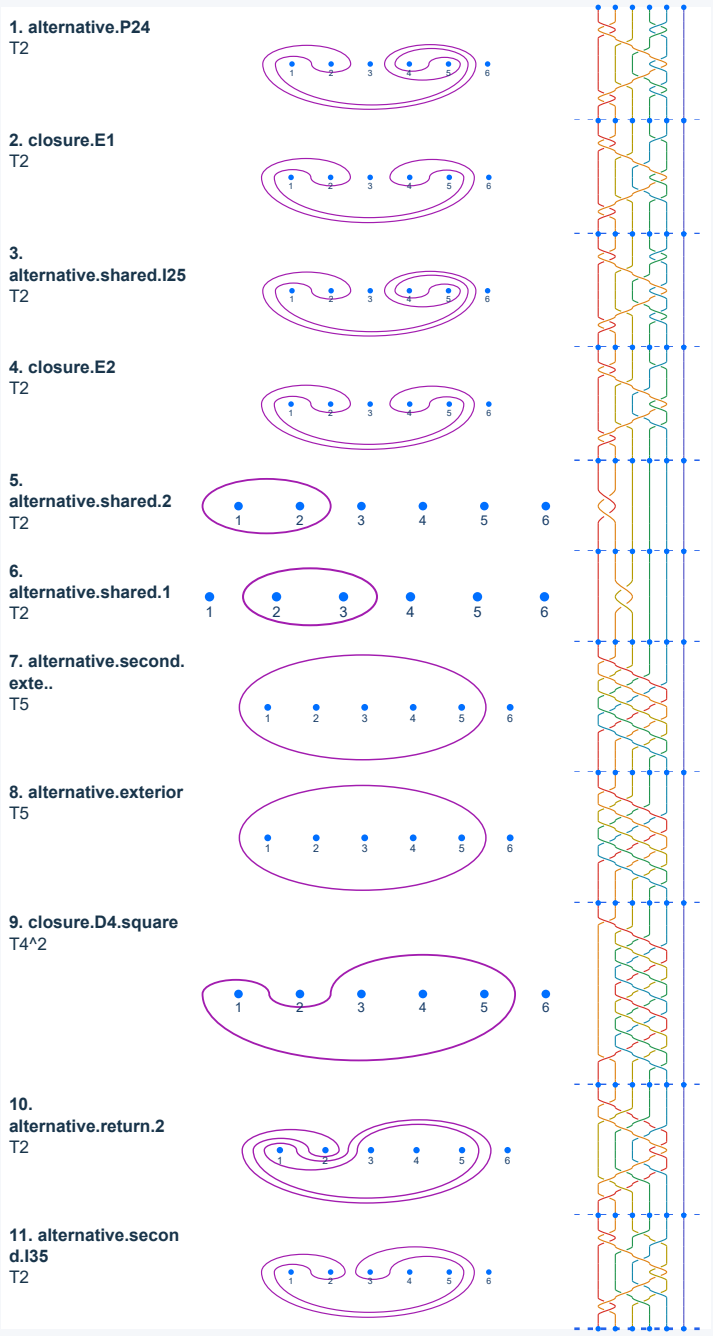
# VII - step 092

Cyclic start at the round four-point square

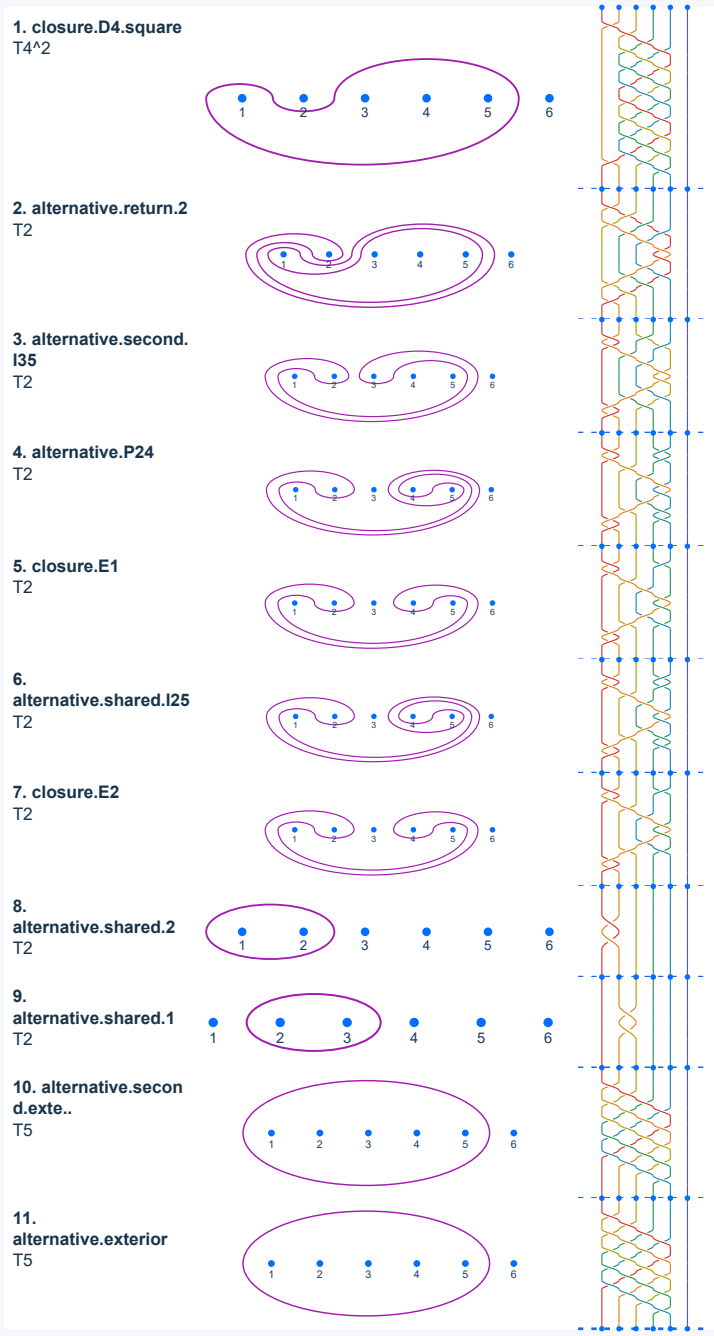
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. Explicit common conjugation; its full braid frame is saved.

## Before




## After



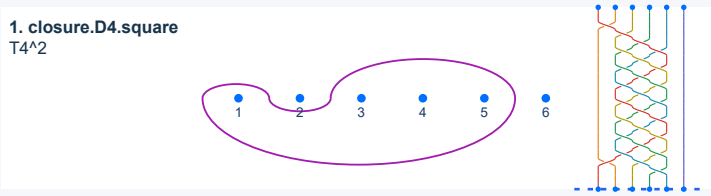
# VII - step 093

Split closure.D4.square into its two positive units

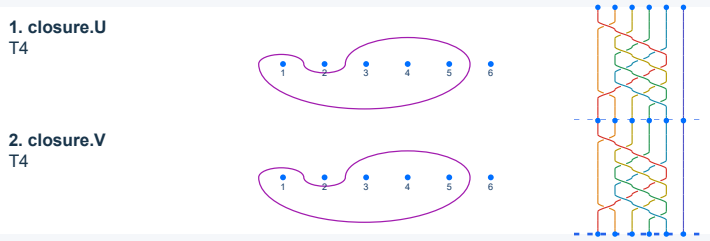
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 10 below. Only the changed interval is shown.

### Before




### After




## VII - step 094

Drag closure.U down

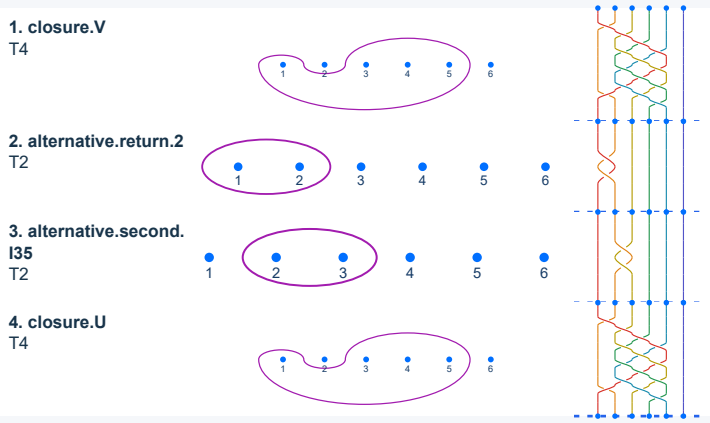
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 8 below. Only the changed interval is shown. Retained adjacent checks: 3.

### Before




### After



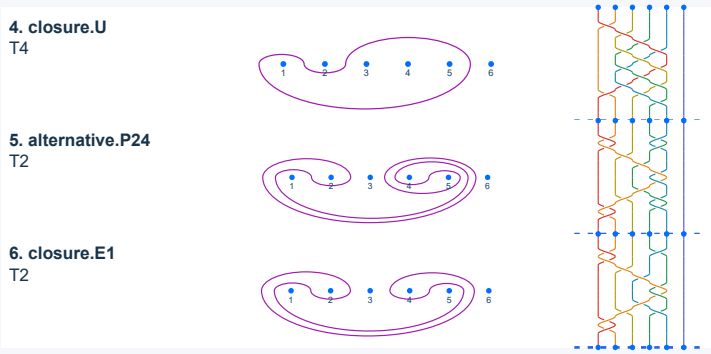
# VII - step 095

Drag closure.U down

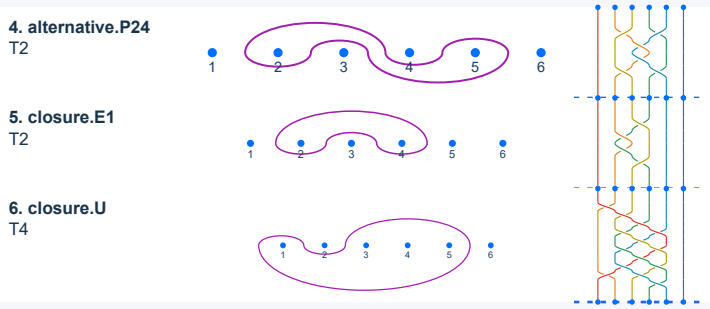
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

3 unchanged displayed factors above; 6 below. Only the changed interval is shown. Retained adjacent checks: 2.

## Before




## After



# VIII. Two daisies, four exteriors and the sphere cap

Actual ordered factor supports and their connected braids. All unchanged context is counted, and every retained move is shown.

# Complete factorization before Two daisies, four exteriors and the sphere cap

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

**1. closure.V**
T4



**2. alternative.return.2**
T2



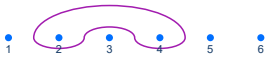
**3. alternative.second.l35**
T2



**4. alternative.P24**
T2



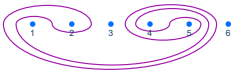
**5. closure.E1**
T2



**6. closure.U**
T4



**7. alternative.shared.l25**
T2



**8. closure.E2**
T2



**9. alternative.shared.2**
T2



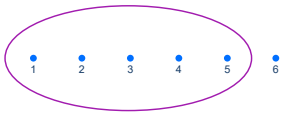
**10. alternative.shared.1**
T2

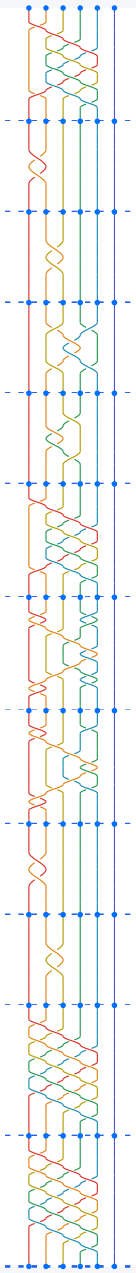


**11. alternative.second.exte..**
T5



**12. alternative.exterior**
T5





# VIII - step 096

First actual transported five-point PC daisy

DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

5 unchanged displayed factors above; 2 below. Only the changed interval is shown.

### Before




### After




## VIII - step 097

Second actual transported five-point PC daisy

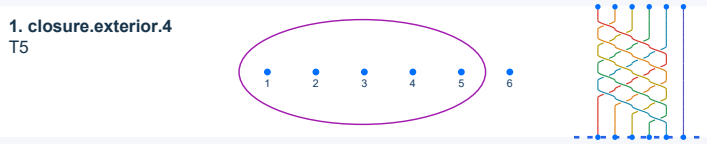
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 3 below. Only the changed interval is shown.

### Before




### After



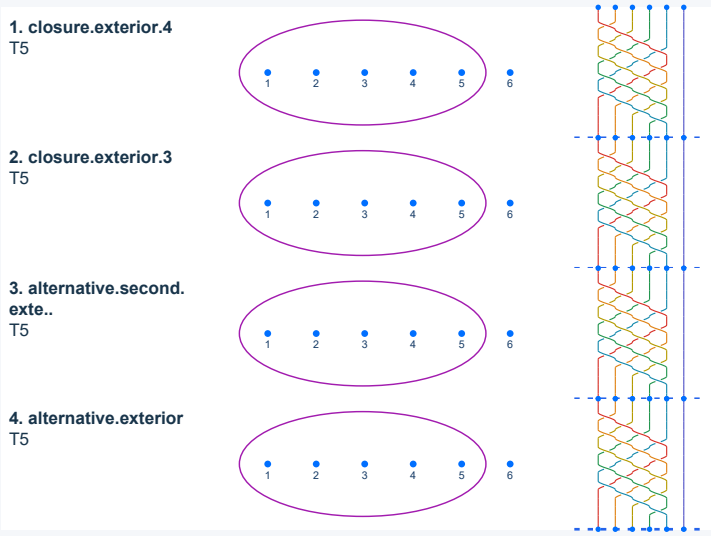
# VIII - step 098

Combine the FOUR identical visible exterior twists

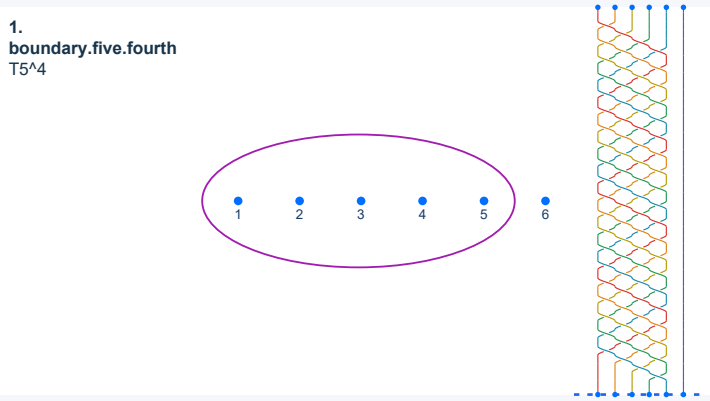
DISK MOVE: exact local free-word, based-action, or supported-class certificate retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

### Before




### After



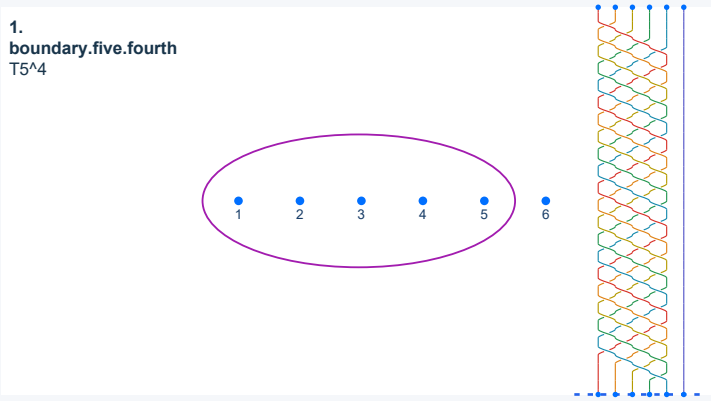

## VIII - step 099

SPHERE MOVE: cap the FOURTH boundary power

SPHERE MOVE: exact sphere action and +I lift checked; disk discrepancy retained.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown. SPHERE MOVE: full disk discrepancy appears in Appendix IX; chosen lift is +I.

### Before




### After

Identity / no factors in this interval.

# IX. Sphere discrepancies and certificates

The following are ALL FOUR sphere operations in this named route. They are not disk braid equalities. Each discrepancy is written in signed generator indices; the exact sphere certificate and +I lift are retained online.

# SPHERE MOVE at step 14

SPHERE LANTERN: absorb the matched pair and the two prefix roots

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

**FULL DISK DISCREPANCY** (66 signed braid letters):

5 -4 2 3 2 3 2 3 4 -5 -4 -2 3 3 2 -5 -2 -3 4 2 3 2 3 2
3 -4 3 2 5 4 2 3 2 5 -3 -3 4 1 2 3 4 5 1 2 3 4 -5 -1
-1 -5 -4 -3 -2 -1 -5 -4 -3 -2 -1 -4 3 3 -5 -2 -3 -2

# SPHERE MOVE at step 16

SPHERE RECHART: use the five-point disk; retain SIX boundary corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

**FULL DISK DISCREPANCY** (482 signed braid letters):

5 4 4 1 2 3 4 5 1 2 3 4 5 1 1 5 -4 -3 -2 -1 -5 -4 -3 -2
-1 -4 3 3 -5 -2 -3 -2 -2 3 4 5 4 5 4 5 4 5 4 5 -3 2
3 2 4 4 -3 2 1 -2 3 -4 -4 -2 -3 -2 3 -4 2 3 2 3 2 3 2 3
2 3 2 3 4 -3 2 -4 -2 -3 2 4 -5 -3 -3 4 3 3 5 -4 -2 5 4 2
3 2 3 2 3 -4 -5 3 2 4 -2 5 3 4 3 4 3 4 -5 2 -5 -4 2 3
2 3 3 -2 1 2 -3 2 3 4 5 4 5 4 5 4 -1 -2 3 -4 3 4 -3 2
1 5 -1 -2 3 4 3 4 3 4 3 4 3 4 3 4 3 4 2 1 -4 2 3 2 3 2
3 2 3 2 3 2 3 4 2 3 2 5 -3 -3 -4 -5 -4 -4 -3 -2 -1 -2 -1 -2
-1 -2 -1 -2 -1 -2 -1 3 -4 -3 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 3 1
1 1 -4 -3 -2 -1 2 3 4 -1 -1 -1 4 4 4 1 1 2 -4 -3 -4 -3 -4 -3
-2 -1 -1 -4 -3 -4 -3 1 2 2 1 -4 -3 -2 -1 2 3 4 -1 -2 -2 -1 3 4
3 -4 -4 -3 -2 -1 -2 -1 -2 -1 3 4 4 1 -3 -2 -3 -2 -4 -3 -2 -3 -2 -3
-2 4 -3 -2 -1 4 -2 3 -4 -3 -3 2 -4 -3 -2 3 4 -2 3 3 4 -3 2 -4
-3 -2 -3 -2 -3 -2 -3 -2 -3 -2 -3 -2 4 -1 -2 -4 -3 -3 -2 3 3 4 2 1
-2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -2 -1 -1 -1 -4 2 1 1 2 3 -4 -4 -3 -2
-1 -1 -2 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3 -2 -1 -4 -3
-2 -1

# SPHERE MOVE at step 17

SPHERE MOVE: cap the SIX framing corrections

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

**FULL DISK DISCREPANCY** (120 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4

# SPHERE MOVE at step 99

SPHERE MOVE: cap the FOURTH boundary power

Exact sphere action certified. Chosen genus-two lift discrepancy: +I. This is not an equality in the ordinary disk braid group.

**FULL DISK DISCREPANCY** (80 signed braid letters):

1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4
1 2 3 4 1 2 3 4

# Certificates and exact scope

Complete named graphical route:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/index.html?phase=namedfull

Full raw certificates:
https://richardbuckman-math.github.io/surface-diagrams/docs/proofs/six-seven-fifty-blocks/certificate.json

The independent six-point sphere cut comparison has six marked points and five consecutive-point arcs. It is separately jointly certified; the individual factor supports in this book are not joint cut complexes.

The original 23-operation closure remains available as a separate chapter. In this complete route its rounding frame is omitted, and two earlier frames combine to sigma4. The two final daisies are matched in their transported chart. All original local and internal named-pattern certificates are retained; the full disk tail stays within support maximum20. Four actual exterior twists remain visible. The goal of an under-twenty complete derivation remains open.

The archived 125-page/177-step PDF is preserved separately. This edition contains the new 99-step named proof, with no action-only boundary-recognition step.

**Independent jointly certified six-point sphere cut comparison**

